

Lectures on Groups and Lie Algebras

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Lecture Notes

Mathematics

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Contents

1	Symmetries	1
2	Basic group theory	4
2.1	Homomorphisms and isomorphisms	4
2.2	Subgroups	7
2.3	Conjugacy classes	9
2.4	Normal subgroups	12
2.5	Extensions	14
3	Representation theory	18
3.1	Group actions on sets	18
3.2	Representations on vector spaces	23
3.2.1	Representations and intertwiners	23
3.2.2	Irreducibility and Schur's lemma	25
3.2.3	Central elements and abelian groups	27
3.2.4	Decomposition of representations	29
3.2.5	Character theory	36
3.3	Representations of S_n	45
3.4	Applications of group theory in quantum mechanics	53
3.4.1	Symmetries of the Hamiltonian	53
3.4.2	Selection rules	58
4	Lie groups	63
4.1	A few basics from differential geometry	64
4.2	Lie algebras	67
5	Representation theory of $\mathfrak{su}(n)$	77
5.1	Representation theory of $\mathfrak{su}(2)$ and $\mathfrak{sl}(2)$	77
5.2	Representation theory of $\mathfrak{su}(3)$ and $\mathfrak{sl}(3, \mathbb{C})$	84
6	Simple Lie algebras	95
6.1	Ideals, simple and semisimple Lie algebras	95
6.2	Cartan subalgebras and the root space decomposition	97
6.3	The Killing form and structure constants	98
6.4	Simple roots, Dynkin diagrams and the classification	102
6.5	Root strings and the Weyl group	105
6.6	Highest weight representations	109

6.7	The Weyl character formula	114
7	Representations of compact Lie groups	117
7.1	Integration on manifolds	117
	Bibliography	126
	Index	127

Chapter 1

Symmetries

In physics symmetries are often first introduced in the context of classical mechanics in order to simplify computations.

Definition 1.0.1: Symmetry

A **symmetry** is an invertible map

$$s : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

such that, whenever $x(t)$ is a solution of the equations of motion, also

$$s \circ x(t)$$

is a solution.

Example.

Consider a free particle, so that $\vec{F} = 0$. Then the equations of motion have solutions

$$x(t) = x_0 + tv_0, \quad x_0, v_0 \in \mathbb{R}^n.$$

For any $T \in \mathbb{R}^n$, the translation

$$s_T : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad x \mapsto x + T$$

is a symmetry.

Example.

In quantum mechanics, a symmetry is typically a unitary operator

$$S : \mathcal{H} \rightarrow \mathcal{H}$$

on the Hilbert space $\mathcal{H}$ such that

$$[S, H] = 0,$$

where $H : \mathcal{H} \rightarrow \mathcal{H}$ is the Hamiltonian.

For

$$\mathcal{H} = L^2(\mathbb{R}), \quad H = -\frac{1}{2m}\partial_x^2,$$

the translation operator

$$S_T : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R}), \quad (S_T f)(x) = f(x + T),$$

and the reflection operator

$$S_R : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R}), \quad (S_R f)(x) = f(-x),$$

are symmetries.

Abstract structure	Concrete realization
group G	representation ρ_V
$g \in G$	$\rho(g) : V \rightarrow V$
$e \in G$ (“do nothing”)	$\rho(e) = \text{id}_V$
$h \circ g \in G$ (symmetry composition)	$\rho(h \circ g) = \rho(h) \rho(g)$
$\exists g^{-1}$ s.t. $g^{-1} \circ g = g \circ g^{-1} = e$	$\rho(g^{-1}) \rho(g) = \rho(g) \rho(g^{-1}) = \text{id}_V$

Definition 1.0.2: Group

A **group** is a set G together with a map

$$\circ : G \times G \rightarrow G, \quad (g, h) \mapsto g \circ h,$$

such that:

1. $\circ$ is associative, that is,

$$(g \circ h) \circ k = g \circ (h \circ k) \quad \forall g, h, k \in G.$$

2. There exists an element $e \in G$ such that

$$e \circ g = g \circ e = g \quad \forall g \in G.$$

3. For every $g \in G$, there exists an element $g^{-1} \in G$ such that

$$g \circ g^{-1} = g^{-1} \circ g = e.$$

Corollary 1.0.3

In a group, the unit element and inverses are unique.

Proof:

Suppose e and e' are both unit elements. Then

$$e = e \circ e' = e'.$$

Hence the unit is unique.

Now let $g \in G$, and suppose h and h' are both inverses of g . Then

$$h = h \circ e = h \circ (g \circ h') = (h \circ g) \circ h' = e \circ h' = h'.$$

Hence the inverse of any element is unique. ■

Definition 1.0.4: Representation

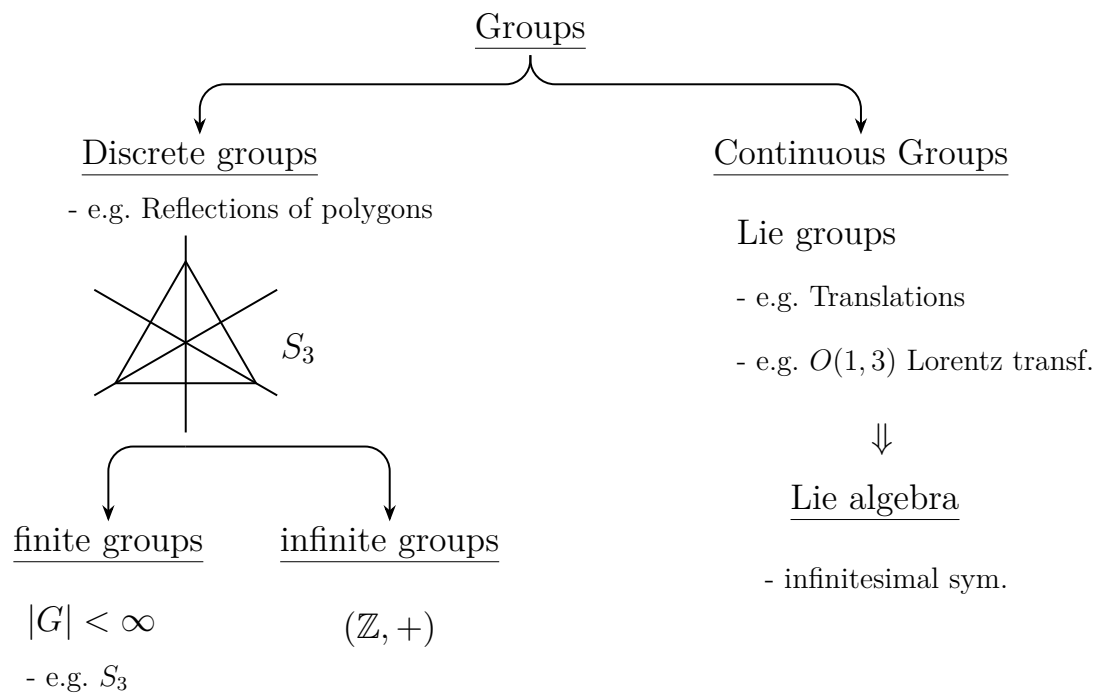
Let $(G, \circ)$ be a group. Usually we simply write G and leave the map $\circ$ implicit. A **representation** of G is a vector space V together with a map

$$\rho : G \rightarrow \text{End}(V)$$

such that:

1. $\rho(e) = \text{id}_V$.
2. $\rho(g)\rho(h) = \rho(g \circ h) \quad \forall g, h \in G$.

We will study these two definitions in this course. The basic structure of the course is as follows:



Chapter 2

Basic group theory

2.1 Homomorphisms and isomorphisms

Definition 2.1.1: Group homomorphism

Let G and H be groups. A **group homomorphism** is a map

$$f : G \rightarrow H$$

such that

$$f(g_1 \circ_G g_2) = f(g_1) \circ_H f(g_2) \quad \forall g_1, g_2 \in G.$$

Corollary 2.1.2

If $f : G \rightarrow H$ is a group homomorphism, then

$$f(e_G) = e_H.$$

Corollary 2.1.3

If $f : G \rightarrow H$ and $\tilde{f} : H \rightarrow K$ are group homomorphisms, then

$$\tilde{f} \circ f : G \rightarrow K$$

is again a group homomorphism.

Corollary 2.1.4

If $f : G \rightarrow H$ is a bijective group homomorphism, then

$$f^{-1} : H \rightarrow G$$

is also a group homomorphism.

Corollary 2.1.5

Every group homomorphism preserves inverses, i.e. if $f : G \rightarrow H$ is a group homomorphism, then

$$f(g^{-1}) = f(g)^{-1} \quad \forall g \in G.$$

Proof:

Since $gg^{-1} = e_G$, we get

$$f(g)f(g^{-1}) = f(gg^{-1}) = f(e_G) = e_H.$$

Similarly,

$$f(g^{-1})f(g) = f(g^{-1}g) = f(e_G) = e_H.$$

Hence $f(g^{-1})$ is the inverse of $f(g)$, so $f(g^{-1}) = f(g)^{-1}$. ■

Definition 2.1.6: Group isomorphism

An invertible group homomorphism is called a **group isomorphism**. If there exists a group isomorphism

$$f : G \rightarrow H$$

then we say that G and H are **isomorphic**.

Example.

Examples of discrete groups:

- $(\mathbb{Z}, +)$.
- $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$.
- $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{C} \setminus \{0\}, \cdot)$.
- Let $n \in \mathbb{N}_{>0}$. Then $\mathbb{Z}/n\mathbb{Z}$ is the quotient of $\mathbb{Z}$ by the equivalence relation

$$a \sim b \quad \Longleftrightarrow \quad a = b + kn \quad \text{for some } k \in \mathbb{Z}.$$

The equivalence class of a is $[a] = \{b \in \mathbb{Z} \mid b \sim a\}$, and $\mathbb{Z}/n\mathbb{Z}$ becomes a group with operation

$$[a] \cdot [b] = [a + b].$$

- The permutation group S_n , for $n \in \mathbb{N}_{>0}$, is the group of all permutations of $\{1, 2, \dots, n\}$. With composition as multiplication, S_n forms a group, and $|S_n| = n!$.

For example, in S_3 let

$$\sigma_1 = \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array}, \quad \sigma_2 = \begin{array}{ccc} 1 & 2 & 3 \\ | & \times & \\ 1 & 2 & 3 \end{array}$$

Then

$$\sigma_2 \circ \sigma_1 = \begin{array}{ccc} 1 & 2 & 3 \\ & \diagdown & \diagup \\ & 1 & 2 \\ & \diagup & \diagdown \\ 1 & 2 & 3 \end{array}.$$

Example.

Examples of continuous groups:

- The general linear group

$$\mathrm{GL}(n, \mathbb{R}) = \{M : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid M \text{ linear, } \det M \neq 0\}.$$

- The orthogonal group

$$\mathrm{O}(n) = \{M : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid M \text{ linear, } M^T M = I\}.$$

Equivalently, $M \in \mathrm{O}(n)$ iff $\det M = \pm 1$.

- More generally, let $T_{ij} \in T\mathbb{R}^{n*} \otimes T\mathbb{R}^{n*}$ be a tensor. Its symmetry group is

$$\mathrm{Sym}(T_{ij}) = \{M^i_j \in \mathrm{GL}(n, \mathbb{R}) \mid T_{ij} = T_{ab} M^a_i M^b_j\}.$$

If $\langle v, w \rangle = T_{ij} v^i w^j$, this is equivalently the group preserving the bilinear form defined by T .

For

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

one gets $\mathrm{Sym}(T) = \mathrm{O}(1, 3)$. For the standard symplectic form

$$T = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

one gets the symplectic group.

- An **affine transformation** is a map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ of the form

$$x \mapsto Mx + T, \quad M \in \mathrm{GL}(n, \mathbb{R}), \quad T \in \mathbb{R}^n.$$

We denote the group of affine transformations by $\mathrm{Aff}(\mathbb{R}^n)$.

- If $B \subseteq \mathbb{R}^n$, define

$$\mathrm{Sym}(B) = \{A \in \mathrm{Aff}(\mathbb{R}^n) \mid A(B) = B\}.$$

For an equilateral triangle Δ , one has $\mathrm{Sym}(\Delta) \cong S_3$.

2.2 Subgroups

Definition 2.2.1: Subgroup

Let G be a group. A **subgroup** $H \leq G$ is a subset $H \subseteq G$ such that H is itself a group:

1. $h \circ_G h' \in H$ for all $h, h' \in H$,
2. if $h \in H$, then $h^{-1} \in H$.

Example.

Examples:

- $\text{SO}(n) \leq \text{O}(n) \leq \text{GL}(n)$.
- $S_2 \leq S_3 \leq S_4$.

This notation is slightly sloppy: strictly speaking, one should specify an injective group homomorphism $H \hookrightarrow G$. For example,

$$H = \{\sigma \in S_4 \mid \sigma(4) = 4\}$$

is isomorphic to S_3 .

Proposition 2.2.2

Let $f : G \rightarrow H$ be a group homomorphism. Then the kernel of f ,

$$\ker f := \{g \in G \mid f(g) = e_H\} \leq G,$$

is a subgroup. Likewise, the image of f ,

$$\text{im } f := \{h \in H \mid \exists g \in G, f(g) = h\} \leq H,$$

is a subgroup.

Proof:

Let $g_1, g_2 \in \ker f$. Then

$$f(g_1 \circ g_2) = f(g_1)f(g_2) = e_H,$$

so $g_1 \circ g_2 \in \ker f$. Also,

$$f(g^{-1}) = f(g)^{-1} = e_H$$

for every $g \in \ker f$, hence $g^{-1} \in \ker f$. Therefore $\ker f$ is a subgroup.

Now let $h_1, h_2 \in \text{im } f$. Then there exist $g_1, g_2 \in G$ such that $f(g_1) = h_1$ and $f(g_2) = h_2$. Hence

$$h_1 h_2 = f(g_1)f(g_2) = f(g_1 \circ g_2) \in \text{im } f.$$

Moreover, if $h = f(g) \in \text{im } f$, then

$$h^{-1} = f(g)^{-1} = f(g^{-1}) \in \text{im } f.$$

Thus $\text{im } f$ is a subgroup. ■

Example.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Then there is a group homomorphism

$$\text{Sym}(\Lambda) \rightarrow \text{O}(n), \quad f(x) = Ax + b \mapsto A.$$

The kernel is the group of translational symmetries of Λ , and the image is called the **point group**.

Example.

There is a group homomorphism

$$\text{sign} : S_n \rightarrow \{\pm 1\}.$$

It records whether a permutation is even or odd: even permutations are sent to 1, odd permutations to -1 .

For example,

$$\sigma = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ & \diagdown & \diagup & \\ & 1 & 2 & \\ & \diagup & \diagdown & \\ 1 & 2 & 3 & 4 \end{array} \Rightarrow \text{sign}(\sigma) = -1,$$

while

$$\tilde{\sigma} = \begin{array}{cccc} 1 & 2 & 3 & 4 \\ & \diagdown & \diagup & \\ & 1 & 2 & \\ & \diagup & \diagdown & \\ 1 & 2 & 3 & 4 \end{array} \Rightarrow \text{sign}(\tilde{\sigma}) = 1.$$

Definition 2.2.3: Alternating group

The **alternating group** A_n is the kernel of the sign homomorphism:

$$A_n := \ker(\text{sign}) \leq S_n.$$

Equivalently, A_n is the subgroup of even permutations.

Example.

For $n = 3$,

$$A_3 = \left\{ \begin{array}{ccc} 1 & 2 & 3 \\ \parallel & \parallel & \parallel \\ 1 & 2 & 3 \end{array}, \begin{array}{ccc} 1 & 2 & 3 \\ & \diagdown & \diagup \\ & 1 & 2 \\ & \diagup & \diagdown \\ 1 & 2 & 3 \end{array}, \begin{array}{ccc} 1 & 2 & 3 \\ & \diagup & \diagdown \\ & 1 & 2 \\ & \diagdown & \diagup \\ 1 & 2 & 3 \end{array} \right\}.$$

Hence A_3 has three elements and is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

2.3 Conjugacy classes

Definition 2.3.1: Conjugation

Let G be a group and $g \in G$. Define the group homomorphism

$$\text{Ad}_g : G \rightarrow G, \quad h \mapsto ghg^{-1}.$$

This is called **conjugation by g** .

Indeed,

$$\text{Ad}_g(e) = geg^{-1} = e,$$

and

$$\text{Ad}_g(hh') = gh'h'^{-1} = ghg^{-1}gh'h'^{-1} = \text{Ad}_g(h)\text{Ad}_g(h').$$

Definition 2.3.2: Conjugate Elements

Let $h, h' \in G$. We say that h and h' are **conjugate** if there exists $g \in G$ such that

$$h = gh'g^{-1} = \text{Ad}_g(h').$$

Proposition 2.3.3

Being conjugate is an equivalence relation on G .

Proof:

- Reflexivity: every element is conjugate to itself, since $h = ehe^{-1}$.
- Symmetry: if $h \sim h'$, so $h = gh'g^{-1}$, then $h' = g^{-1}hg$, hence $h' \sim h$.
- Transitivity: if $h_1 \sim h_2$ and $h_2 \sim h_3$, say $h_1 = g_1h_2g_1^{-1}$ and $h_2 = g_2h_3g_2^{-1}$, then

$$h_1 = g_1g_2h_3(g_1g_2)^{-1},$$

so $h_1 \sim h_3$. ■

Definition 2.3.4: Conjugacy Classes

The equivalence classes of the conjugation relation are called **conjugacy classes**. The set of all conjugacy classes is denoted by $\text{conj}(G)$.

Example.

The conjugacy classes of S_3 are

$$\text{conj}(S_3) = \left\{ \{e\}, \left\{ \begin{array}{ccc} 1 & 2 & \\ & \times & \\ 1 & & 2 \end{array} \right\}, \left\{ \begin{array}{ccc} 3 & & \\ | & & \\ 3 & & \end{array} \right\}, \left\{ \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array} \right\}, \left\{ \begin{array}{ccc} 2 & 3 & \\ & \times & \\ 2 & & 3 \end{array} \right\} \right\},$$

$$\left\{ \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array} \right\}, \left\{ \begin{array}{ccc} 1 & 2 & 3 \\ & \times & \\ 1 & 2 & 3 \end{array} \right\} \right\}.$$

Remark.

A useful way to think about conjugacy classes is that conjugation changes the *description* of an object, but not the underlying structure.

For matrices, if $M \in \text{GL}(n, \mathbb{R})$, then

$$UMU^{-1}$$

is the same linear map written in a different basis.

For permutations, if $\sigma \in S_n$, then

$$\tau\sigma\tau^{-1}$$

simply relabels the elements of $\{1, \dots, n\}$. This is why the conjugacy classes in S_n are determined by cycle type.

Normal subgroups are exactly those subgroups that are stable under conjugation: if $N \trianglelefteq G$, then $gn g^{-1} \in N$ for all $g \in G$ and $n \in N$. This stability is what makes the quotient G/N well defined. This will be made precise in Section 2.4 .

Definition 2.3.5: Abelian Group

A group G is called **abelian** if

$$hg = gh \quad \forall h, g \in G.$$

Corollary 2.3.6

$$\text{conj}(G) = G \quad \Longleftrightarrow \quad G \text{ is abelian.}$$

Definition 2.3.7: Cycle

For $1 \leq k \leq n$, let $a_1, \dots, a_k \in \{1, 2, \dots, n\}$ be pairwise distinct. A **cycle** $(a_1, \dots, a_k)$ is the permutation

$$a_1 \mapsto a_2, \quad a_2 \mapsto a_3, \quad \dots, \quad a_k \mapsto a_1,$$

which acts as the identity on all other elements. We call k the **length** of the cycle.

Example.

To describe $\text{conj}(S_n)$, it is useful to use cycle type.

For $\sigma \in S_n$, the **cycle type** of σ is the list of lengths of the cycles in a disjoint cycle decomposition of σ , written in decreasing order. Thus the cycle type is a partition

$$\lambda_1 \geq \cdots \geq \lambda_k > 0, \quad \sum_{i=1}^k \lambda_i = n.$$

For example, $(1\ 3\ 2)(4\ 5) \in S_5$ has cycle type $(3, 2)$.

Theorem 2.3.8: Conjugacy In S_n

Two permutations $\sigma, \sigma' \in S_n$ are conjugate if and only if they have the same cycle type.

Proof:

Let $\tau, \sigma \in S_n$. Then

$$(\tau\sigma\tau^{-1})(\tau(i)) = \tau(\sigma(i)).$$

So conjugation by τ simply applies τ to the entries in each cycle. In particular,

$$\tau(a_1 \dots a_r)\tau^{-1} = (\tau(a_1) \dots \tau(a_r)).$$

Hence conjugate permutations have the same cycle type.

Conversely, if σ and σ' have the same cycle type, then their disjoint cycles can be matched cycle by cycle. Choosing τ so that it sends the entries of each cycle of σ to the corresponding entries of the matching cycle of σ' , we get

$$\tau\sigma\tau^{-1} = \sigma'.$$

Example.

The conjugacy classes of S_4 are in bijection with the partitions of 4:

$$(1, 1, 1, 1), \quad (2, 1, 1), \quad (3, 1), \quad (2, 2), \quad (4).$$

Representatives are:

- $(1, 1, 1, 1)$: e ,
- $(2, 1, 1)$: a transposition, e.g. $(1\ 2)$,
- $(3, 1)$: a 3-cycle, e.g. $(1\ 2\ 3)$,
- $(2, 2)$: a product of two disjoint transpositions, e.g. $(1\ 2)(3\ 4)$,
- (4) : a 4-cycle, e.g. $(1\ 2\ 3\ 4)$.

2.4 Normal subgroups

Definition 2.4.1: Normal Subgroup

A subgroup $N \leq G$ is called **normal** if

$$gng^{-1} \in N \quad \forall n \in N, \forall g \in G.$$

Example.

In S_3 , the subgroup

$$\overline{N} = \{e, (12)\}$$

is not normal, since

$$(13)(12)(13)^{-1} = (23) \notin \overline{N}.$$

On the other hand,

$$N = \{e, (123), (132)\}$$

is a normal subgroup of S_3 .

Theorem 2.4.2: Quotient Group

Let $N \leq G$ be a normal subgroup. Let G/N be the set of equivalence classes for the relation $g \sim gn$ for $n \in N$. Then G/N is a group with multiplication

$$[g] \cdot [h] = [gh].$$

Proof:

We first check that the multiplication is well defined. If $g' = gn_1$ and $h' = hn_2$ with $n_1, n_2 \in N$, then

$$g'h' = gn_1hn_2 = gh(h^{-1}n_1h)n_2.$$

Since N is normal, $h^{-1}n_1h \in N$, hence $(h^{-1}n_1h)n_2 \in N$. Therefore $g'h' \sim gh$, so $[g'][h'] = [gh]$.

The group axioms are inherited from G : associativity follows from that in G , the neutral element is $[e]$, and the inverse of $[g]$ is $[g^{-1}]$.

The classes $[g] = gN \subseteq G$ are also called **cosets**. ■

Example.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Then the group of translational symmetries of Λ is normal, and

$$\text{Sym}(\Lambda)/\text{Trans}(\Lambda)$$

is the point group.

Another important example is $\mathbb{Z}^2 \leq \mathbb{R}^2$. Then $\mathbb{Z}^2$ is a normal subgroup of $(\mathbb{R}^2, +)$, and the quotient

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2$$

is the 2-torus.

Theorem 2.4.3: Isomorphism Theorem

Let $\varphi : G \rightarrow H$ be a group homomorphism.

1. The kernel $\ker \varphi$ is a normal subgroup of G .
2. The map

$$\Psi : G / \ker \varphi \rightarrow \operatorname{im} \varphi, \quad [g] \mapsto \varphi(g),$$

is a well-defined group isomorphism.

Proof:

Let $n \in \ker \varphi$ and $g \in G$. Then

$$\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g)e_H\varphi(g)^{-1} = e_H,$$

so $gng^{-1} \in \ker \varphi$. Hence $\ker \varphi \trianglelefteq G$.

Now define

$$\Psi([g]) := \varphi(g).$$

To see that Ψ is well defined, let $g' = gn$ with $n \in \ker \varphi$. Then

$$\varphi(g') = \varphi(gn) = \varphi(g)\varphi(n) = \varphi(g).$$

Also,

$$\Psi([g][h]) = \Psi([gh]) = \varphi(gh) = \varphi(g)\varphi(h) = \Psi([g])\Psi([h]),$$

so Ψ is a group homomorphism.

It is surjective by definition of $\operatorname{im} \varphi$. Finally, if $\Psi([g]) = \Psi([g'])$, then

$$\varphi(g) = \varphi(g') \implies \varphi(g^{-1}g') = e_H,$$

hence $g^{-1}g' \in \ker \varphi$, so $g' = g(g^{-1}g')$ and therefore $[g] = [g']$. Thus Ψ is injective. ■

Example.

Examples:

- If $f : G \rightarrow H$ is surjective, then $\operatorname{im} f = H$, so the isomorphism theorem gives

$$G / \ker f \cong H.$$

- Every finite abelian group A is a quotient of $\mathbb{Z}^n$ for some n . Hence

$$A \cong \mathbb{Z}^n / \ker f$$

for a suitable surjective homomorphism $f : \mathbb{Z}^n \rightarrow A$.

- Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. The homomorphism

$$\operatorname{Sym}(\Lambda) \rightarrow \operatorname{O}(n), \quad f(x) = Ax + b \mapsto A,$$

has kernel $\text{Trans}(\Lambda)$. Therefore

$$\text{Sym}(\Lambda)/\text{Trans}(\Lambda) \cong \text{im } f,$$

and $\text{im } f$ is the point group of Λ .

- The determinant defines a surjective homomorphism

$$\det : \text{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}_{>0},$$

where $\mathbb{R}_{>0}$ is viewed as a group under multiplication, and

$$\ker(\det) = \text{SL}(n, \mathbb{R}).$$

Hence

$$\text{GL}(n, \mathbb{R})/\text{SL}(n, \mathbb{R}) \cong \mathbb{R}_{>0}.$$

2.5 Extensions

Definition 2.5.1: Exact Sequence

A sequence of groups and group homomorphisms

$$\cdots \rightarrow G_i \xrightarrow{f_i} G_{i+1} \xrightarrow{f_{i+1}} G_{i+2} \rightarrow \cdots$$

is called **exact** if

$$\text{im}(f_i) = \ker(f_{i+1})$$

at every term.

Remark.

If $N \trianglelefteq G$, then the quotient map

$$\pi : G \rightarrow G/N, \quad g \mapsto [g],$$

fits into the exact sequence

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} G/N \rightarrow 1,$$

where $\iota : N \hookrightarrow G$ is the inclusion. Indeed, $\text{im}(\iota) = N = \ker(\pi)$.

By the isomorphism theorem, any short exact sequence

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$$

identifies N with a normal subgroup of G and H with the quotient G/N .

Definition 2.5.2: Extension

An **extension** of a group H by a group N is a short exact sequence

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1.$$

Definition 2.5.3: Split And Central Extensions

An extension

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$$

is called **split** if there exists a group homomorphism called a **section**

$$s : H \rightarrow G$$

such that

$$\pi \circ s = \text{id}_H.$$

It is called **central** if

$$\iota(n)g = g\iota(n) \quad \forall n \in N, \forall g \in G,$$

i.e. if $\iota(N)$ lies in the center $Z(G)$ of G .

Remark.

A split extension is the group-theoretic analogue of a trivial bundle. Indeed, the map $\pi : G \rightarrow H$ may be viewed as a bundle over H with fiber N , and a splitting $s : H \rightarrow G$ with $\pi \circ s = \text{id}_H$ is a global section. The existence of such a distinguished section means that the extension can be reconstructed from N and H without any twisting; in the simplest case one gets the direct product $N \times H$, and more generally a semidirect product.

Remark.

Central extensions are especially important in quantization. In classical mechanics one often starts with a symmetry group acting on the phase space, but after quantization the corresponding operators on the Hilbert space may realize the symmetry only up to phases. These extra phase factors are encoded by a central extension, typically by $U(1)$ or $\mathbb{R}$. In this sense, central extensions measure the obstruction to lifting a classical symmetry action to an honest linear action in quantum theory.

Example.

Examples:

- The product group $N \times H$ gives an extension

$$1 \rightarrow N \rightarrow N \times H \rightarrow H \rightarrow 1,$$

which is always split. If N is abelian, then this extension is also central.

- The symmetry group of a lattice fits into an extension

$$1 \rightarrow \text{Trans}(\Lambda) \rightarrow \text{Sym}(\Lambda) \rightarrow \text{point group} \rightarrow 1.$$

- The Lorentz group fits into the extension

$$1 \rightarrow \text{SO}^+(3, 1) \rightarrow \text{O}(3, 1) \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow 1,$$

which is split.

- The Poincare group P fits into the extension

$$1 \rightarrow \mathbb{R}^4 \rightarrow P \rightarrow \mathrm{O}(3, 1) \rightarrow 1,$$

which is split.

- The sequence

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 1$$

is a central extension, but it is not split.

Definition 2.5.4: Semidirect Product

Let $\psi : H \rightarrow \mathrm{Aut}(N)$ be a group homomorphism. The **semidirect product** $N \rtimes_{\psi} H$ is the group whose underlying set is $N \times H$, with multiplication

$$(n, h) \cdot (n', h') := (n \psi(h)[n'], hh').$$

In this notation one may think of the relation

$$hn = \psi(h)[n] h.$$

Remark.

There is a canonical split extension

$$1 \rightarrow N \xrightarrow{\iota} N \rtimes_{\psi} H \xrightarrow{\pi} H \rightarrow 1,$$

where

$$\iota(n) = (n, e_H), \quad \pi(n, h) = h, \quad s(h) = (e_N, h).$$

Thus every semidirect product gives a split extension, and conversely every split extension is isomorphic to a suitable semidirect product.

Theorem 2.5.5: Split Extensions And Semidirect Products

Let

$$1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \rightarrow 1$$

be an extension, and let $s : H \rightarrow G$ be a splitting. Then:

1. Conjugation by $s(h)$ defines a group homomorphism

$$\psi : H \rightarrow \mathrm{Aut}(N), \quad h \mapsto (n \mapsto \iota^{-1}(s(h)\iota(n)s(h)^{-1})).$$

2. The group G is isomorphic to the semidirect product

$$N \rtimes_{\psi} H.$$

Proof:

Since $\iota(N) \trianglelefteq G$, conjugation by $s(h)$ preserves $\iota(N)$, hence defines an automorphism of N . This gives the map $\psi : H \rightarrow \text{Aut}(N)$, and it is a group homomorphism because conjugation is compatible with multiplication.

Define

$$\Phi : N \rtimes_{\psi} H \rightarrow G, \quad (n, h) \mapsto \iota(n)s(h).$$

Using the definition of ψ , one checks

$$\Phi((n, h)(n', h')) = \Phi(n, h)\Phi(n', h').$$

Thus Φ is a homomorphism.

It is surjective since every $g \in G$ can be written as $g = \iota(n)s(h)$, where $h = \pi(g)$ and n is determined by the fact that $gs(h)^{-1} \in \ker \pi = \iota(N)$. It is injective because $\iota(n)s(h) = e$ implies $h = e$ after applying π , and then $n = e$. Hence Φ is an isomorphism. ■

Remark.

If an extension is both split and central, then the associated action $\psi : H \rightarrow \text{Aut}(N)$ is trivial, because

$$\iota(n)s(h) = s(h)\iota(n) \quad \forall n \in N, h \in H.$$

Therefore

$$N \rtimes_{\psi} H = N \times H,$$

so every split central extension is in fact a direct product.

Chapter 3

Representation theory

"How groups act on stuff."

3.1 Group actions on sets

Definition 3.1.1: Group Actions On Sets

Let G be a group and X a set.

A **left action** of G on X is a map

$$\triangleright : G \times X \rightarrow X, \quad (g, x) \mapsto g \triangleright x,$$

such that

$$h \triangleright (g \triangleright x) = (hg) \triangleright x, \quad e \triangleright x = x \quad \forall g, h \in G, \forall x \in X.$$

A **right action** of G on X is a map

$$\triangleleft : X \times G \rightarrow X, \quad (x, g) \mapsto x \triangleleft g,$$

such that

$$(x \triangleleft g) \triangleleft h = x \triangleleft (gh), \quad x \triangleleft e = x \quad \forall g, h \in G, \forall x \in X.$$

Equivalently, a left action is the same as a group homomorphism

$$\rho : G \rightarrow \text{Aut}(X), \quad g \mapsto (x \mapsto g \triangleright x),$$

and a right action is the same as an antihomomorphism

$$G \rightarrow \text{Aut}(X), \quad g \mapsto (x \mapsto x \triangleleft g).$$

Corollary 3.1.2

Left and right actions can be converted into each other using inverses. More precisely:

- if $\triangleright$ is a left action of G on X , then

$$x \triangleleft g := g^{-1} \triangleright x$$

defines a right action of G on X ;

- if $\triangleleft$ is a right action of G on X , then

$$g \triangleright x := x \triangleleft g^{-1}$$

defines a left action of G on X .

Proof:

Suppose $\triangleright$ is a left action and define $x \triangleleft g := g^{-1} \triangleright x$. Then

$$(x \triangleleft g) \triangleleft h = h^{-1} \triangleright (g^{-1} \triangleright x) = (h^{-1}g^{-1}) \triangleright x = (gh)^{-1} \triangleright x = x \triangleleft (gh),$$

and

$$x \triangleleft e = e^{-1} \triangleright x = x.$$

So this is a right action. The converse is proved in the same way. ■

Example.

Examples:

- Every group G acts trivially on any set X by

$$g \triangleright x = x \quad \forall g \in G, \forall x \in X.$$

- A group G acts on itself by conjugation:

$$g \triangleright h := ghg^{-1}.$$

Equivalently, this is the homomorphism

$$G \rightarrow \text{Aut}(G), \quad g \mapsto \text{Ad}_g.$$

- If $\Lambda \subseteq \mathbb{R}^n$ is a lattice, then its symmetry group $\text{Sym}(\Lambda)$ acts on Λ .
- The symmetric group S_n acts on the set $\{1, \dots, n\}$.
- The general linear group $\text{GL}(n, \mathbb{R})$ acts on $\mathbb{R}^n$.
- If $H \leq G$ is a subgroup, then G acts on the set of left cosets G/H by left multiplication:

$$g \triangleright [g'] := [gg'].$$

Definition 3.1.3: Orbit And Stabilizer

Let $\rho : G \rightarrow \text{Aut}(X)$ be an action and let $x \in X$.

The **orbit** of x , denoted by $\text{Orb}_G(x)$ or simply $\mathcal{O}_x$, is the set

$$\text{Orb}_G(x) = \{y \in X \mid g \triangleright x = y \text{ for some } g \in G\} \subseteq X.$$

The **stabilizer** of x , denoted by $\text{Stab}_G(x)$ or simply St_x , is the subgroup

$$\text{Stab}_G(x) = \{g \in G \mid g \triangleright x = x\} \leq G.$$

Example.

- Let G act on itself by conjugation. Then

$$\text{Orb}_G(g)$$

is the conjugacy class of g , and

$$\text{Stab}_G(g) = \{h \in G \mid hgh^{-1} = g\}$$

is the centralizer of g .

- The group $\text{SO}(2)$ acts on $\mathbb{R}^2$ by rotations. If $x \neq 0$, then

$$\text{Orb}_{\text{SO}(2)}(x) \cong S^1, \quad \text{Stab}_{\text{SO}(2)}(x) = \{e\}.$$

For the origin one has

$$\text{Orb}_{\text{SO}(2)}(0) = \{0\}, \quad \text{Stab}_{\text{SO}(2)}(0) = \text{SO}(2).$$

Theorem 3.1.4: Orbit-Stabilizer Theorem

Let $\rho : G \rightarrow \text{Aut}(X)$ be an action of a finite group G on a set X . Then for every $x \in X$,

$$|G| = |\text{Orb}_G(x)| |\text{Stab}_G(x)|.$$

Proof:

Write $St_x := \text{Stab}_G(x)$ and $\mathcal{O}_x := \text{Orb}_G(x)$. Since St_x need not be normal in G , the quotient G/St_x is considered only as the set of left cosets.

We claim that

$$\varphi : G/St_x \rightarrow \mathcal{O}_x, \quad [g] \mapsto g \triangleright x,$$

is a bijection.

First, φ is well-defined. If $[g'] = [g]$, then $g' = gh$ for some $h \in St_x$. Hence

$$\varphi([g']) = g' \triangleright x = (gh) \triangleright x = g \triangleright (h \triangleright x) = g \triangleright x = \varphi([g]).$$

The map is surjective by the definition of the orbit. To prove injectivity, suppose

$$g \triangleright x = g' \triangleright x.$$

Applying g^{-1} to both sides gives

$$(g^{-1}g') \triangleright x = x,$$

so $g^{-1}g' \in St_x$. Therefore $g' = g(g^{-1}g')$, and hence

$$[g'] = [g].$$

Thus φ is a bijection.

Consequently,

$$|G/St_x| = |\mathcal{O}_x|.$$

Since every left coset of St_x has $|St_x|$ elements,

$$|G| = |G/St_x| |St_x| = |\mathcal{O}_x| |St_x|.$$

Example.

- a) Let G be a finite group and let $H \leq G$. Then G acts on the set G/H of left cosets by left multiplication:

$$g \triangleright [g'] := [gg'].$$

The orbit of the coset $[e]$ is all of G/H , and its stabilizer is H . Indeed,

$$g \triangleright [e] = [e] \iff [g] = [e] \iff g \in H.$$

Hence the orbit-stabilizer theorem gives

$$|G| = |G/H| |H|.$$

In particular,

$$|H| \mid |G|.$$

Fix $g \in G$ and consider the cyclic subgroup

$$\langle g \rangle = \{g^k \mid k \in \mathbb{Z}\} \leq G.$$

Since G is finite, $\langle g \rangle$ is finite. Its size is called the **order** of g , denoted

$$\text{ord}(g) := |\langle g \rangle|.$$

Equivalently, $\text{ord}(g)$ is the smallest positive integer n such that $g^n = e$, and then

$$\langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}, \quad n = \text{ord}(g).$$

Applying the previous result to the subgroup $\langle g \rangle \leq G$ gives

$$\text{ord}(g) \mid |G|.$$

If $|G| = p$ is prime and $g \neq e$, then

$$\text{ord}(g) \neq 1 \quad \text{and} \quad \text{ord}(g) \mid p.$$

Therefore $\text{ord}(g) = p$, so

$$\langle g \rangle = G.$$

Thus every group of prime order is cyclic, and hence

$$G \cong \mathbb{Z}/p\mathbb{Z}.$$

b) What are the symmetries of a cube? Let

$$G = \text{Sym}([-1, 1]^3)$$

be the full symmetry group of the cube. Let F be the set of its faces. Then G acts on F . If $f \in F$ is one face, then the orbit of f is all of F , so

$$\mathcal{O}_f = F, \quad |\mathcal{O}_f| = 6.$$

The stabilizer of f is the symmetry group of a square face:

$$\text{Stab}_G(f) \cong D_4, \quad |D_4| = 8.$$

Hence the orbit-stabilizer theorem gives

$$|G| = |F| |D_4| = 6 \cdot 8 = 48.$$

Let $\text{Rot} \leq G$ be the subgroup of orientation preserving symmetries. The determinant gives a surjective homomorphism

$$\det : G \rightarrow \{\pm 1\} \cong \mathbb{Z}_2$$

with kernel Rot . Therefore

$$|G| = 2|\text{Rot}|, \quad |\text{Rot}| = 24.$$

There are four body diagonals in the cube. If D is the set of these diagonals, then rotations act on D , giving a homomorphism

$$\text{Rot} \rightarrow \text{Aut}(D) \cong S_4.$$

This action is faithful: a rotation fixing all four body diagonals must be the identity. Since both groups have order 24, we get

$$\text{Rot} \cong S_4.$$

Finally, the map

$$s : \mathbb{Z}_2 \rightarrow G, \quad -1 \mapsto -I_3$$

is a section of $\det$, and $-I_3$ commutes with every symmetry of the cube. Hence

$$G \cong S_4 \times \mathbb{Z}_2.$$

c) Let G act on itself by conjugation:

$$h \triangleright g := hgh^{-1}.$$

For $g \in G$, the orbit is the conjugacy class

$$\text{Orb}_G(g) = \{hgh^{-1} \mid h \in G\},$$

and the stabilizer is the centralizer

$$\text{Stab}_G(g) = \{h \in G \mid hgh^{-1} = g\} = \{h \in G \mid hg = gh\}.$$

Therefore the orbit-stabilizer theorem gives

$$|G| = |\{hgh^{-1} \mid h \in G\}| |\{h \in G \mid hg = gh\}|.$$

3.2 Representations on vector spaces

3.2.1 Representations and intertwiners

Remark.

Recall that a **representation** of a group G on a vector space V is a group homomorphism

$$\rho : G \rightarrow \text{Aut}(V).$$

Thus each $g \in G$ acts on V by a structure-preserving map, namely an invertible linear map.

Definition 3.2.1: Maps Of Representations

Let (V, ρ) and (V', ρ') be representations of a group G .

A **map of representations**, or **intertwiner**, is a linear map

$$f : V \rightarrow V'$$

such that for every $g \in G$, the diagram

$$\begin{array}{ccc} V & \xrightarrow{f} & V' \\ \rho(g) \downarrow & & \downarrow \rho'(g) \\ V & \xrightarrow{f} & V' \end{array}$$

commutes. Equivalently,

$$f \circ \rho(g) = \rho'(g) \circ f \quad \forall g \in G.$$

Example.

- Let V be a vector space. Every group G has the **trivial representation** on V , defined by

$$\rho(g) = 1_V \quad \forall g \in G.$$

- The group $\mathbb{Z}_2 = \{[0], [1]\}$ acts on any vector space V by

$$\rho([0]) = 1_V, \quad \rho([1]) = -1_V.$$

- The orthogonal group $O(n)$ acts on $\mathbb{R}^n$ by its defining representation

$$\rho(A)(v) := Av.$$

- The symmetric group S_n acts on $\mathbb{R}$ through the sign homomorphism:

$$S_n \xrightarrow{\text{sgn}} \mathbb{Z}_2 \rightarrow \text{Aut}(\mathbb{R}), \quad \sigma \mapsto (x \mapsto \text{sgn}(\sigma)x).$$

- The symmetric group S_n acts on $\mathbb{R}^n$ by permuting the standard basis:

$$\rho(\sigma)(e_i) := e_{\sigma(i)}.$$

- Let X be a set equipped with a G -action. The free vector space

$$\mathbb{K}[X] := \text{span}_{\mathbb{K}}\{x \mid x \in X\}$$

carries a representation of G by linear extension:

$$\rho(g) \left(\sum_i \lambda_i x_i \right) := \sum_i \lambda_i (g \triangleright x_i).$$

- Taking $X = G$ with the left multiplication action gives the **regular representation** of G on $\mathbb{K}[G]$:

$$\rho(g)(h) := gh.$$

- The symmetry group of an equilateral triangle,

$$\text{Sym}(\triangle) \cong S_3 \cong D_3,$$

acts on $\mathbb{R}^2$. For example, if (12) is a reflection and (123) is a rotation, then one can choose coordinates such that

$$\rho((12)) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho((123)) = \begin{pmatrix} \cos(2\pi/3) & -\sin(2\pi/3) \\ \sin(2\pi/3) & \cos(2\pi/3) \end{pmatrix}.$$

- If $\Lambda \subseteq \mathbb{R}^n$ is a lattice, then its point group acts on $\mathbb{R}^n$.

3.2.2 Irreducibility and Schur's lemma

Definition 3.2.2: Subrepresentations And Irreducibility

Let (V, ρ) be a representation of a group G .

A **subrepresentation** is a linear subspace $W \leq V$ such that

$$\rho(g)(w) \in W \quad \forall g \in G, \forall w \in W.$$

Equivalently, W is a G -invariant subspace of V .

The representation V is **irreducible** if its only subrepresentations are

$$\{0\} \quad \text{and} \quad V.$$

Proposition 3.2.3

Let

$$f : (V, \rho) \rightarrow (V', \rho')$$

be a map of G -representations. Then

$$\ker(f) \leq V \quad \text{and} \quad \operatorname{im}(f) \leq V'$$

are subrepresentations.

Proof:

Let $v \in \ker(f)$. Then for every $g \in G$,

$$f(\rho(g)(v)) = \rho'(g)(f(v)) = \rho'(g)(0) = 0.$$

Hence $\rho(g)(v) \in \ker(f)$, so $\ker(f)$ is a subrepresentation of V .

Now let $v' \in \operatorname{im}(f)$. Then $v' = f(v)$ for some $v \in V$. For every $g \in G$,

$$\rho'(g)(v') = \rho'(g)(f(v)) = f(\rho(g)(v)).$$

Thus $\rho'(g)(v') \in \operatorname{im}(f)$, so $\operatorname{im}(f)$ is a subrepresentation of V' . ■

Lemma 3.2.4: Schur's Lemma

Let (V, ρ) and (V', ρ') be irreducible representations of a group G . Then every map of G -representations

$$f : V \rightarrow V'$$

is either the zero map or an isomorphism.

Moreover, if V and V' are finite-dimensional complex vector spaces, then for any two isomorphisms of G -representations

$$f, f' : V \rightarrow V'$$

there exists $\lambda \in \mathbb{C}$ such that

$$f' = \lambda f.$$

Proof:

Let $f : V \rightarrow V'$ be a map of G -representations. By the previous proposition, $\ker(f)$ is a subrepresentation of V , and $\operatorname{im}(f)$ is a subrepresentation of V' . Since V and V' are irreducible, we have

$$\ker(f) \in \{\{0\}, V\}, \quad \operatorname{im}(f) \in \{\{0\}, V'\}.$$

If $\operatorname{im}(f) = \{0\}$, then $f = 0$. Otherwise $\operatorname{im}(f) = V'$. In this case $\ker(f) \neq V$, so $\ker(f) = \{0\}$. Hence f is both surjective and injective, and therefore an isomorphism.

Now assume that V and V' are finite-dimensional complex vector spaces, and let

$$f, f' : V \rightarrow V'$$

be isomorphisms of G -representations. Then

$$f^{-1} \circ f' : V \rightarrow V$$

is a map of G -representations. Since V is a finite-dimensional complex vector space, $f^{-1} \circ f'$ has an eigenvalue $\lambda \in \mathbb{C}$. Thus

$$f^{-1} \circ f' - \lambda 1_V$$

is not injective. By the first part, it cannot be an isomorphism, so it must be the zero map. Therefore

$$f^{-1} \circ f' = \lambda 1_V,$$

and multiplying by f gives

$$f' = \lambda f.$$

Example.

The second part of Schur's lemma need not hold over $\mathbb{R}$. Consider the representation of $\mathbb{Z}_4$ on $\mathbb{R}^2$ by rotations:

$$\rho([n]) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^n.$$

Then both

$$1_{\mathbb{R}^2} \quad \text{and} \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

are maps of $\mathbb{Z}_4$ -representations. Indeed, both commute with the action of the generator $[1]$, and hence with the action of every element of $\mathbb{Z}_4$.

However, these two maps are not scalar multiples of each other over $\mathbb{R}$, since the rotation matrix is not of the form $\lambda 1_{\mathbb{R}^2}$ for any $\lambda \in \mathbb{R}$. Thus the complex-vector-space assumption in Schur's lemma is essential for the scalar uniqueness statement.

Remark.

From now on, unless stated otherwise, representations will be taken over complex vector spaces.

3.2.3 Central elements and abelian groups

Corollary 3.2.5

Let (V, ρ) be a finite-dimensional irreducible representation of a group G . Then every element of the center

$$Z(G) := \{c \in G \mid cg = gc \text{ for all } g \in G\}$$

acts by a scalar. In other words, for every $c \in Z(G)$ there exists $\lambda_c \in \mathbb{C}$ such that

$$\rho(c) = \lambda_c 1_V.$$

Proof:

Let $c \in Z(G)$. Then for every $g \in G$,

$$\rho(c) \circ \rho(g) = \rho(cg) = \rho(gc) = \rho(g) \circ \rho(c).$$

Thus $\rho(c) : V \rightarrow V$ is a map of G -representations. Since V is a finite-dimensional complex vector space, $\rho(c)$ has an eigenvalue $\lambda_c \in \mathbb{C}$. Then

$$\rho(c) - \lambda_c 1_V$$

is again a map of G -representations, but it is not injective. By Schur's lemma it cannot be an isomorphism, so it must be the zero map. Hence

$$\rho(c) = \lambda_c 1_V.$$

Example.

Let G be an abelian group. Then

$$Z(G) = G.$$

Hence, if (V, ρ) is a finite-dimensional irreducible representation of G , every element of G

acts by a scalar.

Therefore every linear subspace of V is preserved by the action of G . Since V is irreducible, this is only possible when

$$\dim V = 1.$$

Thus such representations are one-dimensional. Equivalently, they are homomorphisms

$$\rho : G \rightarrow \mathbb{C}^\times.$$

For example, let $G = \mathbb{Z}_n$. If $[1]$ is the generator, then

$$\rho([1])^n = \rho([0]) = 1.$$

Hence $\rho([1])$ must be an n -th root of unity. For $k \in \{0, 1, \dots, n-1\}$, define

$$\rho_k([1]) = e^{2\pi i k/n}.$$

Then

$$\rho_k([a]) = e^{2\pi i k a/n}$$

defines a one-dimensional irreducible representation of $\mathbb{Z}_n$, and these are all of them.

3.2.4 Decomposition of representations

Definition 3.2.6: Constructions From Old Representations

Let G be a group, and let (V_1, ρ_1) and (V_2, ρ_2) be representations of G .

- The **direct sum representation** is

$$(V_1 \oplus V_2, \rho_1 \oplus \rho_2),$$

where

$$(\rho_1 \oplus \rho_2)(g)(v_1, v_2) := (\rho_1(g)v_1, \rho_2(g)v_2).$$

- The **tensor product representation** is

$$(V_1 \otimes V_2, \rho_1 \otimes \rho_2),$$

where on pure tensors

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) := \rho_1(g)v_1 \otimes \rho_2(g)v_2,$$

and then extended linearly.

- The **dual representation** of (V_1, ρ_1) is

$$(V_1^*, \rho_1^*), \quad V_1^* := \text{Hom}(V_1, \mathbb{C}),$$

where

$$(\rho_1^*(g)f)(v) := f(\rho_1(g^{-1})v).$$

- The **Hom representation** is the representation on $\text{Hom}(V_1, V_2)$ defined by

$$\rho(g) : \text{Hom}(V_1, V_2) \rightarrow \text{Hom}(V_1, V_2),$$

with

$$(\rho(g)F)(v) := \rho_2(g)(F(\rho_1(g^{-1})v)).$$

Proposition 3.2.7

The direct sum, tensor product, dual, and Hom constructions above define representations of G .

Proof:

For the direct sum, one has

$$(\rho_1 \oplus \rho_2)(e)(v_1, v_2) = (v_1, v_2),$$

and

$$(\rho_1 \oplus \rho_2)(gh)(v_1, v_2) = (\rho_1(g)\rho_1(h)v_1, \rho_2(g)\rho_2(h)v_2) = ((\rho_1 \oplus \rho_2)(g)(\rho_1 \oplus \rho_2)(h))(v_1, v_2).$$

For the tensor product, the identity element acts as the identity. Moreover, on pure tensors,

$$(\rho_1 \otimes \rho_2)(gh)(v_1 \otimes v_2) = \rho_1(g)\rho_1(h)v_1 \otimes \rho_2(g)\rho_2(h)v_2,$$

which is equal to

$$((\rho_1 \otimes \rho_2)(g)(\rho_1 \otimes \rho_2)(h))(v_1 \otimes v_2).$$

Since pure tensors span $V_1 \otimes V_2$, this proves the representation identity on the whole tensor product.

For the dual representation, let $f \in V_1^*$ and $v \in V_1$. Then

$$(\rho_1^*(e)f)(v) = f(\rho_1(e^{-1})v) = f(v),$$

and

$$(\rho_1^*(gh)f)(v) = f(\rho_1((gh)^{-1})v) = f(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

On the other hand,

$$(\rho_1^*(g)(\rho_1^*(h)f))(v) = (\rho_1^*(h)f)(\rho_1(g^{-1})v) = f(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

Hence $\rho_1^*(gh) = \rho_1^*(g)\rho_1^*(h)$.

Finally, for the Hom representation, the identity element acts as the identity. For $F \in \text{Hom}(V_1, V_2)$ and $v \in V_1$, we compute

$$(\rho(gh)F)(v) = \rho_2(gh)F(\rho_1((gh)^{-1})v) = \rho_2(g)\rho_2(h)F(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

Also,

$$(\rho(g)(\rho(h)F))(v) = \rho_2(g)\left((\rho(h)F)(\rho_1(g^{-1})v)\right) = \rho_2(g)\rho_2(h)F(\rho_1(h^{-1})\rho_1(g^{-1})v).$$

Thus $\rho(gh) = \rho(g)\rho(h)$, so this is a representation. ■

Remark.

Let V_1 and V_2 be finite-dimensional representations of G . Then there is a natural linear isomorphism

$$\Phi : V_2 \otimes V_1^* \rightarrow \text{Hom}(V_1, V_2), \quad v_2 \otimes f \mapsto (v_1 \mapsto f(v_1)v_2).$$

This is an isomorphism of representations. Indeed, for $g \in G$, $v_1 \in V_1$, $v_2 \in V_2$, and $f \in V_1^*$, we have

$$\Phi((\rho_2 \otimes \rho_1^*)(g)(v_2 \otimes f))(v_1) = f(\rho_1(g^{-1})v_1)\rho_2(g)v_2.$$

On the other hand,

$$(\rho(g)\Phi(v_2 \otimes f))(v_1) = \rho_2(g)\left(\Phi(v_2 \otimes f)(\rho_1(g^{-1})v_1)\right) = f(\rho_1(g^{-1})v_1)\rho_2(g)v_2.$$

Hence Φ intertwines the two representations.

Definition 3.2.8: Indecomposable Representation

A representation (V, ρ) of a group G is **indecomposable** if it cannot be written as a direct sum of two nonzero subrepresentations. In other words, there do not exist nonzero subrepresentations $W_1, W_2 \leq V$ such that

$$V = W_1 \oplus W_2.$$

Remark.

Every irreducible representation is indecomposable: if $V = W_1 \oplus W_2$ with W_1, W_2 nonzero subrepresentations, then W_1 and W_2 are proper nonzero subrepresentations of V , so V is not irreducible.

The converse is not true in general: an indecomposable representation can still have nonzero proper subrepresentations.

Example.

- Every one-dimensional representation is indecomposable, since a one-dimensional vector space cannot be written as a direct sum of two nonzero subspaces.
- Consider the representation of $\mathbb{Z}$ on $\mathbb{C}^2$ defined by

$$\rho(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then the line

$$W = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} \subseteq \mathbb{C}^2$$

is a subrepresentation, so the representation is reducible.

However, it is indecomposable. Indeed, a decomposition into two one-dimensional subrepresentations would give a basis of $\mathbb{C}^2$ consisting of eigenvectors of $\rho(1)$. This would make $\rho(1)$ diagonalizable. But

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not diagonalizable; its only eigenspace is W . Therefore this representation is reducible but indecomposable.

Definition 3.2.9: Semisimple Representations

Let (V, ρ) be a representation of a group G . We say that V is **semisimple**, or **completely reducible**, if every subrepresentation $W \leq V$ has a complementary subrepresentation. That is, for every subrepresentation $W \leq V$, there exists a subrepresentation $W' \leq V$ such that

$$V = W \oplus W'.$$

If V is finite-dimensional, this is equivalent to saying that

$$V \cong \bigoplus_i W_i$$

with each W_i irreducible.

Example.

Recall that

$$\mathrm{SU}(2) = \{U \in \mathrm{Mat}(2 \times 2, \mathbb{C}) \mid U^\dagger U = 1, \det(U) = 1\}$$

has a defining two-dimensional representation $V_{1/2} = \mathbb{C}^2$. For this representation one has

$$V_{1/2} \otimes V_{1/2} \cong \mathrm{Sym}^2(V_{1/2}) \oplus \bigwedge^2 V_{1/2}$$

as representations.

More generally, the irreducible representations of $\mathrm{SU}(2)$ are indexed by

$$j \in \frac{1}{2}\mathbb{N}_0,$$

and are given by

$$V_j := \mathrm{Sym}^{2j}(V_{1/2}).$$

The tensor product decomposes according to the Clebsch-Gordan rule

$$V_k \otimes V_\ell \cong \bigoplus_{j=|k-\ell|}^{k+\ell} V_j,$$

where j increases in steps of 1. This is the representation theoretic form of addition of angular momentum in quantum mechanics.

We will come back to this example later when discussing Lie groups.

Remark.

We want to show that, for finite groups, irreducible and indecomposable representations are the same thing. Equivalently, every finite-dimensional representation of a finite group should split as a direct sum of irreducible representations.

The subtle point is that a subspace does not have a preferred complement in general. For Hilbert spaces, however, there is a natural candidate: the orthogonal complement. This motivates the following definition.

Definition 3.2.10: Unitary Representation

A **unitary representation** of a group G is a Hilbert space $\mathcal{H}$ together with a group homomorphism

$$\rho : G \rightarrow U(\mathcal{H}),$$

where $U(\mathcal{H})$ denotes the group of unitary automorphisms of $\mathcal{H}$.

Proposition 3.2.11

Let (V, ρ) be a finite-dimensional representation of a finite group G . Then there exists a positive Hermitian inner product on V such that ρ is unitary. In other words, every finite-dimensional representation of a finite group can be lifted to a unitary representation.

Proof:

Let

$$\langle -, - \rangle$$

be any positive Hermitian inner product on V . Define a new inner product by averaging over the group:

$$\langle v, w \rangle_G := \sum_{g \in G} \langle \rho(g)v, \rho(g)w \rangle.$$

Since G is finite, this sum is finite.

The form $\langle -, - \rangle_G$ is again Hermitian and positive. Indeed, each summand is Hermitian, and for $v \neq 0$ one has

$$\langle v, v \rangle_G = \sum_{g \in G} \langle \rho(g)v, \rho(g)v \rangle > 0,$$

because every $\rho(g)$ is invertible, so $\rho(g)v \neq 0$.

It remains to check that the representation is unitary with respect to this new inner product. Let $h \in G$. Then

$$\langle \rho(h)v, \rho(h)w \rangle_G = \sum_{g \in G} \langle \rho(g)\rho(h)v, \rho(g)\rho(h)w \rangle$$

and hence

$$\langle \rho(h)v, \rho(h)w \rangle_G = \sum_{g \in G} \langle \rho(gh)v, \rho(gh)w \rangle.$$

As g runs through G , so does gh . Therefore, after the change of variables $g' = gh$,

$$\langle \rho(h)v, \rho(h)w \rangle_G = \sum_{g' \in G} \langle \rho(g')v, \rho(g')w \rangle = \langle v, w \rangle_G.$$

Thus every $\rho(h)$ preserves $\langle -, - \rangle_G$, so ρ is unitary. ■

Theorem 3.2.12: Maschke's Theorem

Every finite-dimensional representation of a finite group is completely reducible.

Proof:

Let (V, ρ) be a finite-dimensional representation of a finite group G . By the previous proposition, we may choose a positive Hermitian inner product

$$\langle -, - \rangle_G$$

on V such that ρ is unitary.

Let $W \leq V$ be a subrepresentation. Define its orthogonal complement with respect to this inner product by

$$W^\perp := \{v \in V \mid \langle v, w \rangle_G = 0 \text{ for all } w \in W\}.$$

Since V is finite-dimensional, we have a direct sum decomposition of vector spaces

$$V = W \oplus W^\perp.$$

It remains to show that $W^\perp$ is again a subrepresentation. Let $v \in W^\perp$, $g \in G$, and $w \in W$. Since $\rho(g)$ is unitary,

$$\langle \rho(g)v, w \rangle_G = \langle \rho(g)v, \rho(g)\rho(g^{-1})w \rangle_G = \langle v, \rho(g^{-1})w \rangle_G.$$

Because W is a subrepresentation, $\rho(g^{-1})w \in W$. Since $v \in W^\perp$, this gives

$$\langle \rho(g)v, w \rangle_G = 0.$$

Hence $\rho(g)v \in W^\perp$, so $W^\perp$ is a subrepresentation.

Thus every subrepresentation $W \leq V$ has a complementary subrepresentation $W^\perp$. Therefore V is completely reducible. ■

Corollary 3.2.13

Let G be a finite group. A finite-dimensional representation V of G is indecomposable if and only if it is irreducible. Equivalently, every finite-dimensional representation of G can be written as

$$V \cong \bigoplus_i N_i V_i,$$

where each V_i is irreducible and $N_i \geq 0$ is the multiplicity of V_i in V . Here

$$N_i V_i := \underbrace{V_i \oplus \cdots \oplus V_i}_{N_i \text{ times}}.$$

Proof:

By Maschke's theorem, every finite-dimensional representation of G is completely reducible.

We first prove the direct sum decomposition. If $V = 0$, there is nothing to show. Otherwise, choose a nonzero subrepresentation $W \leq V$ of minimal dimension. Then W is irreducible: any nonzero proper subrepresentation of W would contradict the minimality of W . By complete reducibility, there is a subrepresentation $W' \leq V$ such that

$$V = W \oplus W'.$$

Repeating this argument for W' , and using finite-dimensionality, the process terminates and gives a decomposition of V into irreducible subrepresentations. Grouping isomorphic irreducible summands gives

$$V \cong \bigoplus_i N_i V_i.$$

Now suppose that V is indecomposable. In such a decomposition, there can be only one irreducible summand and it can occur only once; otherwise V would split as a direct sum of two nonzero subrepresentations. Hence V is irreducible.

Conversely, every irreducible representation is indecomposable, as noted above. Therefore indecomposable and irreducible representations agree for finite groups. ■

Example.

Let

$$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$$

be the cyclic group, thought of as n equally spaced points on a circle. Consider the vector space of complex-valued functions

$$\mathcal{O}(\mathbb{Z}_n) := \{f : \mathbb{Z}_n \rightarrow \mathbb{C}\}.$$

The group $\mathbb{Z}_n$ acts on this space by translations:

$$(\rho([a])f)([x]) := f([x + a]).$$

For $k = 0, 1, \dots, n-1$, define

$$f_k([x]) := e^{2\pi i k x / n}.$$

These functions are eigenvectors for every translation, since

$$(\rho([a])f_k)([x]) = f_k([x + a]) = e^{2\pi i k(x+a)/n} = e^{2\pi i k a / n} f_k([x]).$$

Hence the line

$$\mathbb{C}_k := \mathbb{C}f_k$$

is a one-dimensional subrepresentation of $\mathcal{O}(\mathbb{Z}_n)$. Since the functions $f_0, \dots, f_{n-1}$ form a basis of $\mathcal{O}(\mathbb{Z}_n)$, we get the decomposition

$$\mathcal{O}(\mathbb{Z}_n) = \bigoplus_{k=0}^{n-1} \mathbb{C}_k = \bigoplus_{k=0}^{n-1} \mathbb{C}f_k.$$

Thus the decomposition of the regular representation of $\mathbb{Z}_n$ is exactly the discrete Fourier transform.

3.2.5 Character theory

Definition 3.2.14: Characters And Class Functions

Let (V, ρ) be a finite-dimensional representation of a group G . The **character** of V is the function

$$\chi_V : G \rightarrow \mathbb{C}, \quad g \mapsto \text{tr}_V(\rho(g)).$$

A function $f : G \rightarrow \mathbb{C}$ is called a **class function** if it is invariant under conjugation, that is,

$$f(hgh^{-1}) = f(g) \quad \forall g, h \in G.$$

Proposition 3.2.15

Characters are class functions.

Proof:

Let (V, ρ) be a finite-dimensional representation of G . Then for every $g, h \in G$,

$$\chi_V(hgh^{-1}) = \text{tr}_V(\rho(hgh^{-1})) = \text{tr}_V(\rho(h)\rho(g)\rho(h^{-1})).$$

Using cyclicity of the trace, this becomes

$$\text{tr}_V(\rho(h)\rho(g)\rho(h^{-1})) = \text{tr}_V(\rho(g)\rho(h^{-1})\rho(h)) = \text{tr}_V(\rho(g)).$$

Hence

$$\chi_V(hgh^{-1}) = \chi_V(g),$$

so χ_V is invariant under conjugation. ■

Remark.

Since characters are class functions, it is enough to record their values on the conjugacy classes of G . A table whose rows are irreducible characters and whose columns are conjugacy classes is called a **character table**.

Example.

The group S_3 has three conjugacy classes, represented by

$$1, \quad (12), \quad (123).$$

It has three irreducible representations: the trivial representation, the alternating representation, and the standard representation. Their character table is

S_3	1	(12)	(123)
χ_{triv}	1	1	1
χ_{alt}	1	-1	1
χ_{std}	2	0	-1

Example.

For $\mathbb{Z}_3$, every conjugacy class consists of a single element. The irreducible characters are indexed by $k = 0, 1, 2$. Their character table is

$\mathbb{Z}_3$	[0]	[1]	[2]
$k = 0$	1	1	1
$k = 1$	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
$k = 2$	1	$e^{4\pi i/3}$	$e^{8\pi i/3}$

Remark.

Characters behave well with respect to direct sums and tensor products:

$$\chi_{V \oplus W} = \chi_V + \chi_W, \quad \chi_{V \otimes W} = \chi_V \chi_W.$$

The second identity follows from the fact that the trace of a tensor product of linear maps is the product of the traces.

Example.

For the standard representation of S_3 , the character is

$$\chi_{\text{std}} = (2, 0, -1)$$

on the conjugacy classes represented by $1, (12), (123)$. Therefore

$$\chi_{\text{std} \otimes \text{std}} = \chi_{\text{std}} \chi_{\text{std}} = (4, 0, 1).$$

On the other hand,

$$\chi_{\text{std}} + \chi_{\text{alt}} + \chi_{\text{triv}} = (2, 0, -1) + (1, -1, 1) + (1, 1, 1) = (4, 0, 1).$$

Hence the character table suggests the decomposition

$$\text{Std} \otimes \text{Std} \cong \text{Std} \oplus \text{Alt} \oplus \text{Triv}.$$

In particular, the characters in the table are linearly independent.

Definition 3.2.16: Invariant Subspace

Let G be a group and let (V, ρ) be a representation of G . The **subrepresentation of G -invariants** is

$$V^G := \{v \in V \mid \rho(g)(v) = v \text{ for all } g \in G\} \leq V.$$

Remark.

The subspace V^G is the part of V on which G acts trivially. Thus, for finite-dimensional representations of finite groups, one can write

$$V \cong \dim(V^G) \text{Triv} \oplus V',$$

where V' contains no copy of the trivial representation.

In particular, if V_i is irreducible but not trivial, then

$$V_i^G = \{0\}.$$

Proposition 3.2.17

Let G be a finite group and let (V, ρ) be a representation of G . The map

$$N_G : V \rightarrow V, \quad v \mapsto \frac{1}{|G|} \sum_{g \in G} \rho(g)(v)$$

is a projection operator onto V^G . In other words,

$$N_G \circ N_G = N_G, \quad \text{im}(N_G) = V^G.$$

Proof:

First we check that N_G intertwines the G -action. Let $g \in G$ and $v \in V$. Then

$$N_G(\rho(g)(v)) = \frac{1}{|G|} \sum_{h \in G} \rho(hg)(v).$$

As h runs through G , so does $h' = hg$. Therefore

$$N_G(\rho(g)(v)) = \frac{1}{|G|} \sum_{h' \in G} \rho(h')(v) = N_G(v).$$

Also,

$$\rho(g)(N_G(v)) = \rho(g) \left(\frac{1}{|G|} \sum_{h \in G} \rho(h)(v) \right) = \frac{1}{|G|} \sum_{h \in G} \rho(gh)(v).$$

Again, as h runs through G , so does $h' = gh$, and hence

$$\rho(g)(N_G(v)) = N_G(v).$$

Thus $\text{im}(N_G) \subseteq V^G$.

Conversely, if $v \in V^G$, then $\rho(g)(v) = v$ for every $g \in G$. Hence

$$N_G(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g)(v) = \frac{1}{|G|} \sum_{g \in G} v = v.$$

Therefore $V^G \subseteq \text{im}(N_G)$, and so

$$\text{im}(N_G) = V^G.$$

Finally, since $N_G(v) \in V^G$ for every $v \in V$, the previous computation gives

$$N_G(N_G(v)) = N_G(v).$$

Thus $N_G^2 = N_G$, so N_G is a projection operator onto V^G . ■

Corollary 3.2.18

Let G be a finite group and let V be a finite-dimensional representation of G . Then

$$\dim(V^G) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g).$$

In particular, if V is irreducible and not trivial, then

$$\sum_{g \in G} \chi_V(g) = 0.$$

Proof:

By the previous proposition, N_G is a projection onto V^G . Thus, with respect to a decomposition

$$V \cong V^G \oplus \ker(N_G),$$

the operator N_G has matrix

$$\begin{pmatrix} I_{\dim(V^G)} & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore

$$\mathrm{tr}(N_G) = \dim(V^G).$$

On the other hand,

$$\mathrm{tr}(N_G) = \mathrm{tr} \left(\frac{1}{|G|} \sum_{g \in G} \rho(g) \right) = \frac{1}{|G|} \sum_{g \in G} \mathrm{tr}(\rho(g)) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g).$$

This proves the formula. If V is irreducible and not trivial, then $V^G = \{0\}$, and hence the sum of the character values is zero. ■

Remark.

If V and W are representations of G , then $\mathrm{Hom}(V, W)$ is again a representation. Its invariant subspace is

$$\mathrm{Hom}(V, W)^G = \{f : V \rightarrow W \mid f(\rho_V(g)v) = \rho_W(g)f(v) \text{ for all } g \in G, v \in V\}.$$

Thus $\mathrm{Hom}(V, W)^G$ is precisely the space of maps of G -representations from V to W .

Corollary 3.2.19

Let V and W be irreducible finite-dimensional complex representations of a finite group G . Then

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Proof:

By Schur's lemma,

$$\mathrm{Hom}(V, W)^G = \begin{cases} \mathbb{C} & \text{if } V \cong W, \\ \{0\} & \text{otherwise.} \end{cases}$$

Hence

$$\dim \mathrm{Hom}(V, W)^G = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Applying the previous corollary to the representation $\mathrm{Hom}(V, W)$, we get

$$\dim \mathrm{Hom}(V, W)^G = \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}(V, W)}(g).$$

Since

$$\mathrm{Hom}(V, W) \cong V^* \otimes W,$$

its character is

$$\chi_{\mathrm{Hom}(V, W)}(g) = \chi_{V^*}(g) \chi_W(g) = \overline{\chi_V(g)} \chi_W(g).$$

Therefore

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Corollary 3.2.20

The characters of irreducible finite-dimensional complex representations of a finite group G are orthonormal with respect to the scalar product

$$(F, H) := \frac{1}{|G|} \sum_{g \in G} \overline{F(g)} H(g)$$

on the space of class functions on G .

Proof:

This is exactly the previous corollary, written in the notation

$$(\chi_V, \chi_W) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \chi_W(g).$$

Corollary 3.2.21

Every finite-dimensional representation V of a finite group G is completely determined, up to isomorphism, by its character χ_V . More precisely, if

$$V \cong \bigoplus_i N_i V_i$$

is a decomposition into irreducible representations, then

$$N_i = (\chi_{V_i}, \chi_V).$$

Proof:

By Maschke's theorem, write

$$V \cong \bigoplus_i N_i V_i$$

with the V_i irreducible. Then

$$\chi_V = \sum_i N_i \chi_{V_i}.$$

Taking the scalar product with χ_{V_j} , and using the orthonormality of irreducible characters, gives

$$(\chi_{V_j}, \chi_V) = \sum_i N_i (\chi_{V_j}, \chi_{V_i}) = N_j.$$

Thus the character determines all multiplicities N_i , and hence determines V up to isomorphism. ■

Remark.

Since irreducible characters are linearly independent, there can be only finitely many irreducible representations of a finite group up to isomorphism.

Example.

Let

$$D_4 = \langle r, s \mid r^4 = s^2 = 1, \, srs = r^{-1} \rangle$$

be the symmetry group of the square, where r is rotation by 90° , and s is a reflection. Its conjugacy classes are

$$\{1\}, \quad \{r, r^3\}, \quad \{r^2\}, \quad \{s, sr^2\}, \quad \{sr, sr^3\}.$$

The one-dimensional representations are homomorphisms

$$\varphi : D_4 \rightarrow \mathbb{C}^\times.$$

Since

$$\varphi(r) = \varphi(srs) = \varphi(s)\varphi(r)\varphi(s) = \varphi(r)^{-1},$$

we must have $\varphi(r) = \pm 1$. Also, since $s^2 = 1$, one has $\varphi(s) = \pm 1$. Thus the one-dimensional

representations are labelled by the choices

$$(\varphi(r), \varphi(s)) \in \{(1, 1), (1, -1), (-1, 1), (-1, -1)\}.$$

There is also the standard two-dimensional representation

$$D_4 \rightarrow \mathrm{GL}(\mathbb{R}^2),$$

coming from the action of D_4 on the square in the plane. Its character is

$$\chi_{\mathrm{std}} = (2, 0, -2, 0, 0)$$

on the conjugacy classes listed above. Since

$$(\chi_{\mathrm{std}}, \chi_{\mathrm{std}}) = \frac{1}{8}(2^2 + (-2)^2) = 1,$$

the standard representation is irreducible.

The character table is

D_4	1	r	r^2	s	sr
$(1, 1)$	1	1	1	1	1
$(-1, 1)$	1	-1	1	1	-1
$(1, -1)$	1	1	1	-1	-1
$(-1, -1)$	1	-1	1	-1	1
std	2	0	-2	0	0

Here the column labels stand for the conjugacy classes represented by those elements.

Now consider $\mathrm{std} \otimes \mathrm{std}$. Its character is the pointwise product

$$\chi_{\mathrm{std} \otimes \mathrm{std}} = \chi_{\mathrm{std}} \chi_{\mathrm{std}} = (4, 0, 4, 0, 0).$$

The multiplicity of std in this tensor product is

$$(\chi_{\mathrm{std}}, \chi_{\mathrm{std} \otimes \mathrm{std}}) = \frac{1}{8}(2 \cdot 4 + (-2) \cdot 4) = 0.$$

For each one-dimensional character χ , the values on the classes represented by 1 and r^2 are both 1, so

$$(\chi, \chi_{\mathrm{std} \otimes \mathrm{std}}) = \frac{1}{8}(4 + 4) = 1.$$

Therefore

$$\mathrm{std} \otimes \mathrm{std} \cong (1, 1) \oplus (-1, 1) \oplus (1, -1) \oplus (-1, -1).$$

Corollary 3.2.22

Let G be a finite group, and let $V_1, \dots, V_r$ be the irreducible finite-dimensional complex representations of G , up to isomorphism. Then the regular representation decomposes as

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^r \dim(V_i) V_i.$$

In particular,

$$\sum_{i=1}^r \dim(V_i)^2 = |G|.$$

Proof:

Let χ_{reg} be the character of the regular representation $\mathbb{C}[G]$. In the basis $\{e_h\}_{h \in G}$, the action is

$$g \cdot e_h = e_{gh}.$$

If $g = e$, this is the identity on a vector space of dimension $|G|$, so

$$\chi_{\text{reg}}(e) = |G|.$$

If $g \neq e$, then $h \mapsto gh$ has no fixed points. Hence the matrix of g in the basis $\{e_h\}_{h \in G}$ is a permutation matrix with no diagonal entries equal to 1, and therefore

$$\chi_{\text{reg}}(g) = 0.$$

By the previous corollary, the multiplicity of V_i in $\mathbb{C}[G]$ is

$$(\chi_{V_i}, \chi_{\text{reg}}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} \chi_{\text{reg}}(g).$$

Since χ_{reg} vanishes away from id , this becomes

$$(\chi_{V_i}, \chi_{\text{reg}}) = \frac{1}{|G|} \overline{\chi_{V_i}(\text{id})} |G| = \dim(V_i).$$

Thus

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^r \dim(V_i) V_i.$$

Taking dimensions on both sides gives

$$|G| = \dim \mathbb{C}[G] = \sum_{i=1}^r \dim(V_i) \dim(V_i) = \sum_{i=1}^r \dim(V_i)^2.$$

Remark.

The previous corollary also gives a useful relation among the irreducible characters. Since

$$\chi_{\text{reg}} = \sum_{i=1}^r \dim(V_i) \chi_{V_i},$$

and since $\chi_{\text{reg}}(g) = 0$ for every $g \neq \text{id}$, one has

$$0 = \sum_{i=1}^r \dim(V_i) \chi_{V_i}(g) \quad \forall g \neq \text{id}.$$

Thus, once all irreducible characters except one are known, the remaining one is determined on all non-identity conjugacy classes by this equation. Its value at id is its dimension, which is constrained by

$$\sum_{i=1}^r \dim(V_i)^2 = |G|.$$

Theorem 3.2.23: Irreducible Characters Form A Basis

Let G be a finite group. The characters of the irreducible finite-dimensional complex representations of G form an orthonormal basis of the vector space of class functions on G .

In particular, the number of irreducible representations of G , up to isomorphism, is equal to the number of conjugacy classes of G .

Proof:

We already know that the irreducible characters are orthonormal. It remains to show that they span the space of class functions.

Let $\alpha : G \rightarrow \mathbb{C}$ be a class function such that

$$(\alpha, \chi_V) = 0$$

for every irreducible representation V . We will show that $\alpha = 0$.

For an irreducible representation (V, ρ) , define

$$\varphi_{\alpha, V} := \sum_{g \in G} \alpha(g) \rho(g) \in \text{End}(V).$$

Since α is a class function, $\varphi_{\alpha, V}$ commutes with the G -action. Indeed, for $h \in G$,

$$\varphi_{\alpha, V} \rho(h) = \sum_{g \in G} \alpha(g) \rho(gh).$$

Writing $g' = h^{-1}gh$, this becomes

$$\varphi_{\alpha, V} \rho(h) = \sum_{g' \in G} \alpha(hg'h^{-1}) \rho(hg') = \rho(h) \sum_{g' \in G} \alpha(g') \rho(g') = \rho(h) \varphi_{\alpha, V}.$$

Thus $\varphi_{\alpha, V}$ is a map of G -representations $V \rightarrow V$. By Schur's lemma,

$$\varphi_{\alpha, V} = \lambda_V 1_V$$

for some $\lambda_V \in \mathbb{C}$.

Taking traces gives

$$\lambda_V = \frac{1}{\dim(V)} \operatorname{tr}(\varphi_{\alpha,V}) = \frac{1}{\dim(V)} \sum_{g \in G} \alpha(g) \chi_V(g) = \frac{|G|}{\dim(V)} \overline{(\alpha, \chi_{V^*})} = 0.$$

Here V^* is again irreducible, so the last equality follows from the assumption on α . Hence $\varphi_{\alpha,V} = 0$ for every irreducible representation V .

Therefore, for any finite-dimensional representation (W, ρ_W) , one has

$$\sum_{g \in G} \alpha(g) \rho_W(g) = 0,$$

because W is a direct sum of irreducibles.

Apply this to the regular representation $\mathbb{C}[G]$. The operators $\rho_{\text{reg}}(g)$ are linearly independent: indeed,

$$\rho_{\text{reg}}(g)(e_{\text{id}}) = e_g.$$

Hence

$$\alpha(g) = 0 \quad \forall g \in G.$$

Thus $\alpha = 0$, and the irreducible characters span the class functions.

Since the irreducible characters form a basis, their number is the dimension of the space of class functions. A class function is determined by its values on conjugacy classes, so this dimension is the number of conjugacy classes of G . ■

3.3 Representations of S_n

The irreducible representations of S_n are indexed by the conjugacy classes of S_n , and hence by the partitions of n .

Definition 3.3.1: Partitions And Young Diagrams

A **partition** of n is a tuple

$$\lambda = (\lambda_1, \dots, \lambda_k)$$

such that

$$n = \sum_{i=1}^k \lambda_i, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0.$$

We write $\lambda \vdash n$.

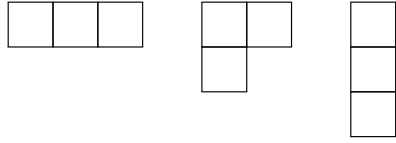
A partition can be represented by a **Young diagram**: draw λ_i boxes in the i -th row, aligned on the left.

Example.

The partitions of 3 are

$$(3), \quad (2, 1), \quad (1, 1, 1).$$

Their Young diagrams are



Thus S_3 has three irreducible representations, corresponding to these three diagrams.

Definition 3.3.2: Associative Algebras

An **associative algebra** $(A, \mu, 1)$ is a vector space A equipped with a linear map

$$\mu : A \otimes A \rightarrow A, \quad a \otimes b \mapsto ab,$$

and an element $1 \in A$ such that

$$(ab)c = a(bc) \quad \forall a, b, c \in A,$$

and

$$1a = a = a1 \quad \forall a \in A.$$

Example.

Let G be a group. The vector space $\mathbb{C}[G]$, with basis

$$\{e_g\}_{g \in G},$$

is an associative algebra with multiplication

$$e_g e_h = e_{gh},$$

and unit

$$1 = e_{\text{id}}.$$

It is called the **group algebra** of G .

Remark.

A representation

$$\rho : G \rightarrow \text{Aut}(V)$$

induces a linear map

$$\hat{\rho} : \mathbb{C}[G] \rightarrow \text{End}(V), \quad e_g \mapsto \rho(g).$$

In particular,

$$\hat{\rho} \left(\frac{1}{|G|} \sum_{g \in G} e_g \right)$$

is the projection onto the trivial component of V .

Definition 3.3.3: Tableaux

A **tableau** T on a given Young diagram with n boxes is a numbering of its boxes by $1, \dots, n$.

Associated to T are two subgroups of S_n :

$$P_T := \{\sigma \in S_n \mid \sigma \text{ preserves the rows of } T\},$$

and

$$Q_T := \{\sigma \in S_n \mid \sigma \text{ preserves the columns of } T\}.$$

Example.

For example, for the tableau

$$T = \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & \\ \hline 6 & 7 & \\ \hline \end{array}$$

one has

$$P_T \cong S_3 \times S_2 \times S_2, \quad Q_T \cong S_3 \times S_3 \times S_1.$$

We will suppress the tableau from the notation and write λ for both a partition and a choice of tableau of shape λ .

Definition 3.3.4: Young Symmetrizers

Let $\lambda \vdash n$, and choose a tableau of shape λ . Define

$$a_\lambda := \sum_{g \in P_\lambda} e_g \in \mathbb{C}[S_n],$$

and

$$b_\lambda := \sum_{g \in Q_\lambda} \text{sign}(g) e_g \in \mathbb{C}[S_n].$$

The **Young symmetrizer** associated to λ is

$$c_\lambda := a_\lambda b_\lambda \in \mathbb{C}[S_n].$$

Example.

For $\lambda = (n)$, the Young diagram has one row. Hence

$$P_\lambda = S_n, \quad Q_\lambda = \{\text{id}\},$$

and so

$$c_\lambda = a_\lambda.$$

In this case

$$\text{im}((\cdot)c_\lambda) \cong \text{Triv}.$$

For $\lambda = (1, \dots, 1)$, the Young diagram has one column. Hence

$$P_\lambda = \{\text{id}\}, \quad Q_\lambda = S_n,$$

and so

$$a_\lambda = 1, \quad c_\lambda = b_\lambda.$$

In this case

$$\text{im}((\cdot)c_\lambda) \cong \text{Alt}.$$

Definition 3.3.5: Representations From Young Symmetrizers

Let $\lambda \vdash n$. Define

$$V_\lambda := \mathbb{C}[S_n]c_\lambda = \{ac_\lambda \mid a \in \mathbb{C}[S_n]\} \leq \mathbb{C}[S_n].$$

This is a representation of S_n by left multiplication:

$$\rho(g) : V_\lambda \rightarrow V_\lambda, \quad ac_\lambda \mapsto e_g ac_\lambda.$$

Example.

For S_3 , the three partitions give

$$V_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}} = \mathbb{C}[e_{\text{id}} + e_{(12)} + e_{(23)} + e_{(13)} + e_{(123)} + e_{(132)}],$$

and

$$V_{\begin{array}{|c|} \hline \square \\ \square \\ \square \\ \hline \end{array}} = \mathbb{C}[e_{\text{id}} - e_{(12)} - e_{(23)} - e_{(13)} + e_{(123)} + e_{(132)}].$$

For the tableau

1	2
3	

one has

$$c_\lambda = (e_{\text{id}} + e_{(12)})(e_{\text{id}} - e_{(13)}) = e_{\text{id}} + e_{(12)} - e_{(13)} - e_{(123)}.$$

Hence

$$V_{\begin{array}{|c|c|} \hline \square & \square \\ \square & \square \\ \hline \end{array}} = \mathbb{C}[S_3]c_\lambda.$$

Proposition 3.3.6

Let T, T' be Young tableaux of the same shape λ . Then there exists a unique $g \in S_n$ such that

$$T' = gT.$$

If P, Q are the row and column subgroups associated to T , and P', Q' are the row and column subgroups associated to T' , then

$$P' = gPg^{-1}, \quad Q' = gQg^{-1}.$$

Hence

$$c_{\lambda, T'} = e_g c_{\lambda, T} e_{g^{-1}}.$$

Therefore

$$V_{\lambda, T} \rightarrow V_{\lambda, T'}, \quad ac_{\lambda, T} \mapsto ac_{\lambda, T} e_{g^{-1}} = ae_{g^{-1}} c_{\lambda, T'},$$

is an isomorphism of S_n -representations.

Moreover,

$$P \cap Q = \{\text{id}\}.$$

Thus, if

$$g = pq = p'q', \quad p, p' \in P, \quad q, q' \in Q,$$

then

$$p'^{-1}p = qq'^{-1} \in P \cap Q.$$

Hence $p = p'$ and $q = q'$, so the decomposition $g = pq$ is unique. Consequently,

$$c_{\lambda, T} = \sum_{g=pq} \text{sign}(q) e_g.$$

Proposition 3.3.7

For each Young tableau, the representation V_λ is irreducible.

Proof:

Let $v \in V_\lambda$. We want to understand $c_\lambda v$. Write

$$v = ac_\lambda.$$

Then $c_\lambda v = c_\lambda ac_\lambda$. Hence $c_\lambda v$ solves the equation

$$e_p x e_q \text{sign}(q) = x \quad \forall p \in P, q \in Q. \tag{1}$$

Write

$$x = \sum_{g \in S_n} n_g e_g.$$

Equation (1) implies

$$n_{pgq} = n_g \text{sign}(q).$$

If $g \neq pq$, then let $T' = gT$. Assume that there are integers $i < j \leq n$ such that i, j are in the same row of T and in the same column of T' . Then $t = (ij) \in P$, and also

$$t \in Q' = gQg^{-1}.$$

Thus there is some $q \in Q$ with $t = gqg^{-1}$, and hence

$$g = tqg^{-1}.$$

Comparing coefficients gives

$$n_g = -n_g, \quad \text{so } n_g = 0.$$

Now let $T, T' = gT$ be two Young tableaux such that no pair i, j is in the same row of T and in the same column of T' . Then there are $p \in P$ and $q' \in Q'$ such that the first rows of pT and $q'T'$ agree. Continuing row by row, we get

$$pT = q'T' = q'gT.$$

Hence

$$p = q'g.$$

Since $q' \in Q' = gQg^{-1}$, we can write $q' = gqg^{-1}$ for some $q \in Q$. Therefore

$$p = gq, \quad g = pq^{-1}.$$

It follows that the only possible non-zero coefficients are those with $g = pq$. Thus every solution of (1) is of the form

$$x = n_1 \sum_{g=pq} e_g \text{sign}(g) = n_1 c_\lambda.$$

Hence

$$c_\lambda V_\lambda \subseteq \mathbb{C}[c_\lambda].$$

Let $W \subseteq V_\lambda$ be a subrepresentation. Then either

$$c_\lambda W = 0 \quad \text{or} \quad c_\lambda W = \mathbb{C}[c_\lambda].$$

In the second case, $c_\lambda \in W$, and hence

$$ac_\lambda \in W \quad \forall a \in \mathbb{C}[S_n].$$

Thus $V_\lambda \subseteq W$, so $W = V_\lambda$.

In the first case, choose a projection operator

$$P_W : \mathbb{C}[S_n] \rightarrow \mathbb{C}[S_n]$$

which projects onto W . It satisfies

$$e_g P_W(a) = P_W(e_g a).$$

For $a = 1$, this gives

$$e_g P_W(1) = P_W(e_g).$$

Hence P_W is right multiplication by $P_W(1)$. Since $P_W(1) \in W$, the assumption $c_\lambda W = 0$ gives

$$P_W(c_\lambda) = c_\lambda P_W(1) = 0.$$

Therefore, for every $a \in \mathbb{C}[S_n]$,

$$P_W(ac_\lambda) = aP_W(c_\lambda) = 0.$$

Since $W \subseteq V_\lambda$ and P_W is the identity on W , we get $W = 0$.

Thus the only subrepresentations are 0 and V_λ . ■

Remark.

The proof shows that

$$c_\lambda V_\lambda \neq 0.$$

Theorem 3.3.8: Classification Of Irreducible Representations Of S_n

For $\lambda \neq \mu$, the representations V_λ and V_μ are not isomorphic.

Hence the V_λ , where $\lambda \vdash n$, form a complete set of irreducible representations of S_n .

Proof:

Assume $\lambda > \mu$, meaning that the first non-zero difference $\lambda_i - \mu_i$ is positive. We show that

$$c_\lambda V_\mu = \{0\}, \quad c_\lambda V_\lambda \neq 0.$$

The second statement was shown in the previous proof. For the first one, it is enough to show that

$$c_\lambda a c_\mu = 0 \quad \forall a \in \mathbb{C}[S_n].$$

Since

$$c_\lambda a c_\mu = a_\lambda b_\lambda a a_\mu b_\mu,$$

it is enough to prove

$$a_\lambda x b_\mu = 0 \quad \forall x \in \mathbb{C}[S_n].$$

It suffices to take $x = e_g$. Then

$$a_\lambda e_g b_\mu = a_\lambda b_{gT_\mu}.$$

Since $\lambda > \mu$, there exist i, j which lie in the same row of T_λ and in the same column of gT_μ .

Let $t = (ij)$. Then $t \in P_\lambda$, so

$$a_\lambda e_t = a_\lambda,$$

while t lies in the column group of gT_μ , so

$$e_t b_{gT_\mu} = -b_{gT_\mu}.$$

Therefore

$$a_\lambda b_{gT_\mu} = a_\lambda e_t e_t b_{gT_\mu} = -a_\lambda b_{gT_\mu}.$$

Hence

$$a_\lambda b_{gT_\mu} = 0.$$

Thus $c_\lambda V_\mu = \{0\}$.

Since $c_\lambda V_\lambda \neq 0$, the representations V_λ and V_μ cannot be isomorphic. We have already shown that every V_λ is irreducible, and there are as many partitions $\lambda \vdash n$ as conjugacy classes in S_n . Therefore these representations classify all irreducible representations of S_n . ■

For

$$i = (i_1, \dots, i_n), \quad \sum_{\alpha=1}^n i_\alpha \alpha = n,$$

we denote by C_i the conjugacy class of S_n consisting of permutations with i_α many α -cycles.

Let $\lambda \vdash n$ be a partition with k rows. We work with polynomials in k variables

$$x_1, \dots, x_k.$$

For $1 \leq j \leq n$, define

$$p_j(x) := x_1^j + \dots + x_k^j,$$

and define

$$\Delta(x) := \prod_{a < b} (x_a - x_b).$$

For a polynomial $Q(x)$ in $x_1, \dots, x_k$, we denote by

$$[Q(x)]_{\ell_1, \dots, \ell_k}$$

the coefficient of

$$x_1^{\ell_1} \dots x_k^{\ell_k}.$$

Here

$$\ell_1 = \lambda_1 + k - 1, \quad \ell_2 = \lambda_2 + k - 2, \quad \dots, \quad \ell_k = \lambda_k.$$

Theorem 3.3.9: Frobenius Formula

With the notation above,

$$\chi_\lambda(C_i) = \left[\Delta(x) \prod_{j=1}^n p_j(x)^{i_j} \right]_{\ell_1, \dots, \ell_k}.$$

Proof:

See (*FultonHarris*) ■

Example.

Let $n = 5$, and let

$$\lambda = (3, 2) = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \\ \hline \end{array}.$$

Consider

$$i = (0, 1, 1, 0, 0),$$

so that

$$(12)(345) \in C_i.$$

Here $k = 2$, hence

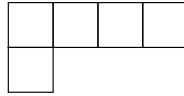
$$\ell_1 = \lambda_1 + 1 = 4, \quad \ell_2 = \lambda_2 = 2.$$

Therefore the Frobenius formula gives

$$\chi_\lambda(C_i) = \left[(x_1 - x_2)(x_1^2 + x_2^2)(x_1^3 + x_2^3) \right]_{4,2} = 1.$$

Remark.

The representation corresponding to



is the standard representation: S_n acts on $\mathbb{C}^n$ by permuting coordinates. This representation is not irreducible, since

$$e_1 + \cdots + e_n \in (\mathbb{C}^n)^{S_n}.$$

The subspace

$$V := \left\{ (x_1, \dots, x_n) \in \mathbb{C}^n \mid \sum_{i=1}^n x_i = 0 \right\}$$

is the standard S_n -representation.

More generally, the representation corresponding to

$$(n - k, 1, \dots, 1)$$

is

$$\bigwedge^k V.$$

If λ' denotes the conjugate partition, then

$$V_\lambda = V_{\lambda'} \otimes \text{Alt}.$$

3.4 Applications of group theory in quantum mechanics

3.4.1 Symmetries of the Hamiltonian

In quantum mechanics, the state space of a particle in $\mathbb{R}^n$ is the Hilbert space

$$\mathcal{H} = L^2(\mathbb{R}^n).$$

Elements of $\mathcal{H}$ are wave functions

$$\psi : \mathbb{R}^n \rightarrow \mathbb{C}$$

with inner product

$$\langle \psi | \varphi \rangle = \int \overline{\psi(x)} \varphi(x) dx.$$

Definition 3.4.1: Affine Group

The **group of affine transformations** of $\mathbb{R}^n$ is

$$\text{Aff}_n := \{\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid \alpha(x) = Ax + T, \ A \in \text{GL}(n, \mathbb{R}), \ T \in \mathbb{R}^n\}.$$

Identifying α with the pair (T, A) acting by

$$(T, A) : x \mapsto Ax + T,$$

composition reads

$$(T, A)(S, B) = (T + AS, AB),$$

and consequently

$$\text{Aff}_n \cong \mathbb{R}^n \rtimes \text{GL}(n, \mathbb{R}).$$

Remark.

There is a natural group homomorphism which remembers only the linear part:

$$\text{Aff}_n \rightarrow \text{GL}(n, \mathbb{R}), \quad (T, A) \mapsto A,$$

The kernel consists of the transformations with $A = \text{id}$, namely

$$x \mapsto x + T.$$

Hence the translations form a normal subgroup of Aff_n .

In quantum mechanics, one is interested in transformations that preserve distances, angles, and the Lebesgue measure. This restricts the linear part from $\text{GL}(n, \mathbb{R})$ to $O(n)$, and hence gives the subgroup

$$\mathbb{R}^n \rtimes O(n) = \{x \mapsto Ax + T \mid A \in O(n), \ T \in \mathbb{R}^n\}.$$

For $\alpha(x) = Ax + T$, one has

$$\alpha^{-1}(x) = A^{-1}(x - T).$$

The corresponding action on a wave function f is by pullback along the inverse:

$$\rho(\alpha)f(x) = f(\alpha^{-1}(x)) = (f \circ \alpha^{-1})(x).$$

For $a, b \in \text{Aff}_n$,

$$\rho(a)\rho(b)f(x) = (f \circ b^{-1} \circ a^{-1})(x) = (f \circ (ab)^{-1})(x) = \rho(ab)f(x).$$

Proposition 3.4.2

The action of $\mathbb{R}^n \rtimes O(n)$ on $L^2(\mathbb{R}^n)$ is unitary.

Proof:

Let $\alpha(x) = Ax + T$ with $A \in O(n)$, and substitute

$$y = \alpha^{-1}(x), \quad x = \alpha(y) = Ay + T.$$

Since $|\det A| = 1$, we have

$$d^n x = d^n y,$$

and therefore

$$\langle \rho(\alpha)f \mid \rho(\alpha)g \rangle = \int \overline{f(\alpha^{-1}(x))} g(\alpha^{-1}(x)) d^n x = \int \overline{f(y)} g(y) d^n y = \langle f \mid g \rangle.$$

Definition 3.4.3: Hamiltonian

A **Hamiltonian** is a self-adjoint, usually unbounded, operator

$$H : \mathcal{H} \rightarrow \mathcal{H}.$$

Definition 3.4.4: Spatial Symmetries of the Hamiltonian

The **group of spatial symmetries** of the Hamiltonian H is the subgroup

$$\text{Sym}(H) = \{a \in \mathbb{R}^n \rtimes O(n) \mid [\rho(a), H] = 0\}.$$

Remark.

Since $\rho(a)H = H\rho(a)$ for every $a \in \text{Sym}(H)$, the Hamiltonian is an intertwiner of the representation of $\text{Sym}(H)$ on $\mathcal{H}$ with itself.

Assume for the moment that the representation of $\text{Sym}(H)$ on $\mathcal{H}$ decomposes as

$$\mathcal{H} = \bigoplus_i N_i V_i,$$

where the V_i are pairwise non-isomorphic irreducible representations and $N_i V_i$ is shorthand for the isotypic component $N_i \otimes V_i$, with N_i the multiplicity space. From Schur's lemma,

$$\langle \psi_i \mid H \varphi_j \rangle = 0 \quad \text{for} \quad \psi_i \in N_i V_i, \quad \varphi_j \in N_j V_j, \quad i \neq j.$$

Remark.

Why does this work here? By the Peter-Weyl theorem, unitary representations of compact groups decompose into irreducible representations. Thus for compact symmetry groups, such as $O(n)$ or $SO(3)$, the direct-sum decomposition above is justified.

If the symmetry group contains a non-compact part, such as translations $\mathbb{R}^n$, the correct analogue is usually a direct integral rather than a direct sum. This is exactly what happens in Fourier theory and in the Bloch decomposition for lattice translations below.

It follows that H is block diagonal with respect to the above decomposition. Moreover, Schur's

lemma forces H to act as the identity on the irreducible factor V_i , hence

$$H = \bigoplus_i H_i \otimes I_{V_i}.$$

Example.

- **Free particle.** The Hamiltonian

$$H = -\frac{\hbar^2}{2m} \nabla^2$$

is invariant under translations and orthogonal transformations, so

$$\text{Sym}(H) = \mathbb{R}^n \rtimes O(n).$$

- **Particle in a potential.** For

$$H = -\frac{\hbar^2}{2m} \nabla^2 + V(x),$$

the multiplication operator V acts on wave functions by

$$(Vf)(x) = V(x)f(x),$$

so that, for $a = \alpha \in \mathbb{R}^n \rtimes O(n)$,

$$\rho(a)(Vf)(x) = V(\alpha^{-1}(x)) f(\alpha^{-1}(x)),$$

while

$$V(\rho(a)f)(x) = V(x) f(\alpha^{-1}(x)).$$

Consequently,

$$[\rho(a), V]f(x) = (V(\alpha^{-1}(x)) - V(x)) f(\alpha^{-1}(x)).$$

Hence a is a symmetry if

$$V(\alpha^{-1}(x)) = V(x) \quad \forall x \in \mathbb{R}^n,$$

and so

$$\text{Sym}(H) = \{a \in \mathbb{R}^n \rtimes O(n) \mid V(ax) = V(x) \ \forall x \in \mathbb{R}^n\}.$$

- **Hydrogen atom.** The Coulomb potential

$$V(x) \sim \frac{1}{\|x\|}$$

is invariant under rotations, since

$$\|Rx\| = \|x\| \quad \forall R \in O(3).$$

The Hilbert space decomposes as

$$\mathcal{H} \cong \bigoplus_{\ell=0}^{\infty} L^2(\mathbb{R}_{>0}, r^2 dr) \otimes \mathcal{H}_{\ell},$$

where $\mathcal{H}_{\ell}$ is the space of spherical harmonics of degree ℓ ,

$$\mathcal{H}_{\ell} = \text{span}\{Y_{\ell}^m : m = -\ell, \dots, \ell\}.$$

This is an irreducible representation of $SO(3)$ of dimension

$$\dim \mathcal{H}_{\ell} = 2\ell + 1.$$

From Schur's lemma,

$$\langle \psi_{\ell, m} | H | \varphi_{\ell', m'} \rangle = 0 \quad \text{if } \ell' \neq \ell.$$

Thus angular momentum labels the irreducible representations.

- **Lattices and Bloch decomposition.** Let $\Delta \subseteq \mathbb{R}^n$ be a lattice, and consider a periodic potential

$$V(x + T) = V(x) \quad \forall T \in \Delta.$$

Then H has at least the symmetry group Δ . Assuming $\Delta \cong \mathbb{Z}^n$, choose a basis $\{a_1, \dots, a_n\}$ of lattice vectors, so that every lattice vector can be written as

$$T = m_1 a_1 + \dots + m_n a_n, \quad m_i \in \mathbb{Z}.$$

Since $\Delta \cong \mathbb{Z}^n$ is abelian, all of its irreducible representations are one-dimensional, i.e. group homomorphisms

$$\mathbb{Z}^n \rightarrow U(1).$$

The group of such homomorphisms is

$$\text{Hom}(\mathbb{Z}^n, U(1)) \cong U(1)^n.$$

A convenient way to write these representations is by choosing plane waves: for $k \in \mathbb{R}^n$, consider

$$e^{2\pi i k \cdot x}.$$

Then

$$e^{2\pi i k \cdot (x-T)} = e^{2\pi i k \cdot x} e^{-2\pi i k \cdot T},$$

so that the character of Δ is

$$\chi_k(T) = e^{-2\pi i k \cdot T}.$$

In particular,

$$\chi_k(a_j) = e^{-2\pi i k \cdot a_j},$$

and the corresponding element of $U(1)^n$ is

$$(e^{-2\pi i k \cdot a_1}, \dots, e^{-2\pi i k \cdot a_n}).$$

Different values of k can yield the same character: k and $k + G$ give the same character

if and only if

$$e^{-2\pi i G \cdot T} = 1 \quad \forall T \in \Delta,$$

that is,

$$G \cdot T \in \mathbb{Z} \quad \forall T \in \Delta.$$

Definition 3.4.5: Reciprocal Lattice and Brillouin Zone

The **reciprocal lattice** is

$$\Delta^* = \{G \in \mathbb{R}^n \mid G \cdot T \in \mathbb{Z} \ \forall T \in \Delta\}.$$

Equivalently, if $\{a_1, \dots, a_n\}$ is a basis of Δ , then the dual basis $\{a_1^*, \dots, a_n^*\}$ is defined by

$$a_i^* \cdot a_j = \delta_{ij},$$

and the possible characters are labeled by

$$k \in \mathbb{R}^n / \Delta^*.$$

This quotient, or a fundamental domain for it, is called the **Brillouin zone**.

Since the Hamiltonian commutes with lattice translations, the Hilbert space decomposes into sectors labeled by k :

$$\mathcal{H} = \int_{\text{BZ}}^{\oplus} \mathcal{H}_k dk,$$

where $\mathcal{H}_k$ is the subspace transforming under translations as χ_k .

3.4.2 Selection rules

In quantum mechanics, one considers both states

$$|\psi\rangle \in \mathcal{H}$$

and operators on $\mathcal{H}$. Observables are represented by Hermitian operators O , so that

$$\langle O\psi \mid \varphi \rangle = \langle \psi \mid O\varphi \rangle.$$

In particular, expectation values

$$\langle \psi \mid O\psi \rangle$$

are real. If O is positive, then

$$\langle \psi \mid O\psi \rangle \geq 0.$$

Suppose G is a symmetry group of the system. Then $\mathcal{H}$ carries a unitary representation

$$\rho : G \rightarrow U(\mathcal{H}).$$

The induced action on operators is by conjugation:

$$O \longmapsto \rho(g)O\rho(g^{-1}).$$

For time evolution, the Heisenberg-picture operator is

$$O(t) = e^{-iHt}O(0)e^{iHt}, \quad t \in \mathbb{R}.$$

Let $|\psi\rangle \in \mathcal{H}$. Then the transformed vector $O|\psi\rangle$ transforms as

$$O|\psi\rangle \mapsto \rho(g)O\rho(g^{-1})\rho(g)|\psi\rangle = \rho(g)(O|\psi\rangle).$$

Abstractly, this means that the evaluation map

$$\text{End}(\mathcal{H}) \otimes \mathcal{H} \longrightarrow \mathcal{H}, \quad O \otimes |\psi\rangle \mapsto O|\psi\rangle,$$

is G -equivariant.

Proposition 3.4.6

Suppose O transforms in an irreducible representation $W \subseteq \text{End}(\mathcal{H})$, and suppose $|\psi_i\rangle$ and $|\psi_j\rangle$ transform in irreducible representations $V_i, V_j \subseteq \mathcal{H}$. If V_j does not occur in the decomposition

$$W \otimes V_i \cong \bigoplus_k N_k,$$

then

$$\langle \psi_j | O\psi_i \rangle = 0.$$

Proof:

Since O transforms in W and $|\psi_i\rangle$ transforms in V_i , the vector $O|\psi_i\rangle$ lies in the representation generated by $W \otimes V_i$. If V_j is not one of the irreducible summands of $W \otimes V_i$, then $O|\psi_i\rangle$ has no component in the V_j -isotypic part of $\mathcal{H}$. Orthogonality of distinct irreducible components gives

$$\langle \psi_j | O\psi_i \rangle = 0.$$

Example.

Consider the ammonia molecule NH_3 . Its spatial symmetry group is S_3 , permuting the three hydrogen atoms. The irreducible representations of S_3 are

$$\text{Triv}, \quad \text{Alt}, \quad \text{Std}.$$

Geometrically, the z -axis transforms trivially, while the x - and y -directions transform in the standard representation:

$$\hat{z} \in \text{Triv}, \quad \hat{x}, \hat{y} \in \text{Std}.$$

We apply the selection rule to the coordinate operator $\hat{y}$, so $W = \text{Std}$. Let $|\psi_i\rangle \in V_i$. Then $\hat{y}|\psi_i\rangle$ transforms in the tensor product

$$\text{Std} \otimes V_i.$$

Hence:

- If $V_i = \text{Triv}$, then

$$\text{Std} \otimes \text{Triv} \cong \text{Std}.$$

Therefore

$$\langle \psi_j | \hat{y} | \psi_i \rangle = 0$$

unless $|\psi_j\rangle$ transforms in Std .

- If $V_i = \text{Alt}$, then

$$\text{Std} \otimes \text{Alt} \cong \text{Std}.$$

Again,

$$\langle \psi_j | \hat{y} | \psi_i \rangle = 0$$

unless $|\psi_j\rangle$ transforms in Std .

- If $V_i = \text{Std}$, then

$$\text{Std} \otimes \text{Std} \cong \text{Triv} \oplus \text{Alt} \oplus \text{Std}.$$

Thus the matrix element

$$\langle \psi_j | \hat{y} | \psi_i \rangle$$

is not ruled out by symmetry.

Example.

Multi-particle states. Suppose $\mathcal{H}$ is the one-particle Hilbert space. For a system of N particles, the natural Hilbert space is

$$\mathcal{H}^{\otimes N}.$$

The symmetric group S_N acts on $\mathcal{H}^{\otimes N}$ by permuting the tensor factors.

In a quantum system of indistinguishable elementary particles, exchanging two particles can only change the state by a phase:

$$e^{i\theta} |\psi\rangle = P_{(12\dots N)} |\psi\rangle.$$

Hence the multi-particle state lies in a one-dimensional representation of S_N . There are only two such representations:

$$\text{Triv} \quad \text{and} \quad \text{Alt}.$$

The trivial representation gives bosons, while the alternating representation gives fermions. Thus

$$\mathcal{H}_{\text{bos}} = \text{Sym}^N(\mathcal{H}) \subseteq \mathcal{H}^{\otimes N}, \quad \mathcal{H}_{\text{ferm}} = \bigwedge^N \mathcal{H} \subseteq \mathcal{H}^{\otimes N}.$$

If $\dim \mathcal{H} = k$, then

$$\dim \text{Sym}^N(\mathcal{H}) = \binom{k+N-1}{N}, \quad \dim \bigwedge^N \mathcal{H} = \binom{k}{N}.$$

For a spin- $\frac{1}{2}$ system,

$$\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2,$$

where $L^2(\mathbb{R}^3)$ is the orbital part and $\mathbb{C}^2 = \text{span}\{|\uparrow\rangle, |\downarrow\rangle\}$ is the spin part. Since spin- $\frac{1}{2}$

particles are fermions, the physical N -particle Hilbert space is

$$\bigwedge^N (L^2(\mathbb{R}^3) \otimes \mathbb{C}^2).$$

For $N = 2$, the spin states decompose as

$$(\mathbb{C}^2)^{\otimes 2} = \text{Sym}^2 \mathbb{C}^2 \oplus \bigwedge^2 \mathbb{C}^2.$$

The symmetric spin states are

$$|\uparrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle, \quad \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle),$$

and the antisymmetric spin state is

$$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

Therefore

$$\mathcal{H} = L^2(\mathbb{R}^3 \times \mathbb{R}^3) \otimes (\text{Sym}^2 \mathbb{C}^2 \oplus \bigwedge^2 \mathbb{C}^2),$$

but the physical subspace is still the fermionic subspace.

For $N = 3$,

$$(\mathbb{C}^2)^{\otimes 3} \cong \mathbb{C}^8.$$

The fully symmetric spin states include

$$|\uparrow\uparrow\uparrow\rangle, \quad |\downarrow\downarrow\downarrow\rangle, \quad \frac{1}{\sqrt{3}}(|\uparrow\uparrow\downarrow\rangle + |\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle),$$

and

$$\text{Sym}^3 \mathbb{C}^2 \cong \text{Triv}^{\oplus 4}.$$

On the other hand,

$$\bigwedge^3 \mathbb{C}^2 = 0$$

by the Pauli exclusion principle.

To find the missing two-dimensional pieces, one uses the irreducible representations of S_3 . For example, for the Young diagram

$$T = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array},$$

the row group and column group are

$$R_T = \{e, (12)\}, \quad C_T = \{e, (13)\}.$$

Hence the Young symmetrizer is

$$\gamma_T = a_T b_T = \left(\sum_{\sigma \in R_T} \sigma \right) \left(\sum_{\tau \in C_T} \text{sgn}(\tau) \tau \right) = e + (12) - (13) - (132).$$

The Specht module

$$S_T := \mathbb{C}[S_3]\gamma_T$$

is the standard representation. Applying Gram-Schmidt to γ_T and $(13)\gamma_T$ gives a basis

$$\gamma_T = \frac{1}{2}(e + (12) - (13) - (132)),$$

$$\gamma_T^\perp = \frac{1}{2\sqrt{3}}(-2e + (12) - (13) - (132)).$$

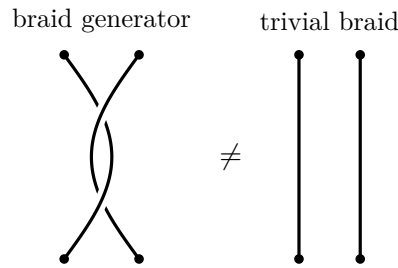
Acting with these elements on $|\uparrow\uparrow\downarrow\rangle$ gives states in the standard representation:

$$\gamma_T|\uparrow\uparrow\downarrow\rangle = \frac{1}{\sqrt{6}}(2|\uparrow\uparrow\downarrow\rangle - |\downarrow\uparrow\uparrow\rangle - |\uparrow\downarrow\uparrow\rangle),$$

$$\gamma_T^\perp|\uparrow\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}}(|\downarrow\uparrow\uparrow\rangle - |\uparrow\downarrow\uparrow\rangle).$$

Remark.

Fermions and bosons only exhaust the possibilities in spacetime dimension at least 3. In two spatial dimensions, exchanges are represented by the braid group B_N , not just by S_N . Thus particles in two dimensions can be anyons.



Chapter 4

Lie groups

"Groups with a notion of continuity."

Example.

- $(\mathbb{R}, +)$ has a notion of closeness $\lim_{n \rightarrow \infty} a_n = b$ and a notion of continuity as we can describe paths $\gamma(-\epsilon, \epsilon) \rightarrow \mathbb{R}$
- $(\mathbb{R}^n, +)$, $U(N)$, $SU(N)$, $SO(N)$, $GL(N, \mathbb{R})$, $GL(N, \mathbb{C})$ are all groups with a notion of closeness and continuity

Consider the 1 dimensional representations of $\mathbb{R}$, i.e. the homomorphisms

$$\rho : (\mathbb{R}, +) \rightarrow (\mathbb{C}^\times, \cdot)$$

Given $\rho(1) = c$, we can immediately conclude the following:

- $\rho(n) = \rho(1 + 1 + \cdots + 1) = \rho(1)^n = c^n$
- $c = \rho(\frac{1}{n} \cot n) = \rho(\frac{1}{n})^n \Rightarrow \rho(\frac{1}{n}) = c^{\frac{1}{n}}$
- $\rho(\frac{m}{n}) = c^{\frac{m}{n}}$

Since $\mathbb{R}$ is a vector space over $\mathbb{Q}$, it has a $\mathbb{Q}$ -basis $\{1, a_i\}_{i \in I}$ with uncountable index set I . Then for a representation ρ , we have $\rho(a_i) \in \mathbb{C}^\times$ and $\rho(\sum_{i \in I} \lambda_i a_i) = \prod_{i \in I} \rho(a_i)^{\lambda_i}$ for $\lambda_i \in \mathbb{Q}$. This means that the representation is determined by the values $\{\rho(a_i)\}_{i \in I}$, which can be chosen arbitrarily. Hence, there are uncountably many 1-dimensional representations of $\mathbb{R}$. Thus representations without continuity conditions are hopeless.

Definition 4.0.1: Matrix Lie groups

A **matrix Lie group** is a closed subgroup of $GL(n, \mathbb{R})$. Closedness is with respect to the usual topology on $\mathbb{R}^{n^2}$.

Remark.

Complex matrix groups e.g. $SU(n)$ can be viewed as a subgroup of $GL(2n, \mathbb{R})$.

Example.

The real numbers with addition $(\mathbb{R}, +) \cong \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} : x \in \mathbb{R} \right\}$ is a matrix Lie group.

4.1 A few basics from differential geometry

Definition 4.1.1: topological space

A **topological space** X is characterized by a collection of subsets called the opensets, $\mathcal{O}(X) \subseteq \mathcal{P}(X)$ such that:

- $\emptyset, X \in \mathcal{O}(X)$
- $\bigcup_{i \in I} U_i \in \mathcal{O}(X)$ for U_i a family of opensets
- $U_1 \cap U_2 \in \mathcal{O}(X)$ for $U_1, U_2 \in \mathcal{O}(X)$

Example.

Let (X, d) be a metric space. Define $U \subseteq X$ to be open if

$$\forall x \in U \quad \exists \epsilon > 0 \text{ such that } B(x, \epsilon) := \{y \in X : d(y, x) < \epsilon\} \subseteq U$$

Definition 4.1.2: continuous map

A map $f : X \rightarrow Y$ between topological spaces is **continuous** if

$$f^{-1}(U) \in \mathcal{O}(X) \quad \forall U \in \mathcal{O}(Y).$$

If f is a bijection whose inverse is also continuous, it is called a **homeomorphism**.

Definition 4.1.3: chart

A **chart** on a topological space X is a pair (U_α, ψ_α) consisting of an open set $U_\alpha \subseteq X$ and a homeomorphism

$$\psi_\alpha : U_\alpha \rightarrow V_\alpha \subseteq \mathbb{R}^n.$$

For two charts (U_α, ψ_α) and (U_β, ψ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the **transition map**

$$\psi_{\alpha\beta} : \tilde{V}_\alpha \rightarrow \tilde{V}_\beta$$

is obtained by restricting ψ_α, ψ_β to the images $\tilde{V}_\alpha, \tilde{V}_\beta$ of the intersection $U_\alpha \cap U_\beta$. Since this is a map between subsets of $\mathbb{R}^n$, the notion of smoothness is well defined.

Definition 4.1.4: atlas

An **atlas** on X is a collection of charts $\{(U_i, \psi_i)\}_{i \in I}$ whose domains form an open cover of X :

$$\bigcup_{i \in I} U_i = X.$$

An atlas is **maximal** if it is not contained in a larger atlas.

Definition 4.1.5: Hausdorff space

A topological space X is called **Hausdorff** if

$$\forall x, y \in X \text{ with } x \neq y \quad \exists U_x, U_y \in \mathcal{O}(X) \text{ such that } x \in U_x, y \in U_y, U_x \cap U_y = \emptyset.$$

Definition 4.1.6: manifold

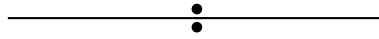
An n -dimensional (smooth) manifold is a topological Hausdorff space equipped with a maximal atlas.

Remark.

Why Hausdorff? We want unique limits. Consider

$$\mathbb{R} \sqcup \mathbb{R} / (x_1 \sim x_2 \text{ for } x_1 \neq 0)$$

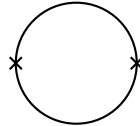
→ “ $\mathbb{R}$ with two zeros”.



This has a maximal atlas but is *not* Hausdorff. Limits are not uniquely defined. Thus we require Hausdorffness to ensure that limits are well-defined.

Example.

- $\mathbb{R}^n$ is a manifold; $\text{id} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an atlas. (This manifold structure is unique, except $\mathbb{R}^4$ which has ∞ many.)
- $S^1 \subset \mathbb{R}^2$ defined by $x^2 + y^2 = 1$ is covered by the four charts



$$\psi_{\pm}^x : \{(x, \pm\sqrt{1-x^2}) \mid x \neq \pm 1\} \longrightarrow (-1, 1), \quad (x, y) \mapsto x$$

$$\psi_{\pm}^y : \{(\pm\sqrt{1-y^2}, y) \mid y \neq \pm 1\} \longrightarrow (-1, 1), \quad (x, y) \mapsto y$$

$\Rightarrow S^1$ is a manifold.

Definition 4.1.7: smooth map

Let M, N be manifolds. A continuous map $f : M \rightarrow N$ is **smooth** if for all $m \in M$ there exists a chart U_m containing m and a chart $U_{f(m)}$ containing $f(U_m)$ such that

$$\psi_{f(m)} \circ f \circ \psi_m^{-1}$$

is smooth. A **diffeomorphism** is a smooth map with smooth inverse.

Definition 4.1.8: derivation

For M a manifold, we write $C^\infty(M)$ for the vector space of smooth functions $f : M \rightarrow \mathbb{R}$. For $x \in M$, a **derivation** at x is a linear map

$$D_x : C^\infty(M) \longrightarrow \mathbb{R}$$

satisfying the Leibniz rule

$$D_x(f \cdot g) = D_x(f) \cdot g(x) + f(x) \cdot D_x(g).$$

Given local coordinates $(x_1, \dots, x_n)$ on a chart U_α containing x , one defines

$$\left. \frac{\partial}{\partial x_i} \right|_x : C^\infty(M) \longrightarrow \mathbb{R}, \quad f \longmapsto \left. \frac{\partial}{\partial x_i} \right|_{\psi_\alpha(x)} (f \circ \psi_\alpha^{-1}).$$

The derivations $\left. \frac{\partial}{\partial x_i} \right|_x$ form a basis of the vector space of derivations at x .

Definition 4.1.9: tangent space

The **tangent space** to M at $x \in M$ is the vector space

$$T_x M := \{\text{derivations at } x\}.$$

Example.

- $T_x \mathbb{R}^n = \text{span} \left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right\}$
- $T_v V = V$ for V a vector space.

Remark.

Tangent vectors as equivalence classes of curves. Consider a smooth curve

$$\gamma(t) : (-\varepsilon, \varepsilon) \longrightarrow M, \quad \gamma(0) = x.$$

Its differential at 0 is a linear map

$$d\gamma|_0 : T_0(-\varepsilon, \varepsilon) \longrightarrow T_x M, \quad \frac{\partial}{\partial t} \longmapsto d\gamma|_0 \left[\frac{\partial}{\partial t} \right].$$

For $F \in C^\infty(M)$ this acts as

$$d\gamma|_0 \left[\frac{\partial}{\partial t} \right] (F) = \left. \frac{\partial}{\partial t} \right|_{t=0} F(\gamma(t)).$$

For $M = V$ a vector space with $T_x V = V$, writing $\gamma(t) = \gamma^1(t)x^1 + \dots + \gamma^n(t)x^n$ gives

$$\frac{\partial}{\partial t} F(\gamma(t)) = \frac{\partial F}{\partial x^1} \frac{\partial \gamma^1}{\partial t} + \dots + \frac{\partial F}{\partial x^n} \frac{\partial \gamma^n}{\partial t},$$

so that

$$d\gamma|_0 \left[\frac{\partial}{\partial t} \right] = \dot{\gamma}(0) \in V = T_x V.$$

Hence a tangent vector at x can equivalently be viewed as the velocity $\dot{\gamma}(0)$ of a curve through x .

Definition 4.1.10: differential

Let $f : M \rightarrow N$ be a smooth map. It induces a linear map

$$df|_x : T_x M \longrightarrow T_{f(x)} N, \quad D_x \longmapsto \left(h \mapsto D_x(h \circ f) \right) \quad \forall h \in C^\infty(N),$$

called the **differential**, or **push-forward**, of f . In local coordinates x^i on M and y^j on N ,

$$\frac{\partial}{\partial x^i} \longmapsto \sum_j \frac{\partial f^j}{\partial x^i} \frac{\partial}{\partial y^j}.$$

4.2 Lie algebras

Definition 4.2.1: Lie group

A **Lie group** is a smooth manifold G equipped with a group structure such that

$$\bullet : G \times G \longrightarrow G \quad \text{and} \quad (\cdot)^{-1} : G \longrightarrow G$$

are smooth maps.

Remark.

The Cartesian product of manifolds is (in most cases) also a manifold.

G acts on itself by conjugation

$$\text{Ad}(g) : G \longrightarrow G, \quad h \longmapsto ghg^{-1},$$

which is a smooth map since Lie groups act smoothly by definition. Taking the differential at the identity gives a linear map

$$d\text{Ad}(g)|_e : T_e G \longrightarrow T_e G,$$

and hence a group homomorphism

$$\text{ad} : G \longrightarrow \text{Aut}(T_e G).$$

Definition 4.2.2: smooth representation

Let G be a Lie group. A **(smooth) representation** of G is a smooth group homomorphism

$$G \longrightarrow \text{Aut}(V)$$

for a vector space V .

Proposition 4.2.3

Let G be a Lie group. The map

$$\text{ad} : G \longrightarrow \text{Aut}(T_e G), \quad g \longmapsto d\text{Ad}(g)|_e,$$

is a representation of G , called the **adjoint action**.

Proof:

Homomorphism. We need $\text{ad}(g) \circ \text{ad}(h) = \text{ad}(gh)$, i.e.

$$d\text{Ad}(g)|_e \circ d\text{Ad}(h)|_e = d\text{Ad}(gh)|_e.$$

This follows from $\text{Ad}_g \circ \text{Ad}_h = \text{Ad}_{gh}$ together with the chain rule: for smooth maps $f : M \rightarrow N$ and $f' : N \rightarrow O$,

$$df'|_{f(m)} \circ df|_m = d(f' \circ f)|_m.$$

Indeed, for a derivation $D_m \in T_m M$ and $h \in C^\infty(O)$,

$$\begin{aligned} (df'|_{f(m)} \circ df|_m)(D_m)[h] &= df'|_{f(m)}(D_m(h \circ f' \circ \cdot)) \\ &= D_m(h \circ f' \circ f) = d(f' \circ f)|_m(D_m)[h]. \end{aligned}$$

Identity. $\text{Ad}(e) = \text{id}_G$, so $\text{ad}(e) = d(\text{id}_G)|_e = \text{id}_{T_e G}$.

Smoothness. Follows from the smoothness of the group operations. ■

Proposition 4.2.4

Let $f : G \rightarrow H$ be a Lie group homomorphism. Then $df|_e : T_e G \rightarrow T_e H$ satisfies

$$df|_e \circ \text{ad}_G(g) = \text{ad}_H(f(g)) \circ df|_e \quad \forall g \in G.$$

Proof:

We first show that $f \circ \text{Ad}_G(g) = \text{Ad}_H(f(g)) \circ f$. For all $g, h \in G$,

$$f(\text{Ad}_G(g)[h]) = f(ghg^{-1}) = f(g)f(h)f(g)^{-1} = \text{Ad}_H(f(g))[f(h)].$$

The statement then follows by differentiating both sides at the identity using $d(\cdot)|_e$. ■

Example.

Let V be a finite-dimensional vector space. The group $\text{Aut}(V)$ is an open subset of $\text{End}(V) \cong \mathbb{R}^{2 \dim V}$, so its tangent space at the identity is

$$T_{\text{id}} \text{Aut}(V) \cong \text{End}(V).$$

The adjoint action of $\text{Aut}(V)$ on $\text{End}(V)$ is

$$\text{ad}(U) : A \mapsto UAU^{-1}.$$

Definition 4.2.5: Lie bracket

Let G be a Lie group. The differential of ad at the identity gives a map

$$d\text{ad}|_e : T_e G \longrightarrow T_{\text{id}} \text{Aut}(T_e G) = \text{End}(T_e G).$$

The **Lie bracket** on $T_e G$ is defined by

$$[X, Y] := d\text{ad}|_e(X)[Y] \in T_e G \quad \forall X, Y \in T_e G.$$

How do we compute $[X, Y]$?

Example.

Consider $G = \text{Aut}(V)$, so $T_e G = \text{End}(V)$. For $X \in \text{End}(V)$, the curve

$$\gamma_X(t) = \mathbb{I} + tX$$

passes through e at $t = 0$ with tangent vector X , since $X[v] = \frac{\partial}{\partial t}|_{t=0} \gamma_X(t)[v]$.

By definition of the Lie bracket,

$$[X, Y]F = d\text{ad}|_e(X)[Y]F = \frac{\partial}{\partial t_1} \Big|_{t_1=0} \text{ad}(\gamma_X(t_1))[Y]F.$$

Since $\text{ad}(U)$ acts by conjugation, $\text{ad}(\gamma_X(t_1))[Y]F = Y[F(\gamma_X(t_1) \cdot \gamma_X(t_1)^{-1})]$. Representing Y by a curve γ_Y ,

$$\begin{aligned} [X, Y]F &= \frac{\partial}{\partial t_1} \Big|_{t_1=0} Y[F(\gamma_X(t_1) (\cdot) \gamma_X(t_1)^{-1})] \\ &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} F(\gamma_X(t_1) \gamma_Y(t_2) \gamma_X(t_1)^{-1}). \end{aligned} \quad (4.1)$$

For $G = \text{GL}(n)$ with $\gamma_X(t) = \mathbb{I} + t_1 X$ and $\gamma_Y(t) = \mathbb{I} + t_2 Y$,

$$\begin{aligned} [X, Y] &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} (\mathbb{I} + t_1 X)(\mathbb{I} + t_2 Y)(\mathbb{I} + t_1 X)^{-1} \\ &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} (\mathbb{I} + t_1 X)(\mathbb{I} + t_2 Y)(\mathbb{I} - t_1 X + \mathcal{O}(t_1^2)) \\ &= XY - YX. \end{aligned}$$

We have shown that any Lie group G gives rise to a tangent space $T_e G$ equipped with a bilinear

bracket

$$[-, -] : T_e G \otimes T_e G \longrightarrow T_e G,$$

and that for matrix Lie groups this bracket is the commutator $[X, Y] = XY - YX$.

Proposition 4.2.6

Let $f : G \rightarrow H$ be a Lie group homomorphism. Then

$$[df|_e X, df|_e Y]_{T_e H} = df|_e [X, Y]_{T_e G}.$$

Proof:

Using formula (4.1),

$$\begin{aligned} [df|_e X, df|_e Y]_{T_e H} F &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} F \left(f(\gamma_X(t_1)) f(\gamma_Y(t_2)) f(\gamma_X(t_1))^{-1} \right) \\ &= \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{t_1=t_2=0} F \left(f \left(\gamma_X(t_1) \gamma_Y(t_2) \gamma_X(t_1)^{-1} \right) \right) \\ &= df|_e [X, Y]_{T_e G} F, \end{aligned}$$

where in the second step we used that f is a group homomorphism, and in the third step we applied formula (4.1) for G . ■

Remark.

For matrix Lie groups $G \leq \text{GL}(n)$, the inclusion $G \hookrightarrow \text{GL}(n)$ is a Lie group homomorphism, so $T_e G \subseteq \text{End}(\mathbb{R}^n)$ and the bracket on $T_e G$ is the restriction of the commutator:

$$[X, Y]_{T_e G} = XY - YX.$$

Theorem 4.2.7: Properties of the Lie bracket

Let G be a Lie group. For all $X, Y, Z \in T_e G$:

1. $[X, Y] = -[Y, X]$ (antisymmetric)
2. $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ (Jacobi identity)

Proof:

Antisymmetry. We show $[X, X] = 0$; antisymmetry then follows from

$$0 = [X + Y, X + Y] = [X, Y] + [Y, X].$$

Assume $\gamma_X(t) \cdot \gamma_X(t') = \gamma_X(t + t')$ for t, t' near 0. Then in local coordinates

$$\frac{\partial}{\partial t} \gamma(t) = \lim_{\varepsilon \rightarrow 0} \frac{\gamma(t + \varepsilon) - \gamma(t)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \gamma(t) \frac{\gamma(\varepsilon) - \gamma(0)}{\varepsilon} = \gamma(t) \cdot X.$$

This is a system of ODEs and hence has a unique solution near $t = 0$, so γ_X exists. Using

the Proposition above,

$$[X, X] F = \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{\substack{t_1=0 \\ t_2=0}} F(\gamma_X(t_1) \gamma_X(t_2) \gamma_X(-t_1)) = \frac{\partial}{\partial t_1} \frac{\partial}{\partial t_2} \Big|_{\substack{t_1=0 \\ t_2=0}} F(\gamma_X(t_2)) = 0,$$

where we used $\gamma_X(t_1) \gamma_X(t_2) \gamma_X(-t_1) = \gamma_X(t_2)$.

Jacobi identity. Consider

$$H : T_e G \rightarrow \text{End}(T_e G), \quad X \mapsto [X, -],$$

which is the differential of $G \rightarrow \text{Aut}(T_e G)$. By the previous proposition,

$$H([X, Y]) := [[X, Y], -] = H(X) \circ H(Y) - H(Y) \circ H(X).$$

Evaluating at $Z \in T_e G$:

$$\begin{aligned} [[X, Y], Z] &= [X, [Y, Z]] - [Y, [X, Z]] \\ &= -[Z, [X, Y]] = [X, [Y, Z]] + [Y, [Z, X]], \end{aligned}$$

hence $0 = [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]]$. ■

Definition 4.2.8: Lie algebra

A **Lie algebra** is a vector space $\mathfrak{g}$ equipped with a linear map

$$[-, -] : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$$

such that for all $X, Y, Z \in \mathfrak{g}$:

- $[X, Y] = -[Y, X]$
- $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ (Jacobi identity)

Definition 4.2.9: Lie algebra homomorphism

A **Lie algebra homomorphism** is a map $f : \mathfrak{g} \rightarrow \mathfrak{g}'$ such that

$$f([X, Y]) = [f(X), f(Y)].$$

Remark.

For any Lie group, $T_e G =: \mathfrak{g}$ is a Lie algebra, and every Lie group homomorphism $G \rightarrow H$ induces a Lie algebra homomorphism $\mathfrak{g} \rightarrow \mathfrak{h}$. In particular, a representation $\rho : G \rightarrow \text{Aut}(V)$ induces a Lie algebra homomorphism $\mathfrak{g} \rightarrow \text{End}(V)$.

Definition 4.2.10: Representation of a Lie algebra

Let $\mathfrak{g}$ be a Lie algebra. A **representation** of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\mathfrak{g} \rightarrow \text{End}(V)$$

for a vector space V .

Remark.

Most definitions for finite groups, such as irreducible representations, carry over unchanged to Lie groups. We will not repeat them here.

Example.

The Lie group $SU(2)$ and its Lie algebra $\mathfrak{su}(2)$.

$$SU(2) := \{U \in GL(2, \mathbb{C}) \mid U^\dagger U = I, \det(U) = 1\}$$

This is not a complex Lie group (real dimension 3), even though the elements are complex.

For $U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $U^\dagger = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}$,

$$U^\dagger U = \begin{pmatrix} \bar{a}a + \bar{c}c & \bar{a}b + \bar{c}d \\ \bar{b}a + \bar{d}c & \bar{b}b + \bar{d}d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad ad - bc = 1.$$

All solutions are of the form

$$\begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (**)$$

hence $\dim SU(2) = 3$.

For every $U \in SU(2)$, there exists $V \in U(2)$ such that

$$VUV^{-1} = \begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix}.$$

Setting $V' = (\det V)^{-1/2}V$ gives $V' \in SU(2)$, so the conjugacy classes are labelled by $\alpha \in U(1)/\mathbb{Z}_2 \cong \mathbb{R}/\pi\mathbb{R}$.

Writing $\alpha = a + ib$ and $\beta = c + id$ with $a, b, c, d \in \mathbb{R}$, condition $(**)$ gives

$$a^2 + b^2 + c^2 + d^2 = 1,$$

so $SU(2) \cong S^3$ as a manifold. With this parameterisation,

$$U = \begin{pmatrix} a + ib & c + id \\ c - id & a - ib \end{pmatrix} = \underbrace{a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{i}} + \underbrace{b \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}}_{\mathbf{j}} + \underbrace{c \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\mathbf{k}} + \underbrace{d \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}}_{\mathbf{l}},$$

so $SU(2)$ is also the group of unit quaternions.

Connection to $SO(3)$. The submanifold $a = 0$ is diffeomorphic to S^2 and preserved under conjugation, giving a map

$$SU(2) = \text{Spin}(3) \longrightarrow SO(3) \quad (\text{not a bijection}),$$

with kernel $\{1, -1\}$, and $SO(3) = SU(2)/\mathbb{Z}_2$ is a double cover. Since $SU(2)$ and $SO(3)$ differ only by -1 , which is not near the identity, the Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are the same.

The Lie algebra $\mathfrak{su}(2)$. $\mathfrak{su}(2)$ consists of traceless skew-Hermitian matrices ($A = -A^\dagger$), with basis

$$X = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

The bracket satisfies $[X, Y] = 2Z$ and cyclic permutations.

Exponential map. For $A \in \text{Mat}(2, \mathbb{C})$, the matrix exponential is $\exp(A) = \sum_{n=0}^{\infty} \frac{1}{n!} A^n$. For $A \in \mathfrak{su}(2)$ (so $A^\dagger = -A$ and $\text{tr}(A) = 0$):

- $\exp(A)^\dagger \exp(A) = \exp(A^\dagger) \exp(A) = \exp\left(\underbrace{A^\dagger + A}_{=0}\right) = \mathbb{I},$
- $\det(\exp(A)) = \exp(\text{tr}(A)) = 1,$

so $\exp: \mathfrak{su}(2) \rightarrow SU(2)$ is a smooth map. Setting $\gamma_A(t) := \exp(tA)$ satisfies $\gamma_A(t+t') = \gamma_A(t)\gamma_A(t')$. Since $(U^\dagger A U)^n = U^\dagger A^n U$, we have $U^\dagger \exp(A) U = \exp(U^\dagger A U)$, so every conjugacy-class representative

$$\begin{pmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{pmatrix} = \exp \begin{pmatrix} ix & 0 \\ 0 & -ix \end{pmatrix}, \quad \alpha = e^{ix},$$

is in the image, hence $\exp: \mathfrak{su}(2) \rightarrow SU(2)$ is surjective.

For a representation $\rho: \mathfrak{su}(2) \rightarrow \text{End}(V)$, one can try to define $\rho(\exp(A)) := \exp(\rho(A))$ for $g = \exp(A)$, but this is not well defined in general.

Remark.

The local solution γ_X extends to all of $\mathbb{R}$ via

$$\hat{\gamma}_X: \mathbb{R} \rightarrow G, \quad \hat{\gamma}_X(t) := \gamma_X\left(\frac{t}{n}\right)^n,$$

which is a Lie group homomorphism.

Definition 4.2.11: 1-parameter subgroup

A **1-parameter subgroup** of G is a Lie group homomorphism $\mathbb{R} \rightarrow G$. These are in bijection with $\mathfrak{g}$.

Definition 4.2.12: Exponential map

Let G be a Lie group with Lie algebra $\mathfrak{g}$. The **exponential map** is

$$\exp : \mathfrak{g} \rightarrow G, \quad X \mapsto \gamma_X(1) \in G.$$

For matrix Lie groups this coincides with the matrix exponential $\exp(A) = \sum_{i=0}^{\infty} \frac{A^i}{i!}$.

Remark.

The differential $d\exp|_0 : \mathfrak{g} \rightarrow \mathfrak{g}$ is the identity. This implies that $\exp$ is a diffeomorphism when restricted to small neighbourhoods of $0 \in \mathfrak{g}$ and $e \in G$.

Remark.

There is a formula for products in G near e in terms of the Lie bracket: the Baker-Campbell-Hausdorff formula.

Proposition 4.2.13

Let $\psi : G \rightarrow H$ be a Lie group homomorphism. Then $\psi \circ \exp_G = \exp_H \circ d\psi|_e$, i.e., the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{d\psi|_e} & \mathfrak{h} \\ \exp_G \downarrow & & \downarrow \exp_H \\ G & \xrightarrow{\psi} & H \end{array}$$

Proof:

Consider $\gamma_X : \mathbb{R} \rightarrow G$. Then $\psi \circ \gamma_X$ is a group homomorphism $\mathbb{R} \rightarrow H$ (as a composition of group homomorphisms) with tangent vector $d\psi|_e[X]$ at $t = 0$. By the bijection between 1-parameter subgroups and $\mathfrak{h}$, it follows that $\psi \circ \gamma_X = \gamma_{d\psi|_e[X]}$, and

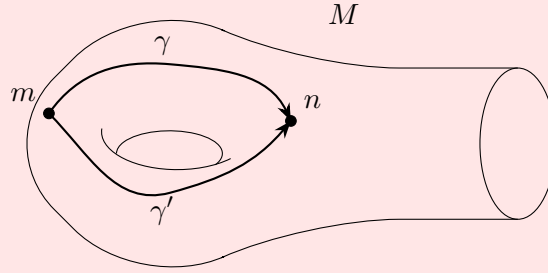
$$\psi(\exp(X)) = \psi(\gamma_X(1)) = \gamma_{d\psi|_e[X]}(1) = \exp(d\psi|_e[X]).$$

Definition 4.2.14: Path-connected and simply connected manifolds

A manifold M is **(path-) connected** if for each pair of points $m, n \in M$ there exists a continuous map $\gamma : [0, 1] \rightarrow M$ with $\gamma(0) = m$ and $\gamma(1) = n$.

M is called **simply connected** if for each pair of paths γ, γ' with $\gamma(0) = \gamma'(0) = m$ and $\gamma(1) = \gamma'(1) = n$, there exists a continuous map $H : [0, 1] \times [0, 1] \rightarrow M$ such that

- $H(0, t) = \gamma(t)$
- $H(1, t) = \gamma'(t)$
- $H(x, 0) = m$
- $H(x, 1) = n$

**Example.**

- $\mathbb{R}^n$ is simply connected.
- S^1 is connected but not simply connected.
- $SU(n)$ for $n \geq 2$ is simply connected.

Theorem 4.2.15: Lie's 1st Theorem

Let G be a connected Lie group. Two Lie group homomorphisms $\psi, \psi' : G \rightarrow H$ are equal if $d\psi|_e = d\psi'|_e$.

Proof:

By the local diffeomorphism property of $\exp$ and the identity $\psi \circ \exp_G = \exp_H \circ d\psi|_e$, the condition $d\psi|_e = d\psi'|_e$ implies $\psi = \psi'$ on a neighbourhood of e . It therefore suffices to show that every $g \in G$ is a product of elements from any neighbourhood U of e .

Let $\Gamma \subseteq G$ be the subgroup generated by U . Since $\Gamma = \bigcup_{h \in \Gamma} U \cdot h$, Γ is a union of open sets and hence open. Letting $\Gamma' = G \setminus \Gamma$, for any $h' \in \Gamma'$ the coset $U \cdot h' \subseteq \Gamma'$ (otherwise $u \cdot h' \in \Gamma$ would imply $h' = u^{-1}(u \cdot h') \in \Gamma$, a contradiction), so

$$\Gamma' = \bigcup_{h' \in \Gamma'} U \cdot h'$$

is likewise open. Thus $G = \Gamma \sqcup \Gamma'$ is a partition into two disjoint open sets.

If $\Gamma' \neq \emptyset$, pick $h \in \Gamma$ and $h' \in \Gamma'$. Since G is path-connected, there exists $\gamma : [0, 1] \rightarrow G$ with $\gamma(0) = h$ and $\gamma(1) = h'$. Then

$$[0, 1] = \gamma^{-1}(\Gamma) \cup \gamma^{-1}(\Gamma')$$

is a partition of $[0, 1]$ into two disjoint open non-empty sets, contradicting the connectedness of $[0, 1]$. Hence $\Gamma' = \emptyset$, i.e. $\Gamma = G$. ■

Theorem 4.2.16: Lie's 2nd Theorem

Let G be a simply connected Lie group. Then there is a one-to-one correspondence between Lie group homomorphisms $\psi : G \rightarrow H$ and Lie algebra homomorphisms $\Psi : \mathfrak{g} \rightarrow \mathfrak{h}$.

Theorem 4.2.17: Lie's 3rd Theorem

For each finite-dimensional real Lie algebra $\mathfrak{g}$ there exists a simply connected Lie group G with Lie algebra $\mathfrak{g}$.

Remark.

G in the above is unique: if G' is another simply connected Lie group with Lie algebra $\mathfrak{g}$, then $\text{id} : \mathfrak{g} \rightarrow \mathfrak{g}$ lifts to a diffeomorphism $G \xrightarrow{\sim} G'$.

Example. $(\mathbb{R}, +) \xrightarrow{\exp} U(1)$ is a Lie algebra isomorphism (both have Lie algebra $\mathbb{R}$), but the only Lie group homomorphism $U(1) \rightarrow \mathbb{R}$ is the constant map at 0, since $U(1) \cong S^1$ is not simply connected.

Corollary 4.2.18

For a simply connected Lie group G there is a bijection between representations of G and representations of $\mathfrak{g}$.

Proof:

A representation $\rho : G \rightarrow \text{Aut}(V)$ is a Lie group homomorphism, which by Lie's 2nd Theorem corresponds bijectively to a Lie algebra homomorphism $\mathfrak{g} \rightarrow \text{End}(V)$, i.e. a representation of $\mathfrak{g}$. ■

Chapter 5

Representation theory of $\mathfrak{su}(n)$

5.1 Representation theory of $\mathfrak{su}(2)$ and $\mathfrak{sl}(2)$

Remark.

Recall that $SU(2)$ and $SO(3)$ share the same Lie algebra (via the double cover $SU(2) \twoheadrightarrow SO(3)$):

$$\mathfrak{su}(2) = \{A \in \text{Mat}(2, \mathbb{C}) \mid \text{tr}(A) = 0, A^\dagger = -A\}.$$

The Pauli matrix

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is Hermitian, so $\sigma_x^\dagger = \sigma_x \neq -\sigma_x$, and hence $\sigma_x \notin \mathfrak{su}(2)$. The associated basis element of $\mathfrak{su}(2)$ is $X := i\sigma_x$, and one has $\sigma_x = -iX$.

Definition 5.1.1: Complexification of a Lie algebra

Let $\mathfrak{g}$ be a real Lie algebra. Its **complexification** is the complex vector space

$$\mathfrak{g}_{\mathbb{C}} := \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C},$$

equipped with the complex-linear extension of the Lie bracket, defined on pure tensors by

$$[X \otimes \lambda, Y \otimes \lambda'] := [X, Y] \otimes \lambda\lambda' \quad \forall X, Y \in \mathfrak{g}, \lambda, \lambda' \in \mathbb{C}.$$

Example.

The complexification of $\mathfrak{su}(2)$ is

$$\mathfrak{sl}(2, \mathbb{C}) = \{A \in \text{Mat}(2, \mathbb{C}) \mid \text{tr}(A) = 0\}.$$

Remark.

Let $\mathfrak{g}$ be a real Lie algebra. Complex representations of $\mathfrak{g}_{\mathbb{C}}$, i.e. complex-linear Lie algebra homomorphisms $\mathfrak{g}_{\mathbb{C}} \rightarrow \text{End}(V)$, are in bijection with real Lie algebra homomorphisms $\mathfrak{g} \rightarrow \text{End}(V)$. The bijection is given by restriction to $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$, with inverse given by complex-linear extension.

To study the representations of $\mathfrak{su}(2)$, we use this bijection and work instead with the complex representations of $\mathfrak{sl}(2, \mathbb{C})$. We choose the standard basis

$$\begin{aligned} H &:= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z = L_z, \\ E &:= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \frac{1}{2}(\sigma_x + i\sigma_y) = L_+, \\ F &:= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \frac{1}{2}(\sigma_x - i\sigma_y) = L_-. \end{aligned}$$

One verifies directly that the commutation relations are

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

Proposition 5.1.2

Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(V)$ be a complex representation, and let $v \in V$ be an eigenvector of $\rho(H)$ with eigenvalue $\lambda \in \mathbb{C}$:

$$\rho(H)v = \lambda v.$$

Then $\rho(E)v$ is either zero or an eigenvector of $\rho(H)$ with eigenvalue $\lambda + 2$, and $\rho(F)v$ is either zero or an eigenvector of $\rho(H)$ with eigenvalue $\lambda - 2$.

Proof:

Since ρ is a Lie algebra homomorphism,

$$\rho(H)\rho(E)v - \rho(E)\rho(H)v = \rho([H, E])v = \rho(2E)v = 2\rho(E)v.$$

Hence

$$\rho(H)\rho(E)v = \rho(E)\rho(H)v + 2\rho(E)v = \lambda\rho(E)v + 2\rho(E)v = (\lambda + 2)\rho(E)v.$$

The same computation using $[H, F] = -2F$ gives

$$\rho(H)\rho(F)v = (\lambda - 2)\rho(F)v.$$

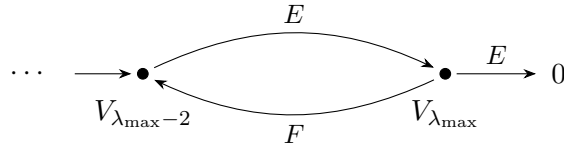
Thus $\rho(E)$ raises and $\rho(F)$ lowers the $\rho(H)$ -eigenvalue by 2. Denote by

$$V_\mu := \ker(\rho(H) - \mu \text{id}_V)$$

the μ -eigenspace of $\rho(H)$. Starting from a single eigenvector $v \in V_\lambda$, let V' be the subrepresentation generated by acting with $\rho(E)$ and $\rho(F)$ on v . Then

$$V' = \bigoplus_{n \in \mathbb{Z}} V_{\lambda+2n}.$$

If V is finite-dimensional, only finitely many summands $V_{\lambda+2n}$ are non-zero. In particular, there exists a maximal eigenvalue $\lambda_{\max}$ such that $\rho(E)$ annihilates $V_{\lambda_{\max}}$:



Definition 5.1.3: Highest weight vector

Let $\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(V)$ be a complex representation. A non-zero vector $v \in V$ is called a **highest weight vector** if

$$\rho(E)v = 0.$$

Its $\rho(H)$ -eigenvalue is called the **highest weight** of v .

Proposition 5.1.4

Let $V = \bigoplus_{n \in \mathbb{Z}} V_{\lambda+2n}$ be an irreducible complex representation of $\mathfrak{sl}(2, \mathbb{C})$, and let $v \in V_{\alpha}$ with $\alpha = \lambda_{\max}$ be a highest weight vector. Then V is generated as a vector space by $\{\rho(F)^k v\}_{k \in \mathbb{N}_0}$.

Proof:

Let $W := \text{span}_{\mathbb{C}}\{\rho(F)^k v \mid k \in \mathbb{N}_0\} \subseteq V$. We show that W is a subrepresentation by checking invariance under each of $\rho(F)$, $\rho(H)$, and $\rho(E)$. Invariance under $\rho(F)$:

$$\rho(F) \cdot \rho(F)^k v = \rho(F)^{k+1} v \in W$$

Invariance under $\rho(H)$. Since $\rho(F)^k v \in V_{\alpha-2k}$

$$\rho(H)\rho(F)^k v = (\alpha - 2k)\rho(F)^k v \in W$$

Invariance under $\rho(E)$. Since $[E, F] = H$, we have

$$\rho(E)\rho(F) = \rho(F)\rho(E) + \rho(H)$$

For $k = 1$:

$$\rho(E)\rho(F)v = \rho(F)\rho(E)v + \rho(H)v = 0 + \alpha v = \alpha v.$$

For general k , apply this identity repeatedly to expand $\rho(E)\rho(F)^k v$:

$$\begin{aligned} \rho(E)\rho(F)^k v &= \rho(F)\rho(E)\rho(F)^{k-1} v + \rho(H)\rho(F)^{k-1} v \\ &= \rho(F)^2 \rho(E)\rho(F)^{k-2} v + \rho(F)\rho(H)\rho(F)^{k-2} v + (\alpha - 2k + 2)\rho(F)^{k-1} v \\ &= \dots \\ &= \rho(F)^k \rho(E)v + (\alpha + (\alpha - 2) + \dots + (\alpha - 2k + 2))\rho(F)^{k-1} v. \end{aligned}$$

Since $\rho(E)v = 0$ and the sum has k terms,

$$\rho(E)\rho(F)^k v = k(\alpha - k + 1)\rho(F)^{k-1} v \in W.$$

Thus W is a subrepresentation. Since $v \neq 0$ we have $W \neq \{0\}$, and since V is irreducible we

conclude $W = V$.

Remark.

The proof above also establishes the formula

$$\rho(E)\rho(F)^k v = k(\alpha - k + 1)\rho(F)^{k-1}v \quad \forall k \in \mathbb{N}_0.$$

Corollary 5.1.5

All weight spaces V_κ are 1-dimensional.

Proof:

The vectors $v, \rho(F)v, \dots, \rho(F)^\alpha v$ form a basis of V (as shown in the proof of the following corollary). Since $\rho(F)^k v \in V_{\alpha-2k}$ and the eigenvalues $\alpha, \alpha-2, \dots, -\alpha$ are pairwise distinct, each eigenspace is spanned by exactly one basis vector, hence 1-dimensional.

Corollary 5.1.6

The highest weight satisfies $\alpha \in \mathbb{N}_0$, and $\dim V = \alpha + 1$.

Proof:

Since V is finite-dimensional and the non-zero vectors among $\{\rho(F)^k v\}_{k \geq 0}$ are linearly independent (they lie in distinct eigenspaces $V_{\alpha-2k}$ of $\rho(H)$), there exists a smallest $k_1 \in \mathbb{N}$ with $\rho(F)^{k_1} v = 0$.

Since $\rho(F)^{k_1} v = 0$, we have $\rho(E)\rho(F)^{k_1} v = 0$. Applying the formula from the remark gives

$$0 = k_1(\alpha - k_1 + 1)\rho(F)^{k_1-1}v.$$

By minimality of k_1 , the vector $\rho(F)^{k_1-1}v \neq 0$. Since $k_1 \geq 1$, both k_1 and $\rho(F)^{k_1-1}v$ are non-zero, so we must have

$$\alpha - k_1 + 1 = 0 \quad \Rightarrow \quad \alpha = k_1 - 1 \in \mathbb{N}_0.$$

The vectors $v, \rho(F)v, \dots, \rho(F)^{k_1-1}v$ are linearly independent and span V by the previous proposition, so $\dim V = k_1 = \alpha + 1$.

Example.

- $\mathfrak{sl}(2, \mathbb{C})$ has a 1-dimensional representation ($\alpha = 0$):

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(\mathbb{C}), \quad X \mapsto 0.$$

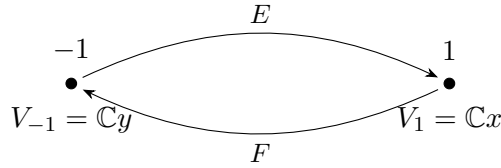
The single weight space is $V_0 = \mathbb{C}$:

$$\bullet \\ V_0$$

- $\mathfrak{sl}(2, \mathbb{C})$ has a 2-dimensional representation ($\alpha = 1$), the **standard representation**:

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(\mathbb{C}^2), \quad X \mapsto X.$$

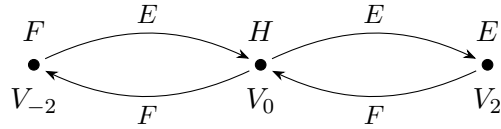
Let x, y be a basis of $\mathbb{C}^2$ with $\rho(H)x = x$ and $\rho(H)y = -y$. Then $V_1 = \mathbb{C}x$ and $V_{-1} = \mathbb{C}y$:



- $\mathfrak{sl}(2, \mathbb{C})$ acts on itself ($\alpha = 2$) by

$$\rho : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \text{End}(\mathfrak{sl}(2, \mathbb{C})), \quad X \mapsto [X, \cdot].$$

The weight spaces are $V_{-2} = \mathbb{C}F$, $V_0 = \mathbb{C}H$, $V_2 = \mathbb{C}E$:



This is called the **adjoint representation**.

Fact.

All finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$ representations are fully reducible.

Definition 5.1.7: Tensor product representation

Let V and W be representations of a Lie algebra $\mathfrak{g}$. The **tensor product representation** on $V \otimes W$ is defined by

$$\rho_{V \otimes W}(X) := \rho_V(X) \otimes \text{id}_W + \text{id}_V \otimes \rho_W(X) \quad \forall X \in \mathfrak{g}.$$

(Verification that this is a representation: see exercise sheets.)

Example.

Isospin (Heisenberg, 1932). When only the strong interaction is considered, there is an approximate symmetry exchanging proton and neutron. This makes $\mathbb{C}[p, n]$ a copy of the standard representation (highest weight $\alpha = 1$), with the proton at weight $+1$ and the neutron at weight -1 . Extending to the three light quarks,

$$\mathbb{C}[u, d, s] \cong V_1 \oplus V_0,$$

with u, d at weights ± 1 and s at weight 0 , where V_n denotes the irreducible representation with highest weight n . Mesons transform in the following multiplets:

- π^+, π^0, π^- form an isospin triplet in V_2 ;
- (K^+, K^0) and $(K^-, \bar{K}^0)$ form isospin doublets in V_1 .

Tensor product $V_1 \otimes V_1$. Weights add under tensor products: $\rho(H)(v \otimes w) = (\lambda_v + \lambda_w)v \otimes w$. For V_1 with basis $\{x, y\}$ at weights $+1, -1$:

$$\begin{array}{ccc}
y \otimes y & y \otimes x, & x \otimes y & x \otimes x \\
\bullet & \bullet \bullet & \bullet & \\
-2 & 0 & 2 &
\end{array}$$

The weight-0 space is 2-dimensional, so $V_1 \otimes V_1 \cong V_2 \oplus V_0$.

Tensor product $V_2 \otimes V_1$. Let $\{|\uparrow\rangle, |0\rangle, |\downarrow\rangle\}$ be a basis of V_2 at weights $+2, 0, -2$. The six tensor products and their weights are:

	$x (+1)$	$y (-1)$
$ \uparrow\rangle (+2)$	$ \uparrow\rangle \otimes x (+3)$	$ \uparrow\rangle \otimes y (+1)$
$ 0\rangle (0)$	$ 0\rangle \otimes x (+1)$	$ 0\rangle \otimes y (-1)$
$ \downarrow\rangle (-2)$	$ \downarrow\rangle \otimes x (-1)$	$ \downarrow\rangle \otimes y (-3)$

The weight spaces have dimensions 1, 2, 2, 1 at weights $+3, +1, -1, -3$, consistent with $V_2 \otimes V_1 \cong V_3 \oplus V_1$.

Theorem 5.1.8: Classification of irreducible representations of $\mathfrak{sl}(2, \mathbb{C})$

The irreducible complex representations of $\mathfrak{sl}(2, \mathbb{C})$ are, up to isomorphism, exactly $\text{Sym}^n(\mathbb{C}^2)$ for $n \in \mathbb{N}_0$, where $\mathbb{C}^2$ carries the standard representation $X \mapsto X$. In particular, $\text{Sym}^n(\mathbb{C}^2)$ is $(n+1)$ -dimensional with highest weight n .

Proof:

Let x, y be the standard basis of $\mathbb{C}^2$. The space $\text{Sym}^n(\mathbb{C}^2)$ is $(n+1)$ -dimensional with basis $\{x^{n-k}y^k\}_{k=0}^n$. Using the Leibniz rule together with $\rho(H)x = x$ and $\rho(H)y = -y$:

$$\rho(H)(x^{n-k}y^k) = (n-k)(+1)x^{n-k}y^k + k(-1)x^{n-k}y^k = (n-2k)x^{n-k}y^k.$$

The highest weight is n , achieved by x^n . Since $\rho(E)x = 0$ in the standard representation, $\rho(E)(x^n) = 0$, so x^n is a highest weight vector. Using $\rho(F)x = y$ and $\rho(F)y = 0$, repeated application gives

$$\rho(F)^k(x^n) = \frac{n!}{(n-k)!} x^{n-k}y^k \neq 0 \quad \text{for } k = 0, 1, \dots, n.$$

By the previous proposition, $\text{Sym}^n(\mathbb{C}^2)$ is generated by x^n and is therefore irreducible.

Conversely, every finite-dimensional irreducible representation has highest weight $\alpha \in \mathbb{N}_0$ and dimension $\alpha + 1$ (by the corollary). Since $\text{Sym}^\alpha(\mathbb{C}^2)$ is irreducible of dimension $\alpha + 1$ with highest weight α , it is the unique such representation up to isomorphism. $\blacksquare$

Remark.

In physics, one is interested in **unitary representations**, i.e. group homomorphisms $G \rightarrow U(n)$. At the Lie algebra level, this means maps into

$$\mathfrak{u}(n) = \{A \in M_n(\mathbb{C}) \mid A^\dagger = -A\},$$

so every generator acts as a skew-adjoint operator. This condition is not preserved by com-

plexification ($\otimes_{\mathbb{R}} \mathbb{C}$): the complexification introduces new elements for which no natural adjointness condition is imposed.

Both $\mathfrak{su}(2)$ and $\mathfrak{sl}(2, \mathbb{R})$ are real forms of $\mathfrak{sl}(2, \mathbb{C})$, but their unitary representation theories are opposite in character:

- $\mathfrak{su}(2)$: all finite-dimensional representations can be made unitary (the associated Lie group $SU(2)$ is compact).
- $\mathfrak{sl}(2, \mathbb{R})$: there are no non-trivial finite-dimensional unitary representations.

The standard basis of $\mathfrak{su}(2)$, expressed in terms of H, E, F , is

$$Z = iH = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad Y = E - F = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X = i(E + F) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

One checks that each satisfies $A^\dagger = -A$. In any unitary $\mathfrak{su}(2)$ -representation, the skew-adjointness of $Z = iH$ forces $\rho(H)$ to be self-adjoint, while the skew-adjointness of Y and X together give

$$\rho(E)^\dagger = \rho(F).$$

Example.

The standard representation on $\mathbb{C}^2$ is unitary: $SU(2) \subset GL(2, \mathbb{C})$ acts by isometries of the standard Hermitian inner product. Since tensor products of unitary representations are again unitary, $\text{Sym}^n(\mathbb{C}^2)$ is unitary for all $n \in \mathbb{N}_0$.

Remark.

Convention. For a representation $\rho : \mathfrak{g} \rightarrow \text{End}(V)$ we write $Xv := \rho(X)(v)$ for $X \in \mathfrak{g}$ and $v \in V$, suppressing ρ from the notation.

The basis $\{F^k v\}_{k=0}^n$ of V_n is orthogonal (the vectors lie in distinct $\rho(H)$ -eigenspaces, and $\rho(H)$ is self-adjoint), but not orthonormal. Using $\rho(E)^\dagger = \rho(F)$, which gives $\langle u | Fw \rangle = \langle Eu | w \rangle$, together with $EF^k v = k(n - k + 1)F^{k-1}v$, one computes

$$\|F^k v\|^2 = \langle F^k v | F \cdot F^{k-1} v \rangle = \langle EF^k v | F^{k-1} v \rangle = k(n - k + 1) \|F^{k-1} v\|^2.$$

Starting from $\|v\|^2 = 1$, this recursion gives $\|F^k v\|^2 = k! \frac{n!}{(n-k)!}$ and defines a positive-definite inner product with respect to which the representation is unitary.

Physics basis. In quantum mechanics, states are labelled by eigenvalues of simultaneously measurable observables. Setting $j := n/2$ (total angular momentum quantum number), the orthonormal basis is defined inductively by

$$|j, j\rangle := v, \quad |j, m-1\rangle := \frac{F|j, m\rangle}{\sqrt{(j+m)(j-m+1)}}, \quad m = -j+1, \dots, j.$$

The second label m (magnetic quantum number, or z-projection of angular momentum) runs over $m = -j, -j+1, \dots, j$. In this basis the generators act by the standard angular

momentum formulas:

$$\begin{aligned} H |j, m\rangle &= 2m |j, m\rangle, \\ E |j, m\rangle &= \sqrt{(j-m)(j+m+1)} |j, m+1\rangle. \end{aligned}$$

5.2 Representation theory of $\mathfrak{su}(3)$ and $\mathfrak{sl}(3, \mathbb{C})$

Remark.

Physics background: the eightfold way. The three lightest quarks — up (u), down (d), and strange (s) — have masses

$$m_u \approx 2.2 \text{ MeV}, \quad m_d \approx 4.7 \text{ MeV}, \quad m_s \approx 93 \text{ MeV}.$$

All three are much lighter than the typical hadronic scale (~ 1 GeV), giving rise to an approximate $SU(3)$ -flavor symmetry acting on $\mathbb{C}[u, d, s]$. Historically, physicists observed that mesons and baryons organize into multiplets matching representations of $\mathfrak{sl}(3)$ — a phenomenon Gell-Mann called the **eightfold way**, named after the 8-dimensional adjoint representation in which the light mesons sit. Some of the predicted particles were initially absent from experiment; quarks were introduced as a mathematical bookkeeping device to explain the multiplet structure, and the missing particles were subsequently discovered. This led to Gell-Mann's Nobel Prize in 1969.

Definition 5.2.1: Cartan subalgebra of $\mathfrak{sl}(3, \mathbb{C})$

The **Cartan subalgebra** of $\mathfrak{sl}(3, \mathbb{C})$ is

$$\mathfrak{h} := \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \mid a + b + c = 0 \right\} \subseteq \mathfrak{sl}(3, \mathbb{C}).$$

Since diagonal matrices commute, $\mathfrak{h}$ is abelian: $[X, Y] = 0$ for all $X, Y \in \mathfrak{h}$. A basis is given by

$$H_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad H_{23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The role of $\mathfrak{h}$ is analogous to that of $\mathbb{C}H$ in $\mathfrak{sl}(2)$: it is the subalgebra whose simultaneous eigenvectors we use to organise representations.

Fact.

In every finite-dimensional $\mathfrak{sl}(3, \mathbb{C})$ -representation, all elements of $\mathfrak{h}$ act by simultaneously diagonalizable operators.

Definition 5.2.2: Weight

Let V be a finite-dimensional $\mathfrak{sl}(3, \mathbb{C})$ -representation. A vector $v \in V$ is a **weight vector** with **weight** $\alpha \in \mathfrak{h}^*$ if

$$\rho(H)v = \alpha(H)v \quad \forall H \in \mathfrak{h}.$$

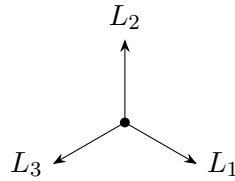
The corresponding **weight space** is $V_\alpha = \{v \in V \mid \rho(H)v = \alpha(H)v \ \forall H \in \mathfrak{h}\}$. By the fact above, $V = \bigoplus_\alpha V_\alpha$ (finite direct sum over $\mathfrak{h}^*$).

Remark.

Define linear functionals $L_1, L_2, L_3 \in \mathfrak{h}^*$ by

$$L_i \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} := a_i.$$

Since every $H \in \mathfrak{h}$ satisfies $a_1 + a_2 + a_3 = 0$, we have $L_1 + L_2 + L_3 = 0$, so $\{L_1, L_2\}$ is a basis of $\mathfrak{h}^*$ with $L_3 = -L_1 - L_2$. The three functionals are symmetric and point in directions 120° apart in $\mathfrak{h}^*$:

**Example.**

The defining 3-dimensional representation $\mathfrak{sl}(3, \mathbb{C}) \rightarrow \text{End}(\mathbb{C}^3)$, $X \mapsto X$, decomposes into weight spaces as

$$\mathbb{C}^3 = \mathbb{C}_{L_1} \oplus \mathbb{C}_{L_2} \oplus \mathbb{C}_{L_3},$$

where e_i has weight L_i : $He_i = a_i e_i = L_i(H)e_i$ for $H = \text{diag}(a_1, a_2, a_3)$.

Remark.

In $\mathfrak{sl}(2)$ we found eigenvectors of $\text{ad}(H)$ (i.e. $[H, E] = 2E$, $[H, F] = -2F$) to build a basis that moves between weight spaces. We do the same for $\mathfrak{sl}(3)$: decompose $\mathfrak{sl}(3, \mathbb{C})$ into weight spaces under the adjoint representation $X \mapsto [X, -]$.

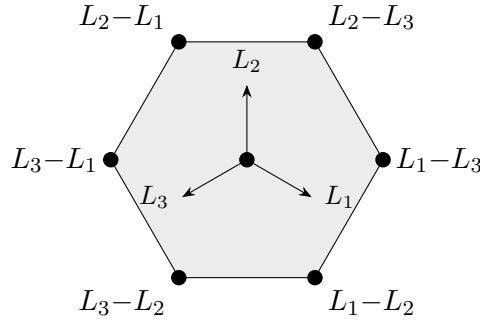
Since $[\mathfrak{h}, \mathfrak{h}] = 0$, the Cartan subalgebra sits at weight 0. The remaining basis elements are the **matrix units** E_{ij} (with 1 in position (i, j) and 0 elsewhere, $i \neq j$). For $H = \text{diag}(a_1, a_2, a_3)$:

$$\begin{aligned} [H, E_{12}] &= \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \\ &= \begin{pmatrix} 0 & a_1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & a_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = (a_1 - a_2) E_{12} = (L_1 - L_2)(H) E_{12}. \end{aligned}$$

The same computation gives $[H, E_{ij}] = (L_i - L_j)(H) E_{ij}$ for all $H \in \mathfrak{h}$, so E_{ij} is a weight vector with weight $L_i - L_j$. This gives the **root space decomposition**

$$\mathfrak{sl}(3, \mathbb{C}) = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{L_i - L_j}, \quad \mathfrak{g}_{L_i - L_j} = \mathbb{C} E_{ij}.$$

The six weights $L_i - L_j$ ($i \neq j$), called **roots**, form a regular hexagon in $\mathfrak{h}^*$. The total dimension $\dim \mathfrak{sl}(3, \mathbb{C}) = 2 + 6 = 8$ matches the eightfold way.



Proposition 5.2.3

Let $X, Y \in \mathfrak{sl}(3, \mathbb{C})$ be weight vectors of weights α and β . Then $[X, Y]$ has weight $\alpha + \beta$, i.e.

$$[X, Y] \in \mathfrak{g}_{\alpha + \beta},$$

where $\mathfrak{g}_\gamma := 0$ whenever γ is not a root.

Proof:

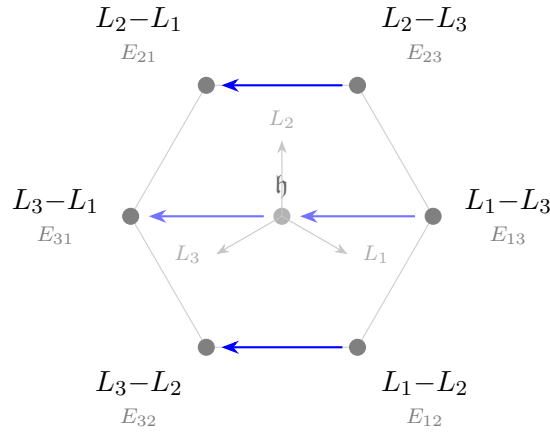
For any $H \in \mathfrak{h}$, the Jacobi identity gives

$$\begin{aligned} [H, [X, Y]] &= -[Y, [H, X]] - [X, [Y, H]] \\ &= -[Y, \alpha(H)X] + [X, \beta(H)Y] \\ &= (\alpha(H) + \beta(H)) [X, Y]. \end{aligned}$$

Since this holds for all $H \in \mathfrak{h}$, the element $[X, Y]$ is a weight vector of weight $\alpha + \beta$. ■

Example.

In the diagram, taking the commutator with E_{ij} moves every weight in the E_{ij} -direction. The diagram below shows the action of $[E_{31}, -]$, which shifts by $L_3 - L_1$, i.e. one step to the left. Moving outside the hexagon gives zero.



Proposition 5.2.4

Let V be a finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$, let $v \in V$ be a weight vector of weight α , and let $X \in \mathfrak{sl}(3, \mathbb{C})$ be of weight β . Then $X(v) \in V$ is of weight $\alpha + \beta$.

Proof:

For any $H \in \mathfrak{h}$,

$$H(X(v)) = X(H(v)) + [H, X](v) = (\alpha + \beta)(H) X(v).$$

Remark.

It follows that the weights appearing in an irreducible representation differ by integer combinations of the roots. The lattice generated by the roots,

$$\Lambda_R \subseteq \mathfrak{h}^*,$$

is called the **root lattice**.

There is no canonical notion of a highest weight, so we must make a choice.

Definition 5.2.5: Highest weight map

A **highest weight map** is a linear map $\ell: \mathfrak{h}^* \rightarrow \mathbb{C}$ such that

$$\operatorname{Re}(\ell(\alpha)) \neq 0 \quad \text{for all } 0 \neq \alpha \in \Lambda_R.$$

A root α is called **positive** if $\ell(\alpha) > 0$, and **negative** otherwise.

To be concrete, for $\mathfrak{sl}(3)$ we pick

$$\ell(a_1 L_1 + a_2 L_2 + a_3 L_3) = a a_1 + b a_2 + c a_3, \quad a + b + c = 0, \quad a > b > c.$$

This particular choice does not matter.

Definition 5.2.6: Highest weight vector ($\mathfrak{sl}(3)$)

Let V be a representation of $\mathfrak{sl}(3, \mathbb{C})$. A weight vector $v \in V_\alpha$ is called a **highest weight vector** if $\operatorname{Re} \ell(\alpha)$ is maximal among all weights of V , or equivalently if the positive root vectors annihilate it:

$$E_{12}v = E_{23}v = E_{13}v = 0.$$

Remark.

The word *weight* is used in several related senses, which are worth keeping apart:

- A **weight** is, abstractly, an element $\alpha \in \mathfrak{h}^*$ of the dual of the Cartan subalgebra — a linear functional on $\mathfrak{h}$.
- The weight of a **vector** v in a representation $\rho: \mathfrak{sl}(3, \mathbb{C}) \rightarrow \operatorname{End}(V)$ is the $\alpha \in \mathfrak{h}^*$ with

$$\rho(H)v = \alpha(H)v \quad \forall H \in \mathfrak{h};$$

such a v is a *weight vector* of weight α . (We usually omit ρ and write $Hv := \rho(H)v$.)

- The **weight space** V_α is the span of all weight vectors of weight α . A weight of the **representation** V is then an α with $V_\alpha \neq 0$, i.e. one that actually occurs in the decomposition $V = \bigoplus_\alpha V_\alpha$.
- In the special case of the **adjoint representation**, where $\mathfrak{sl}(3, \mathbb{C})$ acts on itself by $X \mapsto [X, -]$, the (non-zero) weights are exactly the **roots** $L_i - L_j$, and the corresponding weight vectors are the root vectors E_{ij} .

Proposition 5.2.7

Let V be an irreducible finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$ and $v \in V_\alpha$ a highest weight vector. Then V is generated by the elements obtained from v by successively applying E_{21} , E_{31} , and E_{32} .

Proof:

Since V is irreducible, it suffices to show that the subspace generated in this way is a non-zero subrepresentation; irreducibility then forces it to equal V .

Let $W \leq V$ be the subspace generated by acting on v with E_{21} , E_{31} , and E_{32} . We need to show that acting with the positive root vectors E_{12} , E_{13} , E_{23} also preserves W . Since $E_{13} = [E_{12}, E_{23}]$, it is enough to show $E_{12}(W) \subseteq W$ and $E_{23}(W) \subseteq W$.

Note that W can equally be generated by acting with only E_{21} and E_{32} , since $[E_{21}, E_{32}] = -E_{31}$.

We proceed by induction. Define W_n to be the subspace of W generated by applying words of length at most n in E_{21}, E_{32} to v , with $W_{-1} := 0$ and $W_0 := \mathbb{C}v$. We show that acting with a positive root vector maps W_n into W_{n-1} .

Let $w_n \in W_n$. There are two cases,

$$w_n = E_{21} w_{n-1} \quad \text{or} \quad w_n = E_{32} w_{n-1}, \quad w_{n-1} \in W_{n-1}.$$

In the first case,

$$\begin{aligned} E_{12} E_{21} w_{n-1} &= E_{21} E_{12} w_{n-1} + \underbrace{[E_{12}, E_{21}]}_{\in \mathfrak{h}} w_{n-1} \\ &= E_{21} \underbrace{E_{12} w_{n-1}}_{\in W_{n-2}} + \alpha([E_{12}, E_{21}]) w_{n-1}. \end{aligned}$$

By the induction hypothesis $E_{12} w_{n-1} \in W_{n-2}$, so the first term lies in W_{n-1} ; the second term lies in W_{n-1} as well. Hence $E_{12} w_n \in W_{n-1}$.

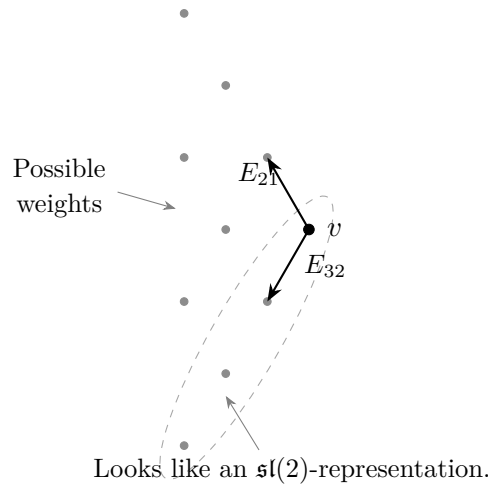
The remaining cases follow in the same way from

$$[E_{23}, E_{12}] = 0, \quad [E_{12}, E_{32}] = 0, \quad [E_{23}, E_{32}] \in \mathfrak{h}.$$

Remark.

Let V be a finite-dimensional representation of $\mathfrak{sl}(3, \mathbb{C})$ and $v \in V_\alpha$ a highest weight vector. Then acting with the negative roots on v generates an irreducible subrepresentation.

The weights reachable from v fill a cone opening in the directions of the negative roots. Along a single root string the picture reduces to a familiar $\mathfrak{sl}(2, \mathbb{C})$ -representation.



Definition 5.2.8: Lie subalgebra

Let $\mathfrak{g}$ be a Lie algebra. A **Lie subalgebra** is a subspace $\mathfrak{h} \leq \mathfrak{g}$ such that

$$[X, Y] \in \mathfrak{h} \quad \forall X, Y \in \mathfrak{h}.$$

Example.

For each pair $1 \leq i < j \leq 3$, the matrices E_{ij}, E_{ji}, H_{ij} span a Lie subalgebra of $\mathfrak{sl}(3, \mathbb{C})$ isomorphic to $\mathfrak{sl}(2, \mathbb{C})$, with E_{ij}, E_{ji}, H_{ij} taking the roles of E, F, H .

Consequently the possible highest weights are labelled by two non-negative integers $(n_{12}, n_{23}) \in$

$\mathbb{N}_0^2$, and the corresponding highest weight is

$$(n_{12} + n_{23}) L_1 + n_{23} L_2 = n_{12} L_1 - n_{23} L_3.$$

In particular, every weight occurring in a finite-dimensional representation is an integer combination $aL_1 + bL_2 + cL_3$ with $a, b, c \in \mathbb{Z}$.

Definition 5.2.9: Weight lattice

The **weight lattice** $\Lambda_W \subseteq \mathfrak{h}^*$ is the lattice generated by L_1, L_2, L_3 . It contains the root lattice:

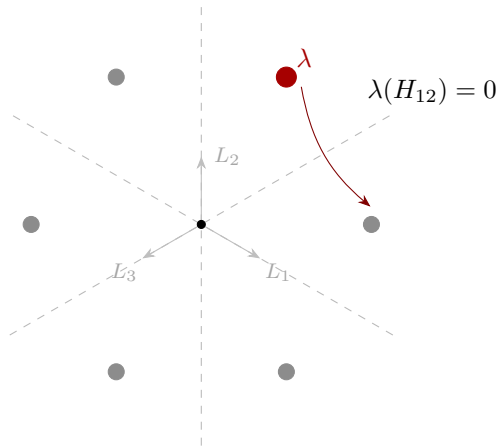
$$\Lambda_R \subseteq \Lambda_W \subseteq \mathfrak{h}^*.$$

Remark.

Restricting a representation to the $\mathfrak{sl}(2, \mathbb{C})$ -subalgebra generated by E_{ij}, E_{ji}, H_{ij} shows, exactly as in the $\mathfrak{sl}(2)$ theory, that its set of weights is symmetric under reflection in the hyperplane

$$\{\lambda \in \mathfrak{h}^* \mid \lambda(H_{ij}) = 0\}.$$

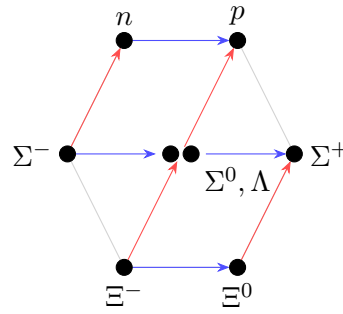
The group generated by these reflections is the **Weyl group**. Every weight diagram is invariant under it, so one completes a weight diagram by reflecting its weights in all the mirror hyperplanes.



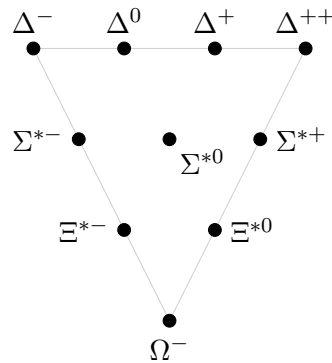
Recall the approximate $SU(3)$ -flavor symmetry rotating the light quarks u, d, s . One might ask why the relevant symmetry is the continuous group $SU(3)$ rather than a discrete group such as $\mathbb{Z}_3$; the answer is that the observed hadron multiplets match precisely the irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$, as the following examples illustrate.

Example.

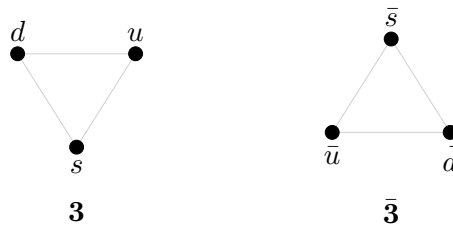
Baryon octet (adjoint representation, 8). The spin- $\frac{1}{2}$ baryons fill the weight diagram of the adjoint representation, a hexagon with a doubly-occupied centre:



Baryon decuplet (10). The spin- $\frac{3}{2}$ baryons fill a triangular weight diagram. The bottom vertex Ω^- was predicted before being discovered experimentally in 1964:



Quarks (3) and antiquarks ($\bar{3}$). The three quarks sit in the fundamental representation, the three antiquarks in its dual:



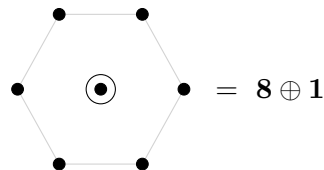
Tensor products. Recall that the tensor product representation acts by

$$\rho_{V \otimes W}(X) = \text{id}_V \otimes \rho_W(X) + \rho_V(X) \otimes \text{id}_W.$$

Combining a quark with an antiquark gives the meson multiplets,

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1},$$

a hexagon (the octet) together with a singlet at the centre:



Combining two quarks gives

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}},$$

where the symmetric sextet $\mathbf{6}$ consists of the diquark states

$$uu, \quad ud, \quad dd, \quad us, \quad ds, \quad ss.$$

These diquarks cannot exist as free particles in nature because of *confinement*.

Finally, $\mathbf{3} \otimes \mathbf{8} = \mathbf{15} \oplus \bar{\mathbf{6}} \oplus \mathbf{3}$ (left as an exercise).

We found that irreducible representations are determined by their highest weight. It remains to ask whether every allowed weight actually occurs as the highest weight of an irreducible representation. This can also be shown using Schur functors, but we take a different route.

Theorem 5.2.10: Classification of irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$

For each highest weight $\alpha = n_1 L_1 - n_2 L_3$ with $n_1, n_2 \in \mathbb{N}_0$, there exists a unique (up to isomorphism) irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with highest weight α .

Proof:

Existence. If V and W are representations with highest weights λ_V and λ_W , then $V \otimes W$ has highest weight $\lambda_V + \lambda_W$ (the tensor product of the two highest weight vectors is again a highest weight vector). Since $\mathbf{3}$ has highest weight L_1 and $\bar{\mathbf{3}}$ has highest weight $-L_3$, the representation

$$\mathbf{3}^{\otimes n_1} \otimes \bar{\mathbf{3}}^{\otimes n_2}$$

has highest weight $n_1 L_1 - n_2 L_3$. Picking a highest weight vector v in it, v generates an irreducible subrepresentation with highest weight $n_1 L_1 - n_2 L_3$. This proves existence.

Uniqueness. Let V and W be irreducible representations, both with highest weight α , and let $v \in V$, $w \in W$ be highest weight vectors. The representation $V \oplus W$ is not irreducible, but $v + w \in V \oplus W$ is again a highest weight vector of weight α . Let $\widetilde{W} \leq V \oplus W$ be the irreducible subrepresentation generated by $v + w$. The two projections

$$\widetilde{W} \rightarrow V, \quad \widetilde{W} \rightarrow W$$

are non-zero homomorphisms of representations, so by Schur's lemma they are isomorphisms. Hence

$$V \cong \widetilde{W} \cong W.$$

Remark.

(Added by transcriber, not part of the lecture.) Representations of a Lie group H classify the particles of any gauge theory with structure group H ; the following makes this precise for the Standard Model (*François*).

Gauge groups. A gauge theory with structure group H is built on a principal H -bundle $\pi: P \rightarrow M$ over spacetime M . The *gauge group*

$$\mathcal{H} = \text{Aut}_v(P) := \{ \Phi: P \xrightarrow{\sim} P \mid \pi \circ \Phi = \pi, \Phi(ph) = \Phi(p)h \}$$

is the group of fiber-preserving automorphisms; each element corresponds to an equivariant map $\gamma: P \rightarrow H$ satisfying $\gamma(ph) = h^{-1}\gamma(p)h$, and $\text{Lie } \mathcal{H} \cong \Omega_{\text{tens}}^0(P, \text{Lie } H)$. Incorporating spacetime diffeomorphisms gives the full symmetry group

$$\mathcal{G} = \text{Diff}(M) \ltimes \mathcal{H},$$

where $\text{Diff}(M)$ acts on $\mathcal{H}$ by conjugation via bundle lifts. Physical observables must be $\mathcal{H}$ -invariant and covariant under $\text{Diff}(M)$. For the Standard Model, $H = SU(3)_c \times SU(2)_L \times U(1)_Y$.

Connections and D^A . A *connection* $A \in \Omega_{\text{eq}}^1(P, \text{Lie } H)$ satisfies $\iota_{X\#} A = X$ for all $X \in \text{Lie } H$, splitting $TP = VP \oplus HP_A$ with horizontal projector $h_A: TP \rightarrow HP_A$. The *exterior covariant derivative* is the composition

$$D^A := \text{hor}_A \circ d : \Omega_{\text{tens}}^k(P, V) \xrightarrow{d} \Omega^{k+1}(P, V) \xrightarrow{\text{hor}_A} \Omega_{\text{tens}}^{k+1}(P, V),$$

where $(\text{hor}_A \alpha)(X_1, \dots, X_{k+1}) := \alpha(h_A X_1, \dots, h_A X_{k+1})$. On tensorial 0-forms, equivariance $(d\psi(X^\#) = -\rho_*(X)\psi)$ gives $D^A \psi = d\psi + \rho_*(A)\psi$. The *curvature*

$$F = dA + \frac{1}{2}[A, A] \in \Omega_{\text{tens}}^2(P, \text{Lie } H), \quad D^A F = 0 \quad (\text{Bianchi}),$$

transforms in the adjoint representation of H under $\mathcal{H}$:

$$A \mapsto \gamma^{-1} A \gamma + \gamma^{-1} d\gamma, \quad F \mapsto \gamma^{-1} F \gamma.$$

Gauge bosons. Each generator of $\text{Lie } H$ gives one gauge boson. The decomposition $\text{Lie } H = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ yields $8 + 3 + 1 = 12$: 8 gluons ($\mathfrak{su}(3)_c$), $W^\pm, W^3$ ($\mathfrak{su}(2)_L$), and B ($\mathfrak{u}(1)_Y$). The Yang–Mills Lagrangian $L_{\text{YM}} = \frac{1}{2} \text{Tr}(F \wedge *F)$ is $\mathcal{H}$ -invariant; a mass term $m^2 \text{Tr}(A \wedge *A)$ is not, since A transforms affinely—gauge invariance forces all gauge bosons massless.

Matter fields. A representation $\rho: H \rightarrow GL(V)$ specifies a matter field: a tensorial 0-form $\psi \in \Omega_{\text{tens}}^0(P, V) \cong \Gamma(P \times_H V)$. The *gauge charge* of ψ is the representation ρ ; minimal coupling to A is automatic via D^A , and $\langle \psi, \not{D}^A \psi \rangle$ is $\mathcal{H}$ -invariant by equivariance. For the quarks of each generation:

$$Q_L \sim (\mathbf{3}, \mathbf{2}, +\frac{1}{6}), \quad u_R \sim (\mathbf{3}, \mathbf{1}, +\frac{2}{3}), \quad d_R \sim (\mathbf{3}, \mathbf{1}, -\frac{1}{3}).$$

The representation content is experimental input, constrained by *anomaly cancellation*: certain traces over the fermion representations must vanish.

Mass generation. The Higgs field $\phi \in \Omega_{\text{tens}}^0(P, \mathbb{C}^2)$ in representation $(\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ of H couples to A via

$$L_{\text{Higgs}} = \langle D^A \phi, *D^A \phi \rangle - V(\phi), \quad V(\phi) = \mu^2 |\phi|^2 + \lambda |\phi|^4.$$

The full Lagrangian $L_{\text{YM}} + L_{\text{Higgs}} + \sum_\psi \langle \psi, \not{D}^A \psi \rangle + L_{\text{Yukawa}}$ is $\mathcal{H}$ -invariant and entirely determined by H and the representations $\{\rho_i\}$; all couplings follow from D^A , and L_{Yukawa} generates fermion masses. Gauge boson mass generation admits two equivalent descriptions.

Spontaneous symmetry breaking. For $\mu^2 < 0$, a nonzero vacuum ϕ_0 has stabilizer $U(1)_{\text{EM}} \subset H$; the residual gauge group is $SU(3)_c \times U(1)_{\text{EM}}$. The $W^\pm$ and Z acquire masses from $|D^A \phi_0|^2$, with the Goldstone modes absorbed as longitudinal polarizations; the photon remains massless.

Dressing field method (AttardFrançoisLazzariniMasson). A **dressing field** for a subgroup $K \subset H$ is a K -valued field $u: P \rightarrow K$ transforming as $u \mapsto \kappa^{-1}u$ under $\kappa \in \mathcal{K} \subset \mathcal{H}$ (not as a connection). The composite fields

$$A^u := u^{-1}Au + u^{-1}du, \quad \psi^u := u^{-1}\psi$$

are $\mathcal{K}$ -invariant by construction. For $K = SU(2)_L$, the polar decomposition $\phi = u[\phi]|\phi|$ provides $u[\phi]$ as a dressing field; the $W^\pm, Z$ masses appear directly from $|D^A \phi|^2$ rewritten in terms of A^u , with $|\phi|$ the physical neutral Higgs. No vacuum, no Goldstone bosons, and no broken symmetry are invoked.

Dressing is not gauge fixing. Gauge fixing imposes a constraint $\chi(A) = 0$, selects one representative per $\mathcal{H}$ -orbit, and requires Faddeev-Popov ghosts. Dressing imposes no constraint: $(A, \phi) \mapsto (A^u, \psi^u)$ is a change of variables on the full field space, globally well-defined (no Gribov copies), with all degrees of freedom dynamical. Both pictures yield the same S-matrix observables; Elitzur's theorem—local gauge symmetry cannot be spontaneously broken—is automatically respected.

Flavor vs. gauge symmetry. The $SU(3)_c$ factor is an exact gauge symmetry: color is a conserved Noether charge and gluons couple identically to all quark flavors. The $SU(3)_{\text{flavor}}$ symmetry studied in this chapter is not a gauge symmetry: it is an accidental near-symmetry of the light-quark mass terms, acts on flavor indices, has no associated connection or gauge bosons, and is explicitly broken by quark mass differences.

Chapter 6

Simple Lie algebras

6.1 Ideals, simple and semisimple Lie algebras

For finite groups, a normal subgroup $H \trianglelefteq G$ gives a quotient G/H , and passing to Lie algebras yields the analogous picture:

$$\begin{array}{ccccc} H & \longrightarrow & G & \longrightarrow & G/H \\ & & \downarrow & & \\ \mathfrak{h} & \longrightarrow & \mathfrak{g} & \longrightarrow & \mathfrak{g}/\mathfrak{h} \end{array}$$

Normality means H is closed under conjugation. Differentiating the conjugation action shows that the corresponding condition on the Lie algebra is the bracket condition $[X, H] \in \mathfrak{h}$ for all $X \in \mathfrak{g}$ and $H \in \mathfrak{h}$. This motivates the following definition.

Definition 6.1.1: Ideal

Let $\mathfrak{g}$ be a Lie algebra. An **ideal** in $\mathfrak{g}$ is a subspace $\mathfrak{h} \subseteq \mathfrak{g}$ such that

$$[X, H] \in \mathfrak{h} \quad \forall X \in \mathfrak{g}, H \in \mathfrak{h}.$$

Remark.

Every ideal is a Lie subalgebra.

Example.

- $0 \subseteq \mathfrak{g}$ and $\mathfrak{g} \subseteq \mathfrak{g}$ are ideals.
- Let G be a Lie group and $H \trianglelefteq G$ a normal Lie subgroup. Then $\mathfrak{h} = T_e H$ is an ideal in $\mathfrak{g} = T_e G$.
- Any subspace of an abelian Lie algebra is an ideal.
- The subspace $\mathfrak{g}' := [\mathfrak{g}, \mathfrak{g}]$, spanned by all brackets $[x, y]$ with $x, y \in \mathfrak{g}$, is an ideal.

Proposition 6.1.2

Let $\mathfrak{h} \leq \mathfrak{g}$ be an ideal. The vector space $\mathfrak{g}/\mathfrak{h}$ is a Lie algebra with bracket

$$[x + \mathfrak{h}, y + \mathfrak{h}] := [x, y] + \mathfrak{h} \quad \forall x, y \in \mathfrak{g}.$$

Proof:

The Lie algebra axioms are inherited from the bracket $[-, -]$ on $\mathfrak{g}$, so it only remains to check that this bracket is well defined. Let $x' = x + H_x$ and $y' = y + H_y$, with $H_x, H_y \in \mathfrak{h}$, be other representatives. Then

$$\begin{aligned} [x', y'] &= [x + H_x, y + H_y] \\ &= [x, y] + \underbrace{[x, H_y]}_{\in \mathfrak{h}} - \underbrace{[y, H_x]}_{\in \mathfrak{h}} + \underbrace{[H_x, H_y]}_{\in \mathfrak{h}}, \end{aligned}$$

where each of the last three terms lies in $\mathfrak{h}$ because $\mathfrak{h}$ is an ideal. Hence $[x', y'] + \mathfrak{h} = [x, y] + \mathfrak{h}$, so the bracket does not depend on the chosen representatives. ■

Remark.

This leads to an “extension” of Lie algebras:

$$0 \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\mathfrak{h} \longrightarrow 0.$$

Definition 6.1.3: Simple Lie algebra

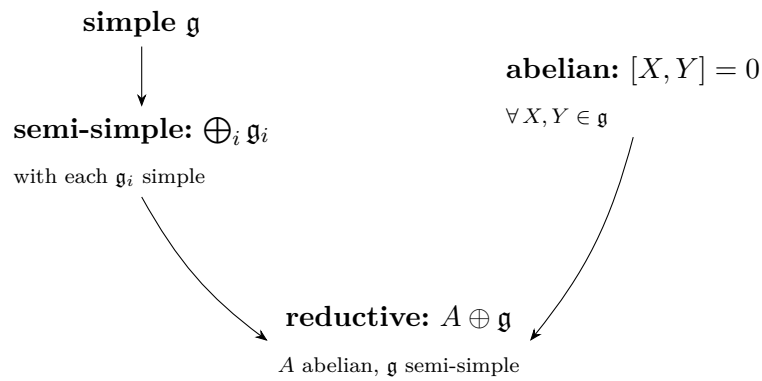
A Lie algebra $\mathfrak{g}$ is **simple** if it is non-abelian and its only ideals are 0 and $\mathfrak{g}$.

Lemma 6.1.4

Let $\mathfrak{g}$ be a simple Lie algebra. Then $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$.

Remark.

Types of Lie algebras.



6.2 Cartan subalgebras and the root space decomposition

Definition 6.2.1: Semi-simple element

An element $X \in \mathfrak{g}$ is called **semi-simple** if $[X, -]: \mathfrak{g} \rightarrow \mathfrak{g}$ is diagonalizable.

Definition 6.2.2: Cartan subalgebra

A **Cartan subalgebra** $\mathfrak{h} \leq \mathfrak{g}$ is a maximal abelian subalgebra whose elements are semi-simple. The dimension of $\mathfrak{h}$ is called the **rank** of $\mathfrak{g}$.

Remark.

This is not unique, but the choice will not matter in the end.

Example.

For $\mathfrak{sl}(N)$ the diagonal matrices form a Cartan subalgebra.

Definition 6.2.3: Root space decomposition

Let $\mathfrak{g}$ be a semi-simple Lie algebra and $\mathfrak{g}_0$ a Cartan subalgebra. The decomposition of $\mathfrak{g}$ into simultaneous eigenspaces for the action of $\mathfrak{g}_0$,

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha, \quad \mathfrak{g}_\alpha = \{X \in \mathfrak{g} \mid [H, X] = \alpha(H)X \ \forall H \in \mathfrak{g}_0\},$$

with $\alpha \in \mathfrak{g}_0^*$, is called the **root space decomposition** of $\mathfrak{g}$. The non-zero α appearing in this decomposition are the **roots** of $\mathfrak{g}$. The set of all roots of $\mathfrak{g}$,

$$\Phi(\mathfrak{g}),$$

is called the **root system** of $\mathfrak{g}$.

Remark.

Facts about roots.

- The roots span $\mathfrak{g}_0^*$.
- The root spaces $\mathfrak{g}_\alpha$ are 1-dimensional.
- If α is a root, the only multiples of α which are also roots are $\pm\alpha$.
- There exists a basis $\{H_i\}_{i=1, \dots, \text{rank}(\mathfrak{g})}$ of $\mathfrak{g}_0$ such that for all roots $\alpha(H_i)$ is an integer. *(This we will show later.)*

Definition 6.2.4: Cartan–Weyl basis

A basis of $\mathfrak{g}$ compatible with the root space decomposition,

$$\{H_i\}_{i=1,\dots,\text{rank}(\mathfrak{g})} \cup \{E^\alpha\}_{\alpha \in \Phi(\mathfrak{g})},$$

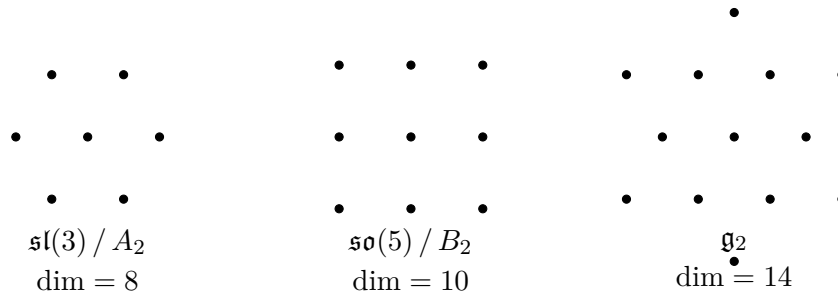
is called a **Cartan–Weyl basis**.

Example.

There are 3 rank-2 simple Lie algebras,

$$\mathfrak{sl}(3), \quad \mathfrak{so}(5) = \{X \in \mathfrak{gl}(5) \mid X^T = -X\}, \quad \mathfrak{g}_2,$$

with root systems



(To fully understand their geometry we will need more theory.)

As for $\mathfrak{sl}(3)$ we find, using the Jacobi identity,

$$\begin{aligned} [H, [E^\alpha, E^\beta]] &= -[E^\beta, [H, E^\alpha]] - [E^\alpha, [E^\beta, H]] \\ &= (\alpha(H) + \beta(H)) [E^\alpha, E^\beta], \end{aligned}$$

so the weight of $[E^\alpha, E^\beta]$ is $\alpha + \beta$.

6.3 The Killing form and structure constants**Definition 6.3.1: Killing form**

Let $\mathfrak{g}$ be a semi-simple Lie algebra. The **Killing form** is

$$\kappa(X, Y) := \text{Tr}_{\mathfrak{g}} (\text{ad}_X \circ \text{ad}_Y), \quad \text{ad}_X = [X, -].$$

It is symmetric because the trace is.

Remark.**Facts.**

- The Killing form is non-degenerate and hence induces an isomorphism $\mathfrak{g} \cong \mathfrak{g}^*$, an inner product on $\mathfrak{g}$, and one on $\mathfrak{g}^*$.
- Its restriction to any Cartan subalgebra $\mathfrak{g}_0$ is also non-degenerate.

Root diagrams are drawn in a way respecting this inner product.

Proposition 6.3.2

Root spaces for roots that do not sum to zero are orthogonal with respect to κ : if $\alpha + \beta \neq 0$, then $\mathfrak{g}_\alpha \perp \mathfrak{g}_\beta$ as subspaces of $\mathfrak{g}$.

Proof:

Using the invariance $\kappa([X, Y], Z) = \kappa(X, [Y, Z])$, for $H \in \mathfrak{g}_0$,

$$\alpha(H) \kappa(E^\alpha, E^\beta) = \kappa([H, E^\alpha], E^\beta) = -\kappa([E^\alpha, H], E^\beta) = -\beta(H) \kappa(E^\alpha, E^\beta).$$

Since $\alpha \neq -\beta$ there is an $H \in \mathfrak{g}_0$ with $\alpha(H) + \beta(H) \neq 0$, and hence $\kappa(E^\alpha, E^\beta) = 0$. ■

Definition 6.3.3: Metric dual t_α of a root

Since κ restricted to $\mathfrak{g}_0$ is non-degenerate, to each root $\alpha \in \mathfrak{g}_0^*$ there is a unique element $t_\alpha \in \mathfrak{g}_0$ such that

$$\kappa(t_\alpha, -) = \alpha(-).$$

Lemma 6.3.4

For each root α ,

$$[E^\alpha, E^{-\alpha}] = \kappa(E^\alpha, E^{-\alpha}) t_\alpha.$$

Proof:

We compute the pairing of both sides with $\kappa(H, -)$ for $H \in \mathfrak{g}_0$:

$$\kappa(H, [E^\alpha, E^{-\alpha}]) = \alpha(H) \kappa(E^\alpha, E^{-\alpha}) = \kappa(t_\alpha, H) \kappa(E^\alpha, E^{-\alpha}),$$

using the invariance $\kappa([X, Y], Z) = \kappa(X, [Y, Z])$ for the first equality and $\kappa(H, t_\alpha) = \alpha(H)$ for the second. Both sides thus have the same pairing with every $\kappa(H, -)$, so by non-degeneracy of κ they are equal. ■

Definition 6.3.5: Inner product on $\mathfrak{g}_0^*$

The Killing form induces an inner product on $\mathfrak{g}_0^*$ via

$$(\alpha, \beta) := \kappa(t_\alpha, t_\beta) = \alpha(t_\beta).$$

Proposition 6.3.6

For each root α , set

$$H^\alpha := \frac{2t_\alpha}{(\alpha, \alpha)},$$

and normalize $E^\alpha, E^{-\alpha}$ such that $\kappa(E^\alpha, E^{-\alpha}) = \frac{2}{(\alpha, \alpha)}$. Then

$$[E^\alpha, E^{-\alpha}] = H^\alpha, \quad [H^\alpha, E^\alpha] = 2E^\alpha, \quad [H^\alpha, E^{-\alpha}] = -2E^{-\alpha},$$

so $(E^{\pm\alpha}, H^\alpha)$ generates an $\mathfrak{sl}(2)$ -subrepresentation.

Proof:

By definition of t_α we have $[t_\alpha, E^{\pm\alpha}] = (\pm\alpha)(t_\alpha) E^{\pm\alpha} = \pm(\alpha, \alpha) E^{\pm\alpha}$. Together with the previous lemma,

$$\begin{aligned} [E^\alpha, E^{-\alpha}] &= \kappa(E^\alpha, E^{-\alpha}) t_\alpha = \frac{2}{(\alpha, \alpha)} t_\alpha = H^\alpha, \\ [H^\alpha, E^\alpha] &= \frac{2}{(\alpha, \alpha)} [t_\alpha, E^\alpha] = \frac{2}{(\alpha, \alpha)} (\alpha, \alpha) E^\alpha = 2E^\alpha, \\ [H^\alpha, E^{-\alpha}] &= \frac{2}{(\alpha, \alpha)} [t_\alpha, E^{-\alpha}] = -\frac{2}{(\alpha, \alpha)} (\alpha, \alpha) E^{-\alpha} = -2E^{-\alpha}. \end{aligned}$$

Remark.

This constrains the representation theory so that we can essentially solve it, similar to $\mathfrak{sl}(3)$.

In physics, Lie algebras $\mathfrak{g}$ are often encoded using a basis $\{T^a\}_{a=1, \dots, \dim \mathfrak{g}}$ of $\mathfrak{g}$. The Lie bracket can then be written explicitly as

$$[T^a, T^b] = \sum_{c=1}^n f^{ab}_c T^c.$$

Definition 6.3.7: Structure constants

The coefficients f^{ab}_c in the expansion

$$[T^a, T^b] = \sum_{c=1}^n f^{ab}_c T^c$$

of the Lie bracket in a basis $\{T^a\}$ are called **structure constants**. Antisymmetry of the bracket and the Jacobi identity translate into

$$f^{ab}_c = -f^{ba}_c \quad \text{and} \quad \sum_{d=1}^n \left(f^{ab}_d f^{dc}_e + f^{bc}_d f^{da}_e + f^{ca}_d f^{db}_e \right) = 0.$$

Example.

- $\mathfrak{sl}(2) = \text{span}(\sigma_1, \sigma_2, \sigma_3)$ with

$$[\sigma_i, \sigma_j] = 2i \varepsilon_{ijl} \sigma_l \quad \implies \quad f^{ij}_l = 2i \varepsilon_{ijl}.$$

- $\mathfrak{gl}(n)$ has the basis $\{E_{ij}\}$ with commutation relations

$$\begin{aligned} [E_{ij}, E_{kl}] &= \delta_{jk} E_{il} - \delta_{li} E_{kj} \\ \implies f^{(ij)(kl)}_{(mn)} &= \delta_{jk} \delta_{im} \delta_{ln} - \delta_{li} \delta_{km} \delta_{jn}. \end{aligned}$$

We want to compute the structure constants for the Cartan–Weyl basis $\{H^i\} \cup \{E^\alpha\}$.

Proposition 6.3.8

In the Cartan–Weyl basis the structure constants are

$$\begin{aligned} f^{ij}_k &= f^{ij}_\alpha = 0, & f^{i\alpha}_\beta &= \alpha^i \delta_{\alpha\beta} \quad \text{with } \alpha^i := \alpha(H^i), \\ f^{\alpha\beta}_i &= \tilde{\alpha}_i \delta_{\alpha, -\beta}, & f^{\alpha\beta}_\gamma &= \begin{cases} e_{\alpha, \beta} \delta_{\alpha+\beta, \gamma} & \text{if } \alpha + \beta \in \Phi, \\ 0 & \text{else.} \end{cases} \end{aligned}$$

Proof:

Since the root spaces are one-dimensional, the non-trivial brackets are

$$\begin{aligned} [E^\alpha, E^\beta] &= e_{\alpha, \beta} E^{\alpha+\beta} && \text{for } \alpha + \beta \in \Phi, \quad e_{\alpha, \beta} \in \mathbb{C}, \\ [H^i, H^j] &= 0, \\ [H^i, E^\alpha] &= \alpha(H^i) E^\alpha = \alpha^i E^\alpha, \\ [E^\alpha, E^{-\alpha}] &= \kappa(E^\alpha, E^{-\alpha}) t_\alpha = \sum_i \tilde{\alpha}_i H^i && \text{(defining } \tilde{\alpha}_i), \\ [E^\alpha, E^\beta] &= 0 && \text{if } \alpha + \beta \neq 0 \text{ and } \alpha + \beta \notin \Phi. \end{aligned}$$

Reading off the coefficients of the basis elements yields the structure constants. ■

If we express the Killing form in this basis we find, using Exercise 1 on Sheet 10,

$$\kappa^{ab} = \begin{cases} \sum_{\alpha \in \Phi} \alpha^i \alpha^j & \text{for } a = i, \quad b = j, \\ \left(2(\alpha, \tilde{\alpha}) + \sum_\gamma e_{\alpha, \gamma} e_{-\alpha, \alpha+\gamma} \right) \delta_{\alpha, -\beta} & \text{for } a = \alpha, \quad b = \beta, \\ 0 & \text{otherwise.} \end{cases} \quad (*)$$

Corollary 6.3.9

The inner product induced on $\mathfrak{g}_0^*$ is, for $\lambda, \mu \in \mathfrak{g}_0^*$,

$$(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \sum_{\alpha \in \Phi} (\alpha, \lambda) (\alpha, \mu).$$

In particular,

$$(\lambda, \lambda) = \sum_{\alpha \in \Phi} (\alpha, \lambda)^2 = \sum_{\alpha \in \Phi} \alpha(t_\lambda)^2.$$

Proof:

Writing $t_\lambda = \sum_i \tilde{\lambda}_i H^i$ and using the Cartan block of $(*)$,

$$(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \sum_{\alpha \in \Phi} \sum_{i,j} \tilde{\lambda}_i \alpha^i \alpha^j \tilde{\mu}_j = \sum_{\alpha \in \Phi} \alpha(t_\lambda) \alpha(t_\mu) = \sum_{\alpha \in \Phi} (\alpha, \lambda) (\alpha, \mu),$$

where $\sum_i \tilde{\lambda}_i \alpha^i = \sum_i \tilde{\lambda}_i \alpha(H^i) = \alpha(t_\lambda) = (\alpha, \lambda)$. ■

6.4 Simple roots, Dynkin diagrams and the classification

Recall that there exists a basis $\{H^i\}$ of $\mathfrak{g}_0$ such that $\alpha(H^i) = \alpha^i \in \mathbb{Z}$. By the Cartan block of $(*)$,

$$\kappa(H^i, H^i) = \sum_{\alpha \in \Phi} (\alpha^i)^2 > 0.$$

Hence, on the real dual vector space spanned by the roots of $\mathfrak{g}$, the pairing $(\cdot, \cdot)$ is positive definite.

Definition 6.4.1: Root space

The real vector space spanned by the roots of $\mathfrak{g}$, on which the induced pairing $(\cdot, \cdot)$ is positive definite, is called the **root space** of $\mathfrak{g}$.

Remark.

The term “root space” is used for two different objects, which should not be confused:

1. the 1-dimensional eigenspaces $\mathfrak{g}_\alpha$ appearing in the root space decomposition $\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$;
2. the real vector space just defined, on which $(\cdot, \cdot)$ is positive definite. It is the dual of the real span of the basis $\{H^i\}$ of $\mathfrak{g}_0$, and it is the space in which the roots and their diagrams live.

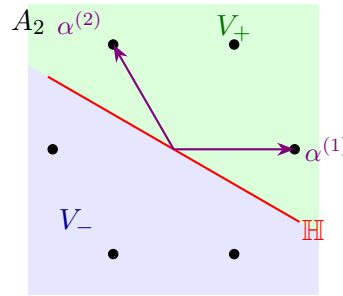
Definition 6.4.2: Positive and negative roots

Let $\Phi \subseteq \mathfrak{g}_0^*$ be a root system associated to a semi-simple Lie algebra, and let $\mathbb{H}$ be a hyperplane in the root space which does not contain any roots. It divides the root space into two halves V_+ and V_- . The roots in V_+ are called **positive roots** (with respect to the choice $(\mathbb{H}, V_+)$) and denoted by Φ_+ ; the roots in V_- are called **negative roots** and denoted by Φ_- . In particular,

$$\alpha \in \Phi_+ \implies -\alpha \in \Phi_-.$$

Example.

For A_2 , a hyperplane $\mathbb{H}$ avoiding all roots splits the root space into V_+ and V_- . The two positive roots adjacent to $\mathbb{H}$ are the simple roots $\alpha^{(1)}, \alpha^{(2)}$; the third positive root is their sum.



We can take as the functional ℓ (from the previous chapter) the distance to $\mathbb{H}$.

Definition 6.4.3: Simple root

A **simple root** is a positive root which cannot be written as a linear combination of other positive roots with positive coefficients.

Remark.**Facts about simple roots.**

- They are the positive roots which are closest to $\mathbb{H}$.
- The elements $H^{\alpha^{(i)}}$ for the simple roots generate $\mathfrak{g}_0$.
- There are $r = \text{rank}(\mathfrak{g}) = \dim(\mathfrak{g}_0^*)$ simple roots.
- We denote by Φ_S the set of simple roots.

Definition 6.4.4: Cartan matrix

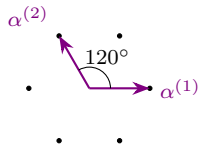
Let $\Phi_S = (\alpha^{(1)}, \dots, \alpha^{(r)})$ be the set of simple roots of a semi-simple Lie algebra $\mathfrak{g}$. The **Cartan matrix** of $\mathfrak{g}$ is the $r \times r$ matrix

$$A^{ij} := 2 \frac{(\alpha^{(i)}, \alpha^{(j)})}{(\alpha^{(j)}, \alpha^{(j)})},$$

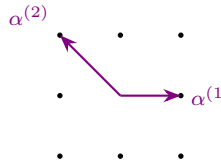
where $(\alpha^{(i)}, \alpha^{(j)}) = \cos(\angle(\alpha^{(i)}, \alpha^{(j)})) |\alpha^{(i)}| |\alpha^{(j)}|$.

Example.

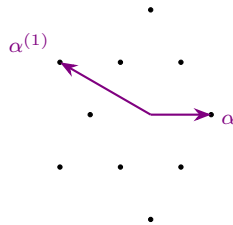
The Cartan matrices of the three rank-2 simple Lie algebras. The simple roots are the violet arrows; the angle between them and their relative lengths determine the matrix.



$$A_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$



$$A_{B_2} = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$



$$A_{\mathfrak{g}_2} = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

We can represent the data of the Cartan matrix by a **Dynkin diagram**: we draw a node for each simple root,

$$\begin{array}{ccccccc} \bullet & & \bullet & & \dots & & \bullet \\ \alpha^{(1)} & & \alpha^{(2)} & & & & \alpha^{(r)} \end{array}$$

and we draw $A^{ij} A^{ji}$ (no sum!) many edges between $\alpha^{(i)}$ and $\alpha^{(j)}$. We indicate by $>$ which of the two roots is longer (the arrow points towards the shorter root).

Example.

The Dynkin diagrams of the three rank-2 simple Lie algebras:

$$A_2 : \bullet \text{---} \bullet \qquad B_2 : \bullet \rightrightarrows \bullet \qquad \mathfrak{g}_2 : \bullet \rightrightarrows \bullet$$

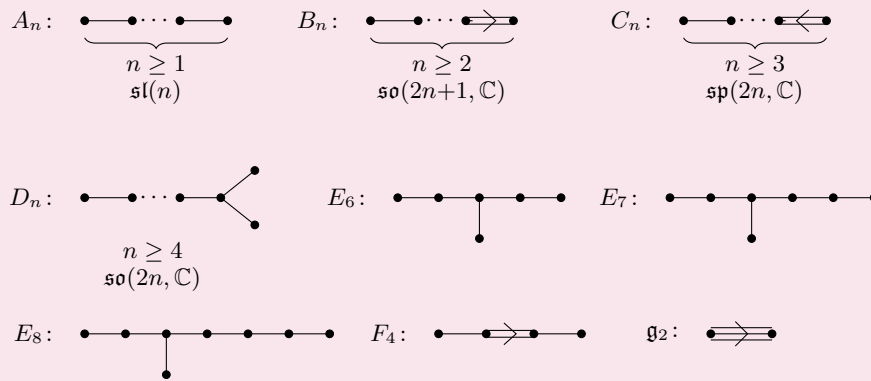
Remark.

Facts about the Cartan matrix A .

- (1) $A^{ii} = 2$.
- (2) $A^{ij} = 0 \iff A^{ji} = 0$.
- (3) $A^{ij} \in \{0, -1, -2, -3\}$ for $i \neq j$.
- (4) $n^{ij} := A^{ij} A^{ji} = \frac{4(\alpha^{(i)}, \alpha^{(j)})^2}{(\alpha^{(i)}, \alpha^{(i)})(\alpha^{(j)}, \alpha^{(j)})} \in \{0, 1, 2, 3\}$, corresponding respectively to the angles $90^\circ, 120^\circ, 135^\circ, 150^\circ$ between $\alpha^{(i)}$ and $\alpha^{(j)}$.

Theorem 6.4.5: Classification of simple Lie algebras

Simple Lie algebras are fully characterized by their root system, or equivalently their corresponding Dynkin diagram. The list of all allowed diagrams is:

**Proof:**

We will not prove this. For the proof, see (Bröckertom Dieck, Chapter 5). ■

6.5 Root strings and the Weyl group

However, we will explain how the Cartan matrix determines the Lie algebra. Given a collection of simple roots, all other roots are integer linear combinations of these.

Definition 6.5.1: Height of a root

Given a collection of simple roots, every root is an integer linear combination of them. The **height** of a root is the sum of the integer coefficients appearing in this combination.

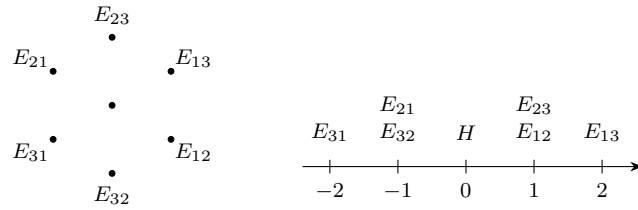
This induces a grading

$$\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_{(j)},$$

where $\mathfrak{g}_{(j)}$ is the space of elements of height j (in particular $\mathfrak{g}_{(0)} = \mathfrak{g}_0$ is the Cartan subalgebra).

Example.

For $\mathfrak{sl}(3)$, with simple roots E_{12} and E_{23} , the root space decomposition is graded by height:

**Remark.**

There is always a root of highest height, the **highest root** θ , with $(\theta, \theta) \geq (\alpha, \alpha)$ for all roots α . Sometimes one normalizes the inner product $(\cdot, \cdot)$ such that $(\theta, \theta) = 2$.

Recall that for each root α we get an $\mathfrak{sl}_\alpha(2)$ Lie subalgebra $(E^\alpha, E^{-\alpha}, H^\alpha)$. We can use $\mathfrak{sl}(2)$ representation theory to constrain and construct the whole of $\mathfrak{g}$. Let α, β be roots and consider the root string.

Definition 6.5.2: Root string

For roots α, β , the **root string** of β through α is

$$\bigoplus_{m \in \mathbb{Z}} \mathfrak{g}_{m\alpha + \beta}.$$

Proposition 6.5.3

The root string $\bigoplus_{m \in \mathbb{Z}} \mathfrak{g}_{m\alpha + \beta}$ is an irreducible $\mathfrak{sl}_\alpha(2)$ -module.

Proof:

It is an $\mathfrak{sl}_\alpha(2)$ -module by construction. Each summand $\mathfrak{g}_{m\alpha + \beta}$ is an H^α -weight space, all of distinct weight, and is either 1- or 0-dimensional. A module with distinct one-dimensional weight spaces is irreducible. ■

Lemma 6.5.4

For two simple roots $\alpha^{(i)} \neq \alpha^{(j)}$, the difference $\alpha^{(i)} - \alpha^{(j)}$ is never a root.

Proof:

If $\alpha^{(i)} - \alpha^{(j)}$ were a root, then either it or its negative is positive, and correspondingly

$$\alpha^{(i)} = \alpha^{(j)} + (\alpha^{(i)} - \alpha^{(j)}) \quad \text{or} \quad \alpha^{(j)} = \alpha^{(i)} - (\alpha^{(i)} - \alpha^{(j)})$$

would express a simple root as a sum of two positive roots, contradicting simplicity. ■

Hence $E^{\alpha^{(i)}}$ and $E^{\alpha^{(j)}}$ are lowest weight vectors for each other's $\mathfrak{sl}(2)$. We also know their weights from the Dynkin diagram:

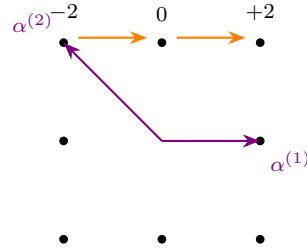
$$\alpha^{(i)}(H^{\alpha^{(j)}}) = 2 \frac{(\alpha^{(i)}, \alpha^{(j)})}{(\alpha^{(j)}, \alpha^{(j)})} = A^{ij},$$

which are negative integers. This allows us to construct the full root diagram from the simple

roots.

Example.

For B_2 , starting from the simple roots $\alpha^{(1)}$ (short) and $\alpha^{(2)}$ (long), the $\mathfrak{sl}_{\alpha^{(1)}}(2)$ -string through $\alpha^{(2)}$ generates the remaining positive roots; the labels are the $H^{\alpha^{(1)}}$ -weights.



Remark.

The root system is symmetric around the hyperplanes defined by $\beta(H^{\alpha^{(i)}}) = 0$ in $\mathfrak{g}_0^*$.

Definition 6.5.5: Weyl group

Let $\mathfrak{g}$ be a simple Lie algebra with simple roots $\alpha^{(i)}$. The **Weyl group** $W(\mathfrak{g})$ is the subgroup of the linear automorphisms of the root space generated by the reflections

$$s_{\alpha^{(i)}}(\beta) = \beta - 2 \frac{(\beta, \alpha^{(i)})}{(\alpha^{(i)}, \alpha^{(i)})} \alpha^{(i)}.$$

For each simple root $\alpha^{(i)}$ we obtain an $\mathfrak{sl}(2)$ -triple $E^{\pm\alpha^{(i)}}, H^{\alpha^{(i)}}$. From the representation theory of $\mathfrak{sl}(2)$ we then know the following relations:

- Serre relation

$$\underbrace{[E^{\pm\alpha^{(i)}}, [\dots, [E^{\pm\alpha^{(i)}}, E^{\alpha^{(j)}}] \dots]]}_{1-A^{ij}} = 0,$$

- $[H^{\alpha^{(i)}}, H^{\alpha^{(j)}}] = 0,$
- $[E^{\alpha^{(i)}}, E^{-\alpha^{(j)}}] = \delta_{ij} H^{\alpha^{(i)}},$
- $[H^{\alpha^{(i)}}, E^{\pm\alpha^{(j)}}] = \pm A^{ij} E^{\pm\alpha^{(j)}}.$

These are called the **Chevalley relations**.

Fact.

These are the only relations in $\mathfrak{g}$. Hence $\mathfrak{g}$ is generated by the $E^{\pm\alpha^{(i)}}, H^{\alpha^{(i)}}$ modulo the Chevalley relations.

Example.

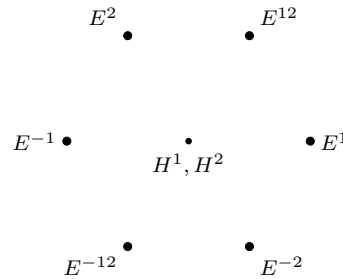
Consider A_2 with Cartan matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

and Chevalley generators $E^{\pm 1}, E^{\pm 2}, H^1, H^2$. The two remaining (non-simple) root vectors are obtained by bracketing,

$$E^{+12} := [E^1, E^2], \quad E^{-12} := [E^{-1}, E^{-2}],$$

corresponding to the roots $\pm(\alpha^{(1)} + \alpha^{(2)})$.



Every other bracket follows from the Jacobi identity and the Chevalley relations. For example,

$$\begin{aligned} [E^{-1}, E^{+12}] &= [E^{-1}, [E^1, E^2]] \\ &= -[E^2, [E^{-1}, E^1]] - [E^1, \underbrace{[E^2, E^{-1}]}_{=0}] \\ &= [E^2, H^1] \\ &= -A^{12} E^2 = E^2. \end{aligned}$$

Carrying this out for all pairs yields the full commutation table of the Cartan–Weyl basis (entries are [row, column]; the lower triangle follows by antisymmetry):

$[\cdot, \cdot]$	H^1	H^2	E^1	E^{-1}	E^2	E^{-2}	E^{12}	E^{-12}
H^1	0	0	$2E^1$	$-2E^{-1}$	$-E^2$	E^{-2}	E^{12}	$-E^{-12}$
H^2		0	$-E^1$	E^{-1}	$2E^2$	$-2E^{-2}$	E^{12}	$-E^{-12}$
E^1			0	H^1	E^{12}	0	0	E^{-2}
E^{-1}				0	0	E^{-12}	E^2	0
E^2					0	H^2	0	$-E^{-1}$
E^{-2}						0	$-E^1$	0
E^{12}							0	$-(H^1 + H^2)$
E^{-12}								0

6.6 Highest weight representations

Proposition 6.6.1

Let (V, ρ) be a representation of a simple Lie algebra $\mathfrak{g}$. Then there exists a decomposition

$$V = \bigoplus_{\alpha \in \mathfrak{g}_0^*} V_\alpha$$

such that

$$H(v_\alpha) = \alpha(H) v_\alpha \quad \forall H \in \mathfrak{g}_0 \text{ and } v_\alpha \in V_\alpha.$$

Proof:

The $H^{\alpha^{(i)}}$ form a basis of $\mathfrak{g}_0$. From the representation theory of $\mathfrak{sl}(2)_{\alpha^{(i)}} \subseteq \mathfrak{g}$ we know that each $H^{\alpha^{(i)}}$ is diagonalizable. As they mutually commute, they are simultaneously diagonalizable, which yields the claimed decomposition into common eigenspaces. ■

Definition 6.6.2: Weight and weight vector

The elements of V_α are said to have **weight** α , and the non-zero ones are called **weight vectors**.

Proposition 6.6.3

Let v_α be a weight vector of weight α and $X \in \mathfrak{g}_\beta$. Then the element Xv_α has weight $\alpha + \beta$.

Proof:

$$\begin{aligned} H(Xv_\alpha) &= [H, X] v_\alpha + X(Hv_\alpha) \\ &= (\alpha + \beta)(H) Xv_\alpha, \end{aligned}$$

using $[H, X] = \beta(H) X$ (as $X \in \mathfrak{g}_\beta$) and $Hv_\alpha = \alpha(H) v_\alpha$. ■

In particular every representation decomposes as $V = \bigoplus_\beta V_\beta$ over its weights β , and from the $\mathfrak{sl}(2)$ -theory the eigenvalues $\beta(H^{\alpha^{(i)}})$ are integers. Recalling

$$H^{\alpha^{(i)}} = \frac{2t_{\alpha^{(i)}}}{(\alpha^{(i)}, \alpha^{(i)})}, \quad \kappa(t_{\alpha^{(i)}}, -) = \alpha^{(i)}(-),$$

this integrality condition reads

$$\beta(H^{\alpha^{(i)}}) = \frac{2(\beta, \alpha^{(i)})}{(\alpha^{(i)}, \alpha^{(i)})} \in \mathbb{Z}.$$

Definition 6.6.4: Weight lattice

The set of weights β with $\beta(H^{\alpha^{(i)}}) \in \mathbb{Z}$ for all i forms a lattice, the **weight lattice** Λ_W of $\mathfrak{g}$.

Definition 6.6.5: Fundamental weights

The weight lattice is generated by the **fundamental weights** $w^{(i)}$, defined by

$$w^{(i)}(H^{\alpha^{(j)}}) = \delta_{ij}.$$

The simple roots are expressed through the fundamental weights by the Cartan matrix,

$$\alpha^{(i)} = \sum_j A^{ij} w^{(j)}.$$

To check this, note that $w^{(i)}(H^{\alpha^{(j)}}) = \delta_{ij}$ gives $(\alpha^{(k)}, w^{(j)}) = \delta_{jk} (\alpha^{(k)}, \alpha^{(k)})/2$, so

$$\left(\alpha^{(k)}, \sum_j A^{ij} w^{(j)} \right) = A^{ik} \frac{(\alpha^{(k)}, \alpha^{(k)})}{2} = (\alpha^{(k)}, \alpha^{(i)}).$$

Since $(\cdot, \cdot)$ is non-degenerate and the $\alpha^{(k)}$ form a basis, this forces $\sum_j A^{ij} w^{(j)} = \alpha^{(i)}$. Inverting the Cartan matrix yields

$$w^{(i)} = \sum_j (A^{-1})^{ij} \alpha^{(j)}.$$

Example.

For A_2 we have

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \implies A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

so that

$$w^{(1)} = \frac{1}{3}(2\alpha^{(1)} + \alpha^{(2)}), \quad w^{(2)} = \frac{1}{3}(\alpha^{(1)} + 2\alpha^{(2)}).$$

For B_2 we have

$$A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix},$$

so that

$$w^{(1)} = \alpha^{(1)} + \frac{1}{2}\alpha^{(2)}, \quad w^{(2)} = \alpha^{(1)} + \alpha^{(2)}.$$

Assume the hyperplane $\mathbb{H}$ does not intersect the weight lattice Λ_W . Then we can define $\ell(\beta)$ as the distance of a weight β to $\mathbb{H}$.

Definition 6.6.6: Highest weight vector

A **highest weight vector** $v \in V$ is a weight vector such that ℓ of its weight is maximal.

Theorem 6.6.7

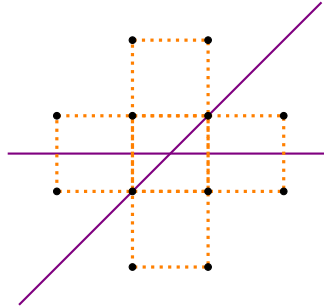
Let $\mathfrak{g}$ be a simple Lie algebra.

- The possible highest weights are the linear combinations of the fundamental weights $w^{(i)}$ with non-negative integer coefficients. These are called the **dominant weights**.
- For each dominant weight w there exists a unique irreducible finite-dimensional representation of $\mathfrak{g}$ with highest weight w .
- This representation is generated by acting with the negative root vectors on a highest weight vector.

Example.

We can again reconstruct the shape of the weight diagram using the Weyl group.

- For $B_2 = \mathfrak{so}(5)$ we get: $2w_1$ gives the adjoint representation, and $w_1 + w_2$ gives a representation of dimension 16, with weight diagram



It has dimension 16.

- For D_n , $\bullet \cdots \bullet \begin{smallmatrix} \nearrow \\ \searrow \end{smallmatrix}$, there are n fundamental weights corresponding to the n nodes. The two end points of the fork correspond to the two chiral spinor representations in $2n$ dimensions.
- For B_n , $\bullet \cdots \bullet \Rightarrow \bullet$, the representation corresponding to the last vertex is the spinor representation in $2n + 1$ dimensions.
- For A_n , the representations corresponding to the fundamental weights $w_1, \dots, w_n$ are

$$\mathbb{C}^{n+1}, \quad \Lambda^2 \mathbb{C}^{n+1}, \quad \Lambda^3 \mathbb{C}^{n+1}, \quad \dots, \quad \Lambda^n \mathbb{C}^{n+1}.$$

The corresponding weights are $w_i = L_1 + \cdots + L_i$, where

$$L_i(\text{diag}(a_{11}, \dots, a_{n+1, n+1})) = a_{ii}.$$

Hence the dominant weights are of the form $\sum_{i=1}^n \lambda_i L_i$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, which we represent in terms of Young diagrams:

$$\begin{array}{c} \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} \lambda_1 \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array} \lambda_2 \\ \vdots \\ \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \lambda_n \end{array}$$

For $\mathfrak{sl}(3)$ (A_2) we find

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} = \mathbf{3}, \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \bar{\mathbf{3}}, \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array} = \mathbf{8}.$$

(These are related to Schur functors.)

Definition 6.6.8: Quadratic Casimir operator

Let T^a be a basis of $\mathfrak{g}$ and T_a the dual basis with respect to the Killing form κ . Let (V, ρ) be a finite-dimensional representation of $\mathfrak{g}$. The endomorphism

$$C_2 = \sum_a \rho(T_a) \circ \rho(T^a) : V \rightarrow V$$

is called the **quadratic Casimir operator** for V .

In the Cartan–Weyl basis we have

$$C_2 = \sum_{i \in \Phi_S} t_{w(i)} H^{\alpha(i)} + \sum_{\alpha \in \Phi} \frac{(\alpha, \alpha)}{2} E^{-\alpha} E^{\alpha} = \sum_{i \in \Phi_S} t_{w(i)} H^{\alpha(i)} + \sum_{\alpha \in \Phi_+} (\alpha, \alpha) E^{-\alpha} E^{\alpha} + \sum_{\alpha \in \Phi_+} t_{\alpha},$$

where Φ_S denotes the simple roots, and we used

$$\kappa(E^{\alpha}, E^{-\alpha}) = \frac{2}{(\alpha, \alpha)}, \quad H^{\alpha} := \frac{2t_{\alpha}}{(\alpha, \alpha)}.$$

Example.

For $\mathfrak{sl}(2)$ we have

$$C_2 = \frac{1}{2} H^2 + EF + FE.$$

Fact.

- C_2 commutes with the action of $\mathfrak{g}$. Hence it acts by a scalar in each irreducible representation.
- Let V be a highest weight module of weight ω . We can compute this scalar by acting on a highest weight vector v :

$$C_2 v = \left(\underbrace{\sum_i \omega(t_{w(i)}) \omega(H^{\alpha(i)})}_{=(\omega, \omega)} + 0 + \sum_{\alpha \in \Phi_+} \omega(t_{\alpha}) \right) v = (\omega, \omega + 2\rho) v,$$

with $\rho = \frac{1}{2} \sum_{\alpha \in \Phi_+} \alpha$. The underbraced sum equals (ω, ω) because we sum over a dual basis, and the middle term vanishes since $E^{\alpha} v = 0$ for all positive roots α . *Up to normalization this is the angular momentum for $\mathfrak{sl}(2)$.*

We now explain how to compute the dimensions of the weight spaces. Let V be a representation of $\mathfrak{g}$, decomposed as $V = \bigoplus_{\mu} V_{\mu}$, and set

$$m_{\mu} := \dim V_{\mu} \quad \text{for all } \mu.$$

For each positive root α we have $[E^\alpha, E^{-\alpha}] = H^\alpha$. We start by computing

$$\begin{aligned} \operatorname{tr}_{V_\mu}(E^{-\alpha}E^\alpha) &= \underbrace{\operatorname{tr}_{V_\mu}(E^\alpha E^{-\alpha})}_{=\operatorname{tr}_{V_{\mu-\alpha}}(E^{-\alpha}E^\alpha)} - \underbrace{\operatorname{tr}_{V_\mu}(H^\alpha)}_{=\mu(H^\alpha)m_\mu}, \end{aligned}$$

so that

$$\operatorname{tr}_{V_{\mu-\alpha}}(E^{-\alpha}E^\alpha) = \operatorname{tr}_{V_\mu}(E^{-\alpha}E^\alpha) + \mu(H^\alpha)m_\mu.$$

Moving up until we reach the largest $\mu + j\alpha$ on which E^α acts by 0, we get

$$\operatorname{tr}_{V_\mu}(E^{-\alpha}E^\alpha) = \sum_{j \geq 1} (\mu + j\alpha)(H^\alpha)m_{\mu+j\alpha}.$$

Note that if we sum over all positive roots α and multiply by (α, α) , this is the off-diagonal part of C_2 . Since C_2 acts by $(\omega, \omega + 2\rho)$, we find

$$\sum_{\alpha \in \Phi_+} (\alpha, \alpha) \operatorname{tr}_{V_\mu}(E^{-\alpha}E^\alpha) + m_\mu(\mu, \mu + 2\rho) = (\omega, \omega + 2\rho)m_\mu,$$

and therefore, using $(\alpha, \alpha)(\mu + j\alpha)(H^\alpha) = 2(\mu + j\alpha, \alpha)$,

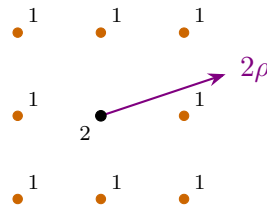
$$2 \sum_{\alpha \in \Phi_+} \sum_{j \geq 1} (\mu + j\alpha, \alpha)m_{\mu+j\alpha} = m_\mu[(\omega, \omega + 2\rho) - (\mu, \mu + 2\rho)].$$

This gives the **Freudenthal recursion formula**

$$m_\mu = \frac{2 \sum_{\alpha \in \Phi_+, j \geq 1} (\mu + j\alpha, \alpha)m_{\mu+j\alpha}}{(\omega, \omega + 2\rho) - (\mu, \mu + 2\rho)}$$

Example.

We apply this to the adjoint representation of $\mathfrak{so}(5)$, with highest weight $\omega = w_1 + w_2$. Its weight diagram consists of the roots (each with multiplicity 1) together with the zero weight (multiplicity 2):



Writing weights in the orthonormal basis e_1, e_2 , the highest root is $\omega = e_1 + e_2$ and $2\rho = 3e_1 + e_2$, with positive roots $e_1, e_2, e_1 - e_2, e_1 + e_2$. We compute the multiplicity m_0 of the zero weight (only $j = 1$ contributes, with $m_\alpha = 1$):

$$m_0 = \frac{2 \sum_{\alpha \in \Phi_+} (\alpha, \alpha)m_\alpha}{(\omega, \omega + 2\rho)} = \frac{2(1 + 1 + 2 + 2)}{(e_1 + e_2, 4e_1 + 2e_2)} = \frac{2 \cdot 6}{6} = 2.$$

6.7 The Weyl character formula

The weight lattice Λ_W is an abelian group, so we may form its group algebra $\mathbb{C}[\Lambda_W]$. We write e^λ for the basis element corresponding to $\lambda \in \Lambda_W$ and think of these as formal exponentials,

$$e^\lambda \cdot e^{\lambda'} = e^{\lambda+\lambda'}, \quad e^\lambda e^{-\lambda} = 1.$$

Definition 6.7.1: Character

For a finite-dimensional representation $V = \bigoplus_\lambda V_\lambda$ we define the element

$$\chi_V := \sum_\lambda m_\lambda e^\lambda \in \mathbb{C}[\Lambda_W].$$

The corresponding function

$$\chi_V : \mathfrak{g}_0 \rightarrow \mathbb{C}, \quad H \mapsto \sum_\lambda m_\lambda e^{\lambda(H)},$$

is called the **character** of V . One can also regard χ_V as a function on $\mathfrak{g}_0^*$.

Example.

Let $V = V_k$ be a highest weight representation of $\mathfrak{sl}(2)$ of weight $k \in \mathbb{N}_{\geq 0}$. Its weights are $k, k-2, \dots, -k$, each with multiplicity 1, so by the geometric series

$$\chi_{V_k}(tH) = \sum_{i=0}^k e^{(k-2i)t} = \frac{e^{(k+1)t} - e^{-(k+1)t}}{e^t - e^{-t}} = \frac{\sinh((k+1)t)}{\sinh(t)}.$$

Remark.

- For a finite-dimensional representation $V = \bigoplus_{\lambda \in \mathfrak{g}_0^*} V_\lambda$ and $H \in \mathfrak{g}_0$ we have

$$\mathrm{tr}_V(\exp(\rho(H))) = \sum_\lambda m_\lambda \exp(\lambda(H)) = \chi_V(H).$$

- For any element $w \in W(\mathfrak{g})$ we have $m_\lambda = m_{w(\lambda)}$ (this follows again from the fact that $\mathfrak{sl}(2)$ -representations are symmetric).

Thinking of the characters as functions on $\mathfrak{g}_0^*$ via $\alpha \mapsto \sum_\lambda m_\lambda \exp((\lambda, \alpha))$, they are Weyl-invariant:

$$\chi_V(w(\alpha)) = \sum_\lambda m_\lambda \exp((\lambda, w(\alpha))) = \sum_\lambda m_{w^{-1}(\lambda)} \exp((w^{-1}(\lambda), \alpha)) = \chi_V(\alpha).$$

Definition 6.7.2: Sign of a Weyl group element

For an element $w \in W(\mathfrak{g})$ we define $\mathrm{sign}(w) \in \{\pm 1\}$ as $\det(w)$, where we view w as an orthogonal automorphism of the root space.

Theorem 6.7.3: Weyl character formula

For a highest weight module V_Λ with $\Lambda \in \Lambda_W$ we have

$$\chi_{V_\Lambda}(\mu) = \frac{\sum_{w \in W} \text{sign}(w) \exp((w(\Lambda + \rho), \mu))}{\sum_{w \in W} \text{sign}(w) \exp((w(\rho), \mu))},$$

where the denominator can be written as

$$\sum_{w \in W} \text{sign}(w) \exp((w(\rho), \mu)) = \prod_{\alpha \in \Phi_+} \left(\exp\left(\frac{1}{2}(\alpha, \mu)\right) - \exp\left(-\frac{1}{2}(\alpha, \mu)\right) \right).$$

Corollary 6.7.4

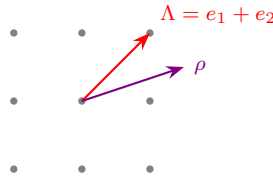
Weyl dimension formula. Setting $\mu = 0$,

$$\dim(V_\Lambda) = \chi_{V_\Lambda}(0) = \prod_{\alpha \in \Phi_+} \frac{(\Lambda + \rho, \alpha)}{(\rho, \alpha)}.$$

Example.

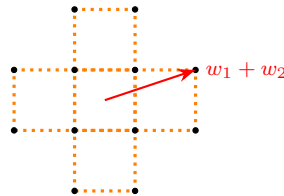
Weyl dimension formula for $\mathfrak{so}(5) = B_2$, with $\rho = \frac{3}{2}e_1 + \frac{1}{2}e_2$ and positive roots $e_1, e_2, e_1 + e_2$.

- **Adjoint representation.** Here $\Lambda = e_1 + e_2$, so $\Lambda + \rho = \frac{5}{2}e_1 + \frac{3}{2}e_2$, and



$$\dim = \prod_{\alpha \in \Phi_+} \frac{(\frac{5}{2}e_1 + \frac{3}{2}e_2, \alpha)}{(\frac{3}{2}e_1 + \frac{1}{2}e_2, \alpha)} = \frac{5}{3} \cdot \frac{3}{2} \cdot \frac{2}{e_1 + e_2} \cdot \frac{1}{e_1 - e_2} = 10.$$

- **The representation with highest weight $w_1 + w_2 = \frac{3}{2}e_1 + \frac{1}{2}e_2$.** Here $\Lambda + \rho = 3e_1 + e_2$, and



$$\dim = \prod_{\alpha \in \Phi_+} \frac{(3e_1 + e_2, \alpha)}{(\frac{3}{2}e_1 + \frac{1}{2}e_2, \alpha)} = 2 \cdot 2 \cdot 2 \cdot 2 = 16.$$

Fact.

Some facts about characters:

- $\chi_{V \oplus W} = \chi_V + \chi_W$,
- $\chi_{V \otimes W} = \chi_V \cdot \chi_W$,
- they determine the representation uniquely.

Chapter 7

Representations of compact Lie groups

For finite groups, we defined an inner product of maps:

$$\langle F, H \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{F(g)} H(g).$$

For Lie groups the sum is replaced by an integral, $\sum \rightarrow \int$. This does not converge in general, but for compact Lie groups we obtain a well-defined measure:

$$\langle F, H \rangle = \int_G \overline{F(g)} H(g) d\mu.$$

7.1 Integration on manifolds

Definition 7.1.1: Metric

A **metric** on M is a family of non-degenerate inner products

$$g_m : T_m M \otimes T_m M \rightarrow \mathbb{R}$$

for $m \in M$, depending smoothly on $m \in M$. Smooth in this context means in local coordinates

$$g(x) \left(v^i \frac{\partial}{\partial x^i}, w^j \frac{\partial}{\partial x^j} \right) = \sum_{i,j} g(x)_{ij} v^i w^j$$

with the $g(x)_{ij}$ being smooth functions of x . A metric is called **Euclidean** if it is positive definite at each point.

An integral is defined by

$$\int_M f d\text{vol}(g).$$

In local coordinates,

$$\int_U f(x^i) \sqrt{\det(g(x^i))} \underbrace{dx^1 \cdots dx^n}_{\text{Lebesgue measure on } U \subseteq \mathbb{R}^n}.$$

To extend this to all of M we need to cover M . There are two options:

- Cover M with a collection of charts and use a partition of unity. Let $\{U_i\}_{i \in I}$ be a cover. A **partition of unity** is a collection of functions $f^i : M \rightarrow \mathbb{R}$ such that each f^i is 0 outside U^i and $\sum_i f^i = 1$. Then

$$\int_M f \, d\text{vol}(g) = \sum_i \int f^i f \sqrt{\det(g(x^i))} \, dx^1 \cdots dx^n.$$

This makes sure to eliminate over-counting.

- More practical: find charts such that the complement is of lower dimension, hence of measure 0.

Example.

- The circle S^1 : $\bigcirc_X$ removing a point gives a chart $S^1 \setminus \{X\} \xrightarrow[\theta]{} (0, 2\pi) \subset \mathbb{R}$. Since X is lower-dimensional, it does not matter whether we include it or not, and

$$\int_{S^1} f \, d\text{vol}(g) = \int_0^{2\pi} f(\theta) \sqrt{g(\theta)} \, d\theta.$$

- The three-sphere $S^3 \cong \text{SU}(2)$:

$$\int_{S^3} f \, d\text{vol}(g) = \int_0^{2\pi} \int_0^\pi \int_0^\pi f(\theta, \phi, \chi) \sqrt{\det(g(\theta, \phi, \chi))} \, d\theta \, d\phi \, d\chi.$$

This is for a general manifold. But we need metrics that are compatible with the group structure.

Definition 7.1.2: Left- and right-invariant metric

Let G be a Lie group and m a metric on G . We say m is **left invariant** if

$$m_h(V_h, V'_h) = m_{g \cdot h}((l_g)_* V_h, (l_g)_* V'_h),$$

where $(l_g)_*$ is the pushforward (differential) of left multiplication $l_g : x \mapsto g \cdot x$. It is **right invariant** if the same holds for right multiplication.

Equivalently, for a left-invariant metric the integral is left-invariant:

$$\int_G l_g^* f \, d\text{vol}(m) = \int_G f \, d\text{vol}(m),$$

where $l_g^* f = f \circ l_g$ is the pullback along left multiplication. For a left-invariant metric we furthermore have

$$m_g(V_g, V'_g) = m_e((l_{g^{-1}})_* V_g, (l_{g^{-1}})_* V'_g), \quad (*)$$

so a left-invariant metric is determined by its value at the identity,

$$m_e : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathbb{R}.$$

On the other hand, $(*)$ can be used to define a left-invariant metric from an inner product on $\mathfrak{g}$.

If G is compact, then

$$\int_G 1 \, d\text{vol}(m) < \infty,$$

and hence we can normalize m such that

$$\int_G 1 \, d\text{vol}(m) = 1.$$

Theorem 7.1.3: Haar measure

There exists a unique left-invariant (normalized) measure on G . It is called the **Haar measure** $d\mu$. For compact G it is also right invariant.

Remark.

A general topological space is compact if for every cover $\{U_i\}_{i \in I}$ there exists a finite subcover. For submanifolds of $\mathbb{R}^n$ this is equivalent to being closed and bounded.

Example.

- $U(1) = \{e^{i\phi} \in \mathbb{C} \mid \phi \in [0, 2\pi)\}$, with Haar measure

$$d\mu = \frac{1}{2\pi} d\phi.$$

- Via the exponential map $\exp : \mathfrak{g} \rightarrow G$: for $SU(2)$,

$$g(\phi, \theta, \psi) := \exp\left(\frac{i\phi}{2\pi} \sigma_3\right) \exp\left(\frac{i\theta}{2\pi} \sigma_2\right) \exp\left(\frac{i\psi}{2\pi} \sigma_1\right),$$

with $\phi \in [0, 2\pi)$, $\theta \in [0, \pi)$, $\psi \in [0, 4\pi)$, and Haar measure

$$d\mu = \frac{1}{16\pi^2} \sin(\theta) d\phi d\theta d\psi.$$

- $GL(n)$ (non-compact), with left-invariant measure

$$d\mu = |\det \gamma|^{-n} \prod_{i,j} dE_{ij}.$$

Theorem 7.1.4: Weyl's unitarity trick

Let G be a compact Lie group and (V, ρ) a finite-dimensional complex representation.

- V admits a G -invariant scalar product.
- V is completely reducible, i.e. for a subrepresentation $V_1 \subseteq V$ there exists $V_2 \subseteq V$ such that $V_1 \oplus V_2 \cong V$ as G -representations.

Proof:

Let $\langle -, - \rangle : \overline{V} \otimes V \rightarrow \mathbb{C}$ be a scalar product on V . Define

$$\langle v, w \rangle_G := \int_G \langle \rho(g)v, \rho(g)w \rangle d\mu.$$

This is G -invariant. For a subrepresentation $V_1 \subseteq V$ we set $V_2 = V_1^\perp$ (the orthogonal complement with respect to $\langle -, - \rangle_G$), so that

$$V_1 \oplus V_2 = V.$$

(See Maschke's theorem, 3.2.12.)

Remark.

- Every $SU(2)$ representation can be made unitary. Hence the corresponding $\mathfrak{sl}(2)$ -representation is unitary, so H acts by a self-adjoint operator and is therefore diagonalizable in such a finite-dimensional representation.
- Each simple Lie algebra $\mathfrak{g}$ has a real form such that the corresponding Lie group is compact; it is characterized by the Killing form $\kappa(-, -)$ being negative definite on it. Hence each finite-dimensional representation of $\mathfrak{g}$ is completely reducible.

Definition 7.1.5: Character

Let (V, ρ) be a representation of a compact Lie group G . The **character** of V is the function

$$\chi_V : G \rightarrow \mathbb{C}, \quad g \mapsto \text{tr}_V(\rho(g)).$$

Remark.

The characters satisfy:

- $\chi_V(hgh^{-1}) = \chi_V(g)$, hence χ_V is a class function:

$$\chi_V \in \text{Cl}(G) := \{f \in L^2(G) \mid f(hgh^{-1}) = f(g)\}.$$

- A finite-dimensional representation is fully determined by its character.
- $\chi_{V \otimes W}(g) = \chi_V(g) \chi_W(g)$.
- $\chi_{V \oplus W}(g) = \chi_V(g) + \chi_W(g)$.

Example.

- $U(1)$. Since $U(1)$ is abelian, its irreducible representations are 1-dimensional, and they correspond to Lie group homomorphisms $U(1) \rightarrow U(1)$. What are all smooth such homomorphisms? By Lie's first theorem they are completely characterized by Lie algebra maps $\mathbb{R} \xrightarrow{a} \mathbb{R}$; but $U(1)$ is not simply connected, so we must ask whether the map descends:

$$\begin{array}{ccc}
 \mathbb{R} & & \\
 \downarrow & \searrow & \\
 \mathrm{U}(1) & \dashrightarrow & \mathrm{U}(1)
 \end{array}$$

Here the vertical arrow is the covering map $x \mapsto e^{ix}$, the diagonal arrow is $x \mapsto \exp(ixa)$, and the dashed arrow is the sought homomorphism $\mathrm{U}(1) \rightarrow \mathrm{U}(1)$. Such a homomorphism exists if and only if

$$a \cdot (x + 2\pi) = a \cdot x \bmod 2\pi \iff a \in \mathbb{Z}.$$

Hence all irreducible $\mathrm{U}(1)$ -representations are labelled by $\mathbb{Z}$, with characters

$$\chi_a(e^{ix}) = e^{ixa}.$$

- $\mathrm{SU}(2)$. Representations are labelled by a non-negative integer k (the highest weight). The conjugacy classes are represented by

$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} = \exp(i\theta H), \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \theta \in [0, \pi).$$

From the earlier $\mathfrak{sl}(2)$ computation,

$$\chi_{V_k}(\theta) = \frac{\sin((k+1)\theta)}{\sin(\theta)}.$$

For a class function $f \in \mathrm{Cl}(\mathrm{SU}(2))$, integration reduces to (a special case of the *Weyl integration formula*)

$$\int_{\mathrm{SU}(2)} f \, d\mu = \frac{2}{\pi} \int_0^\pi f(\theta) \sin^2(\theta) \, d\theta.$$

In particular the characters are orthonormal:

$$\langle \chi_{V_k}, \chi_{V_{k'}} \rangle = \int_{g \in \mathrm{SU}(2)} \overline{\chi_{V_k}} \chi_{V_{k'}}(g) \, d\mu = \frac{2}{\pi} \int_0^\pi \sin((k+1)\theta) \sin((k'+1)\theta) \, d\theta = \delta_{k,k'}.$$

Theorem 7.1.6

Let G be a compact Lie group. The characters χ_V for all irreducible finite-dimensional representations of G form an orthonormal *Hilbert basis* with respect to

$$\langle F, H \rangle = \int_G \overline{F(g)} H(g) \, d\mu.$$

(Without proof.)

Remark.

Hilbert basis: it is not necessary that an element is a finite sum of basis elements, but there always exists a convergent expansion in the basis.

Example.

For $\mathrm{SU}(2)$, $V_2 \otimes V_1 \cong V_3 \oplus V_1$. We showed this before using the weights; now again with characters. We have

$$\chi_{V_2 \otimes V_1}(\theta) = \frac{\sin(3\theta) \sin(2\theta)}{\sin(\theta)^2},$$

and hence, using $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ and $\sin(3\theta) \cos(\theta) = \frac{1}{2}(\sin(4\theta) + \sin(2\theta))$,

$$\begin{aligned} \langle \chi_{V_k}, \chi_{V_2 \otimes V_1} \rangle &= \frac{2}{\pi} \int_0^\pi \frac{\sin((k+1)\theta) \sin(3\theta) \sin(2\theta)}{\sin(\theta)} d\theta \\ &= \frac{4}{\pi} \int_0^\pi \sin((k+1)\theta) \sin(3\theta) \cos(\theta) d\theta \\ &= \frac{2}{\pi} \int_0^\pi \left(\sin((k+1)\theta) \sin(4\theta) + \sin((k+1)\theta) \sin(2\theta) \right) d\theta \\ &= \delta_{3,k} + \delta_{1,k}. \end{aligned}$$

This confirms $V_2 \otimes V_1 \cong V_3 \oplus V_1$.

A torus can be written as $\mathbb{T}^n = \mathrm{U}(1)^{\times n}$.

Definition 7.1.7: Maximal torus

For a simple Lie group G , a **maximal torus** is an abelian subgroup $H \leq G$ which is isomorphic to $\mathbb{T}^n$ and maximal with this property. The corresponding $\mathfrak{h} \subseteq \mathfrak{g}$ is a Cartan subalgebra (after complexification $\otimes \mathbb{C}$).

Remark.

The space of conjugacy classes of G is isomorphic to H/W , the maximal torus modulo the Weyl group. For $\mathrm{SU}(2)$ we found $\mathrm{U}(1)/\mathbb{Z}_2$.

Example.

(*Fourier transform.*) A class function on $\mathrm{U}(1)$ is a function $f : \mathrm{U}(1) \rightarrow \mathbb{C}$, equivalently a function $f : \mathbb{R} \rightarrow \mathbb{C}$ with $f(x + 2\pi) = f(x)$. With

$$\chi_k(x) = e^{ikx},$$

we recover the Fourier series

$$f(x) = \sum_{k \in \mathbb{Z}} \langle \chi_k, f \rangle e^{ikx}, \quad \langle \chi_k, f \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{-ikx} f(x) dx.$$

What about $L^2(G)$? Yes, this leads to the *non-abelian Fourier transform*. For each irreducible representation (V, ρ) of G we fix a basis $e_1, \dots, e_n$ with $n = \dim V$. In this basis

$$\rho(g) = \{\rho(g)_{ij}\}_{i,j=1,\dots,n}.$$

Definition 7.1.8: Matrix coefficient functions

The functions $\rho(g)_{ij} : G \rightarrow \mathbb{C}$ are called **matrix coefficient functions**.

Theorem 7.1.9: Peter–Weyl theorem

The matrix coefficient functions, multiplied by $\frac{1}{\sqrt{n}}$, form an orthonormal basis of $L^2(G)$.

Remark.

There is a map

$$\begin{aligned} V^* \otimes V &\rightarrow L^2(G), \\ f \otimes v &\mapsto (g \mapsto f(\rho^V(g) v)). \end{aligned}$$

In this formulation, the Peter–Weyl theorem says that

$$\bigoplus_{V \text{ irrep.}} V^* \otimes V \rightarrow L^2(G)$$

is a Hilbert space isomorphism. Explicitly, any (square-integrable) function $f : G \rightarrow \mathbb{C}$ can be written as

$$f(g) = \sum_V \sum_{i,j=1}^{\dim(V)} \hat{f}(V, i, j) \rho_{ij}^V(g),$$

with

$$\hat{f}(V, i, j) = \int_{g \in G} \dim(V)^{-\frac{1}{2}} \overline{\rho_{ij}^V(g)} f(g) d\mu.$$

Remark.

This is useful for solving PDEs, e.g. the heat equation. We pick a bi-invariant metric on G (e.g. $-\kappa$ extended to G) and form the Laplacian

$$\Delta f(x) = \underbrace{m^{ij} \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j}}_{\Delta} f(x), \quad \Delta : L^2(G) \rightarrow L^2(G), \quad \Delta u = -\operatorname{div}(\operatorname{grad}(u)).$$

The heat equation reads

$$\frac{\partial}{\partial t} u(t, g) + \Delta u(t, g) = 0.$$

The action $G \curvearrowright L^2(G)$ differentiates to $\mathfrak{g} \curvearrowright L^2(G)$ (“ $X \triangleright f = \frac{\partial}{\partial x} f(g)$ ”), and Δ acts as the Casimir C_2 , which acts by multiplication. Hence

$$\partial_t u(t, g) = -C_2 u(t, g), \quad u : G \rightarrow \mathbb{C}.$$

On V_Λ we have $C_2 v = (\Lambda, \Lambda + 2\rho) v$ for $v \in V_\Lambda$, so the equation has solutions

$$u_{V_\Lambda}^{ij}(t, g) = e^{-t(\Lambda, \Lambda + 2\rho)} \rho_{ij}^{V_\Lambda}(g).$$

For an arbitrary initial condition $u_0(g) = \sum_{\Lambda, i, j} \tilde{u}_0(\Lambda, i, j) \rho_{ij}^{V_\Lambda}(g)$, the solution is

$$u(t, g) = \sum_{\Lambda, i, j} \tilde{u}_0(\Lambda, i, j) e^{-t(\Lambda, \Lambda + 2\rho)} \rho_{ij}^{V_\Lambda}(g).$$

Example.

2-dimensional Yang–Mills theory with gauge group G a compact Lie group, on a closed surface (for instance a genus-2 surface).

Canonical quantization. To each spatial slice S^1 we assign a Hilbert space. The classical configuration space is the moduli space of flat G -connections on S^1 modulo gauge transformations,

$$\mathrm{Bun}_G^\nabla(S^1)/\mathrm{Gauge} \cong G/G,$$

the adjoint quotient of G by conjugation. Hence

$$\mathcal{H} = "L^2(\mathcal{F}(S^1))" = L^2(G/G) \cong \mathrm{Cl}(G),$$

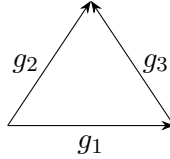
the space of class functions.

Hamiltonian. The Hamiltonian is proportional to the quadratic Casimir,

$$\mathbb{H} : \mathcal{H} \rightarrow \mathcal{H}, \quad \mathbb{H} = \frac{e^2}{2} C_2,$$

with e the coupling constant. By the character theory above it acts diagonally: on the irreducible character χ_{V_Λ} it acts by the scalar $\frac{e^2}{2}(\Lambda, \Lambda + 2\rho)$.

Discretization. Triangulate the surface and assign a group element $g_E \in G$ to each edge E . To each triangle Δ of area a , with boundary holonomy $g_2^{-1}g_3^{-1}g_1$,



we assign the heat kernel, and integrate over all edges with the Haar measure:

$$Z = \int_{G^{|E|}} \prod_{\Delta} \left(\sum_V \dim(V) e^{-C_2(V)a} \chi_V(g_2^{-1}g_3^{-1}g_1) \right) d\mu.$$

This is independent of the chosen triangulation, so the continuum limit is well defined.

(The remainder of this section situates the example in its topological context and goes slightly beyond the board sketch.)

Gluing the triangles over a closed orientable surface Σ_g of genus g and total area $A = \sum_{\Delta} a$, triangulation independence collapses the lattice integral to a single sum over irreducible representations,

$$Z(\Sigma_g) = \sum_V (\dim V)^{2-2g} e^{-\frac{e^2 A}{2} C_2(V)},$$

where $2 - 2g = \chi(\Sigma_g)$ is the Euler characteristic *Witten*. This is pure representation theory: the ingredients $\dim V$ and $C_2(V)$ are exactly the Peter–Weyl and character data of G . In the zero-area (topological) limit $A \rightarrow 0$ the exponential drops out and

$$Z(\Sigma_g) = \sum_V (\dim V)^{2-2g}$$

becomes a topological invariant of Σ_g — the partition function of a 2-dimensional topological quantum field theory.

Theorem 7.1.10: 2-dimensional TQFTs and Frobenius algebras

A 2-dimensional topological quantum field theory is a symmetric monoidal functor $Z : 2\text{Cob} \rightarrow \text{Vect}_{\mathbb{C}}$ out of the category 2Cob of 2-dimensional cobordisms. Such functors are equivalent to commutative Frobenius algebras: the object $Z(S^1)$ carries a natural commutative Frobenius algebra structure, and this assignment is an equivalence of categories *Kock*.

For 2-dimensional Yang–Mills the relevant Frobenius algebra is precisely $Z(S^1) = \text{Cl}(G)$, the class functions, with convolution as multiplication and the L^2 -pairing $\langle F, H \rangle = \int_G \overline{F(g)} H(g) d\mu$ as the Frobenius form. The genus- g invariant $\sum_V (\dim V)^{2-2g}$ is its associated cobordism invariant, and this is the point at which category theory — the category 2Cob — enters explicitly.

Remark.

Specializing G to a *finite* group reproduces (untwisted) Dijkgraaf–Witten theory with gauge group G : the same class-function / Frobenius-algebra structure now counts G -bundles on Σ_g weighted by their automorphisms. Its three-dimensional counterpart is governed by the quantum double (Drinfeld double) $D(G)$, whose braided category of representations has as simple objects the *anyons* underlying topological quantum computing *Kitaev*. This is where the categorical machinery is genuinely headed; see also *MüllerSchweigert* for a construction of three-dimensional abelian Dijkgraaf–Witten theories.

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Index

Definitions are collected automatically from the definition boxes.

A

abelian group	10
adjoint action	68
adjoint representation	
of a Lie algebra	81
affine group	53
affine transformation	6
alternating group	8
associative algebra	46
atlas	64
maximal	64

B

Baker-Campbell-Hausdorff formula	74
Brillouin zone	58

C

Cartan matrix	103
Cartan subalgebra	84, 97
Cartan-Weyl basis	97
Casimir operator	112
central extension	15
centralizer	20
Character	114, 120
character	36, 114, 120
class function	120
orthonormality	121
Weyl character formula	115
character table	36
chart	64
class function	36, 120
Clebsch-Gordan rule	32
completely reducible	32
complexification	77
conjugacy classes	9
conjugate elements	9
conjugation	9
continuous map	64
coset	12

cycle	10
cycle type	11

D

derivation	65
diffeomorphism	65
differential (of a smooth map)	67
direct sum representation	29
dominant weight	111
double cover	73
dual representation	29

E

exact sequence	14
exponential map	73
extension	14

F

Frobenius algebra	125
Frobenius formula	52
fundamental weight	110
Fundamental weights	110

G

group	2
group action	
left	18
right	18
group algebra	46
group homomorphism	4
group isomorphism	5

H

Haar measure	119
Hamiltonian	55
Hausdorff space	64
height of a root	105
highest weight map	87

Highest weight vector 110
 highest weight vector 110
 Hilbert basis 121
 Hom representation 29
 homeomorphism 64

I

ideal 95
 indecomposable representation 30
 inner product on Cartan dual 99
 intertwiner 23
 invariant subspace 37
 irreducible representation 25
 isomorphic groups 5
 isomorphism theorem 13

J

Jacobi identity 71

K

Killing form 98

L

lattice
 weight lattice 110
 Left- and right-invariant metric 118
 Lie algebra 71
 Lie algebra homomorphism 71
 Lie algebra representation 71
 Lie bracket 69
 Lie group 67
 Lie subalgebra 89
 Lie's theorems
 first 75
 second 76
 third 76

M

manifold 65
 Maschke's theorem 33
 matrix coefficient function 122
 Matrix coefficient functions 122
 matrix Lie group 63
 Maximal torus 122
 maximal torus 122
 measure
 Haar measure 119
 Metric 117
 metric 117
 Euclidean 117
 left invariant 118

right invariant 118

N

normal subgroup 12

O

one-parameter subgroup 73
 orbit 20
 orbit-stabilizer theorem 20
 order (of a group element) 21

P

partition 45
 partition of unity 118
 path-connected manifold 74
 Peter-Weyl theorem 123
 point group 8
 positive roots 102
 push-forward 67

Q

Quadratic Casimir operator 112
 quadratic Casimir operator 112
 quaternion
 unit 72
 quotient group 12

R

rank 97
 reciprocal lattice and Brillouin zone 58
 regular representation 24
 representation 3
 smooth 68
 root 86, 97
 negative 87, 103
 positive 87, 103
 root lattice 87
 root space 102
 root space decomposition 86, 97
 root string 106
 root system 97

S

Schur's lemma 26
 section (of an extension) 15
 semi-simple element 97
 semidirect product 16
 semisimple representations 31
 Sign of a Weyl group element 114
 simple Lie algebra 96

simple root 103
 simply connected manifold 75
 skew-Hermitian matrix 73
 smooth map 65
 spatial symmetries of the Hamiltonian 55
 $\text{Spin}(3)$ 73
 split extension 15
 stabilizer 20
 standard representation 80
 structure constants 100
 $\text{SU}(2)$ 72
 $\mathfrak{su}(2)$ 72
 subgroup 7
 subrepresentation 25
 symmetry 1

T

t-alpha 99
 tableau 46
 tangent space 66
 tensor product representation 29
 Lie algebra 81
 topological quantum field theory 125
 topological space 64
 torus
 maximal torus 122
 transition map 64
 trivial representation 24

U

unitarity trick 119
 unitary representation 32, 82

W

weight 109
 $\mathfrak{sl}(3)$ 85
 dominant weight 111
 fundamental weight 110
 highest weight 79, 110
 highest weight vector 79, 88
 weight lattice 110
 weight vector 109
 Weight and weight vector 109
 Weight lattice 109
 weight lattice 90
 weight space 85
 weight vector
 $\mathfrak{sl}(3)$ 85
 Weyl character formula 115
 Weyl dimension formula 115
 Weyl group 90, 107
 sign 114
 Weyl's unitarity trick 119

Y

Young diagram 45
 Young symmetrizer 47